

BASIC GENETIC MECHANISMS

DNA, Chromosomes, and Genomes

CHAPTER

4

Life depends on the ability of cells to store, retrieve, and translate the genetic instructions required to make and maintain a living organism. This *hereditary information* is passed on from a cell to its daughter cells at cell division and from one generation of an organism to the next through the organism's reproductive cells. The instructions are stored in every living cell's genes, the information-containing units in its genome that determine the characteristics of a species as a whole and of the individuals within it.

As soon as genetics emerged as a science at the beginning of the twentieth century, scientists became intrigued by the nature of the hereditary information. They knew that it was copied and transmitted from cell to daughter cell millions of times to produce the many generations of a multicellular organism, and that it survives the process essentially unchanged. What form of molecule could be capable of such accurate and almost unlimited replication and also be able to exert precise control, directing multicellular development as well as the daily life of every cell? What kind of instructions does the hereditary information contain? And how can the enormous amount of information required for the development and maintenance of an organism fit within the tiny space of a cell?

The answers to some of these questions began to emerge in the 1940s. At this time researchers discovered, from studies in simple fungi, that genes consist largely of instructions for making proteins. Proteins are phenomenally versatile macromolecules that perform most cell functions. As we saw in Chapter 3, they serve as building blocks for cell structures and form the enzymes that catalyze most of the cell's chemical reactions. They also regulate gene expression (Chapter 7), and they enable cells to communicate with each other (Chapter 15) and to move (Chapter 16). The properties and functions of cells and organisms are determined to a great extent by the proteins that they are able to make, and these are determined by their genes.

Painstaking observations of cells and embryos in the late nineteenth century set the stage for experiments that led to the realization that the hereditary information is carried on *chromosomes*—threadlike structures in the nucleus of a eukaryotic cell that become visible by light microscopy as the cell begins to

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Figure 4–1 Chromosomes in cells. (A) Two adjacent plant cells photographed through a light microscope. The DNA has been stained with a fluorescent dye (DAPI) that binds to it. The DNA is present in chromosomes, which become visible as distinct structures in the light microscope only when they become compact, sausage-shaped structures just before cell division. This can be easily seen in the cell on the *left*, where for clarity a single chromosome has been shaded *dark brown*. The cell on the *right* contains identical chromosomes, but they cannot be clearly distinguished at this phase in the cell's life cycle, because they are in a more extended conformation. (B) Schematic diagram of the outlines of the two cells along with their chromosomes. (A, courtesy of Peter Shaw.)

divide (**Figure 4–1**). Later, when biochemical analysis became possible, chromosomes were found to consist of deoxyribonucleic acid (DNA) and protein, with both being present in roughly the same amounts. For many decades, the DNA was thought to be merely a structural element. However, the other crucial advance made in the 1940s was the identification of DNA as the likely carrier of hereditary information. This breakthrough in our understanding of cells came from studies of inheritance in bacteria (**Figure 4–2**). But still, as the 1950s began, both how proteins could be specified by instructions in the DNA and how this information could be copied for transmission from cell to cell seemed completely mysterious. The puzzle was suddenly solved in 1953, when James Watson and Francis Crick derived the answers from their model for the structure of a DNA molecule. As outlined in Chapter 1, the determination of the double-helical structure of DNA immediately solved the problem of how the information in this molecule might be copied, or *replicated*. It also provided the first clues as to how a molecule of DNA might use the sequence of its subunits to encode the instructions for making proteins. Today, the fact that DNA is the genetic material is so fundamental to biological thought that it is difficult to appreciate the enormous intellectual gap that was filled by this sudden breakthrough discovery.

We begin this chapter by describing the structure of DNA. We see how, despite its chemical simplicity, the structure and chemical properties of DNA make it ideally suited as the raw material of genes. We then consider how the chromosomal proteins arrange and package this DNA. The packaging has to be done in an orderly fashion so that the chromosomes can be replicated and apportioned correctly between the two daughter cells at each cell division. And it must also allow access to chromosomal DNA, both for the enzymes that constantly repair DNA damage and for the specialized proteins that direct the expression of its many genes.

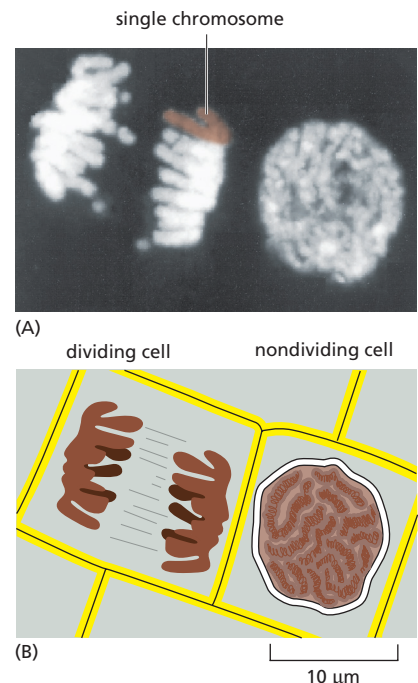
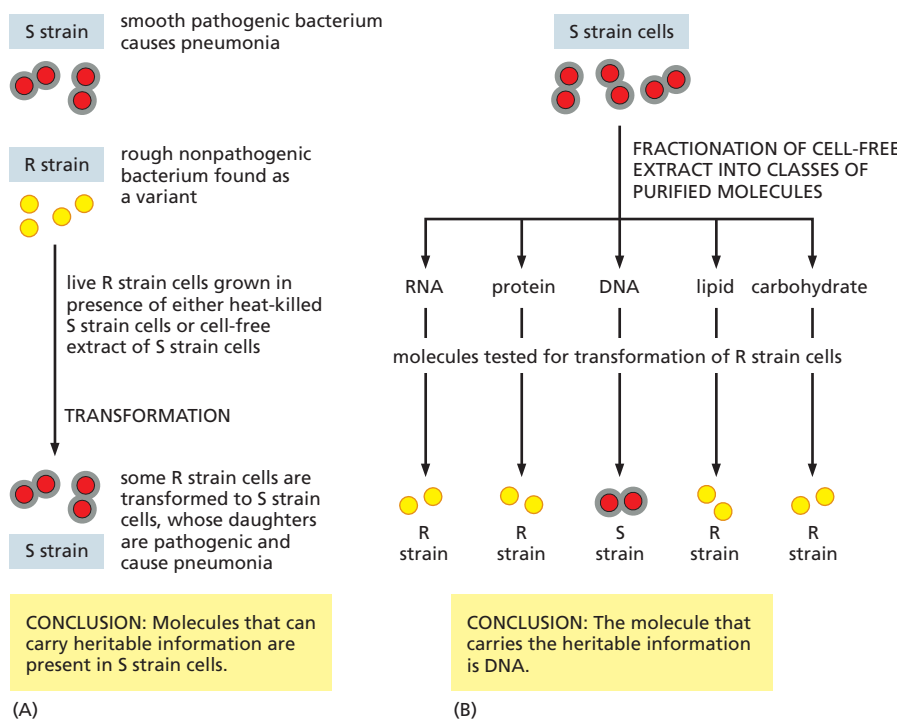


Figure 4–2 The first experimental demonstration that DNA is the genetic material. These experiments, carried out in the 1920s (A) and 1940s (B), eventually showed that adding purified DNA to a bacterium changed the bacterium's properties and that this change was faithfully passed on to subsequent generations. Two closely related strains of the bacterium *Streptococcus pneumoniae* differ from each other in both the appearance of their colonies grown on an agar surface and their pathogenicity. One strain appears smooth (S) and causes death when injected into mice, and the other appears rough (R) and is nonlethal. (A) An initial experiment shows that some substance present in the S strain can change (or transform) the R strain into the S strain and that this change is inherited by subsequent generations of bacteria. (B) This later experiment, in which the R strain has been incubated with various types of biological molecules that were purified from the S strain, identifies the active substance as DNA.



In the past few decades, there has been a revolution in our ability to determine the exact order of subunits in a DNA molecule, as described in Chapter 8. As a result, we now know the sequence of the entire *human genome*—the 3.1 billion nucleotide pairs that provide the information for producing a human adult from a fertilized egg. We also have the DNA sequences for many thousands of other organisms. Detailed analyses of these sequences are providing exciting insights into the process of genome evolution, and it is with this subject that the chapter ends.

This is the first of four chapters that deal with basic genetic mechanisms—the ways in which the cell maintains, replicates, and expresses the hereditary information carried in its DNA. In the next chapter (Chapter 5), we shall discuss the mechanisms by which the cell accurately replicates and repairs DNA; we also describe how DNA sequences can be rearranged through the process of genetic recombination. Gene expression—the process through which the information encoded in DNA is interpreted by the cell to guide the synthesis of proteins—is the main topic of Chapter 6. In Chapter 7, we describe how this gene expression is controlled by the cell to ensure that each of the many thousands of proteins and RNA molecules encrypted in its DNA is manufactured at the proper time and place in the life of the cell.

THE STRUCTURE AND FUNCTION OF DNA

Biologists in the 1940s had difficulty in conceiving how DNA could be the genetic material. The molecule seemed too simple: a long polymer composed of only four types of nucleotide subunits, which resemble one another chemically. In the early 1950s, DNA was examined by x-ray diffraction analysis, a technique for determining the three-dimensional atomic structure of a molecule (discussed in Chapter 8). The x-ray diffraction results indicated that DNA was composed of two strands of the polymer wound into a helix. The observation that DNA was double-stranded provided one of the major clues that led to the Watson-Crick model for DNA structure that, as soon as it was proposed in 1953, made DNA's potential for replication and information storage immediately apparent.

A DNA Molecule Consists of Two Complementary Chains of Nucleotides

A **deoxyribonucleic acid (DNA)** molecule consists of two long polynucleotide chains composed of four types of nucleotide subunits. Each of these chains is known as a *DNA strand*. The two strands run antiparallel to each other, and *hydrogen bonds* between the base portions of the nucleotides hold the two strands together (**Figure 4-3**). As we saw in Chapter 2 (Panel 2-6, pp. 104–105), a nucleotide is composed of a five-carbon sugar to which a phosphate group and a nitrogen-containing base are attached. In the case of the nucleotides in DNA, the sugar is deoxyribose attached to a phosphate group (hence the name deoxyribonucleic acid), and the base may be either *adenine* (A), *cytosine* (C), *guanine* (G), or *thymine* (T). The nucleotides are covalently linked together in a chain through the sugars and phosphates, which thus form a “backbone” of alternating sugar-phosphate-sugar-phosphate (**Figure 4-4**). Because only the base differs in each of the four types of nucleotide subunit, each polynucleotide chain in DNA is analogous to a sugar-phosphate necklace (the backbone), from which hang four types of beads (the bases A, C, G, and T). These same symbols (A, C, G, and T) are commonly used to denote either the four bases or the four entire nucleotides; that is, the bases with their attached sugar and phosphate groups.

The way in which the nucleotides are linked together gives a DNA strand a chemical polarity. If we think of each sugar as a block with a protruding knob (the 5' phosphate) on one side and a hole (the 3' hydroxyl) on the other (see **Figure 4-3**), each completed chain, formed by interlocking knobs with holes, will have all of its subunits lined up in the same orientation. Moreover, the two ends of the chain will be easily distinguishable, as one has a hole (the 3' hydroxyl) and

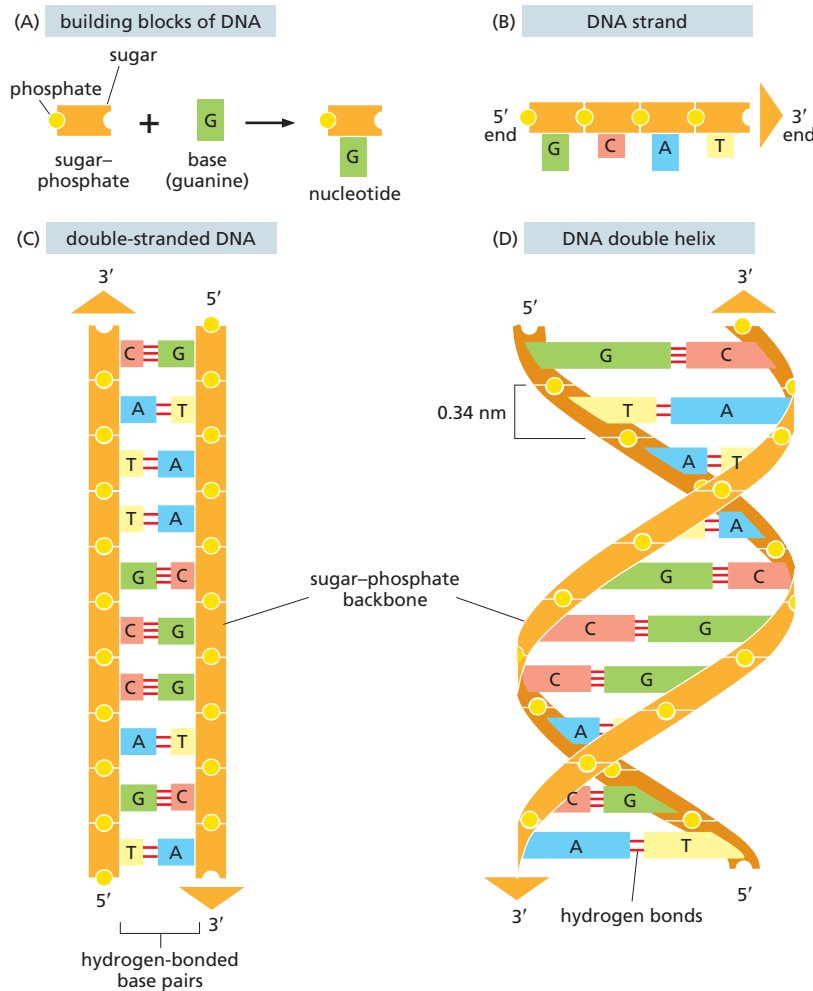


Figure 4-3 DNA and its four nucleotide building blocks. (A) Each nucleotide is composed of a sugar-phosphate covalently linked to a base—guanine (G) in this figure. (B) The nucleotides are covalently linked together into polynucleotide chains, with a sugar-phosphate backbone from which the bases—adenine, cytosine, guanine, and thymine (A, C, G, and T)—extend. (C) A DNA molecule is composed of two polynucleotide chains (complementary DNA strands) held together by hydrogen bonds between the paired bases—denoted here by either two (A-T) or three (G-C) red lines. The arrows on the DNA strands indicate the polarities of the two strands, which run antiparallel to each other (with opposite chemical polarities) in the DNA molecule. (D) Although the DNA molecule is shown straightened out in panel C, in reality it is wound into a double helix, as shown here. For details, see the figures that follow.

the other a knob (the 5' phosphate) at its terminus. This polarity of a DNA strand is indicated by referring to one end as the *3' end* (pronounced “3 prime end”) and the other as the *5' end* (pronounced “5 prime end”), names derived from the orientation of the deoxyribose sugar (see Figure 4-4). With respect to DNA's information-carrying capacity, the chain of nucleotides in a DNA strand, being both directional and linear, can be read in much the same way as the letters on this page.

The three-dimensional structure of DNA—the DNA **double helix**—arises from the chemical and structural features of its two polynucleotide chains. Because the DNA strands are held together by hydrogen-bonding between the bases on the two different strands, all the bases are on the inside of the double helix, and the sugar-phosphate backbones are on the outside (see Figure 4-3). In each case, a bulkier two-ring base (a purine; see Panel 2-6, pp. 104–105) is paired with a single-ring base (a pyrimidine): A always pairs with T, and G with C (**Figure 4-5A**). This *complementary base-pairing* enables the **base pairs** to be packed in the energetically most favorable arrangement in the interior of the double helix. In this arrangement, each base pair is of similar width, thus holding the sugar-phosphate backbones a constant distance apart along the DNA molecule. To maximize the efficiency of base-pair packing, the two sugar-phosphate backbones wind around each other to form a right-handed double helix, with one complete turn every 10.4 base pairs (**Figure 4-5B** and **Figure 4-6**).

The members of each base pair can fit together within the double helix only if the two strands of the helix are **antiparallel**; that is, only if the polarity of one strand is oriented opposite to that of the other strand (see Figures 4-3 and 4-5). As a consequence of DNA's structure and base-pairing requirements, each strand of a

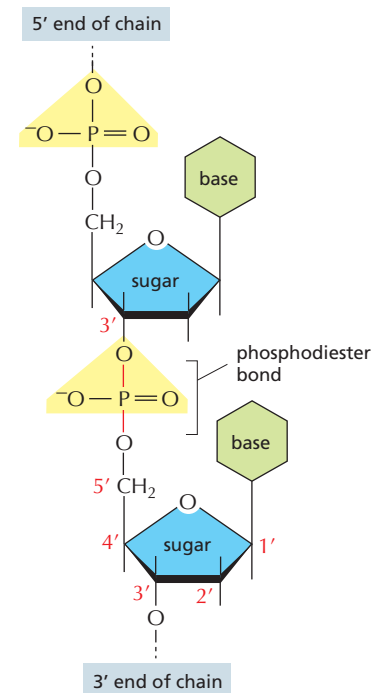


Figure 4-4 The nucleotide subunits within a DNA strand are held together by a phosphodiester bond. This bond connects one sugar to the next. The chemical differences in the ester linkages—between the 5' carbon of one sugar and the 3' carbon of the other—give rise to the polarity of the resulting DNA strand, as indicated. For simplicity, only two nucleotides are shown here.

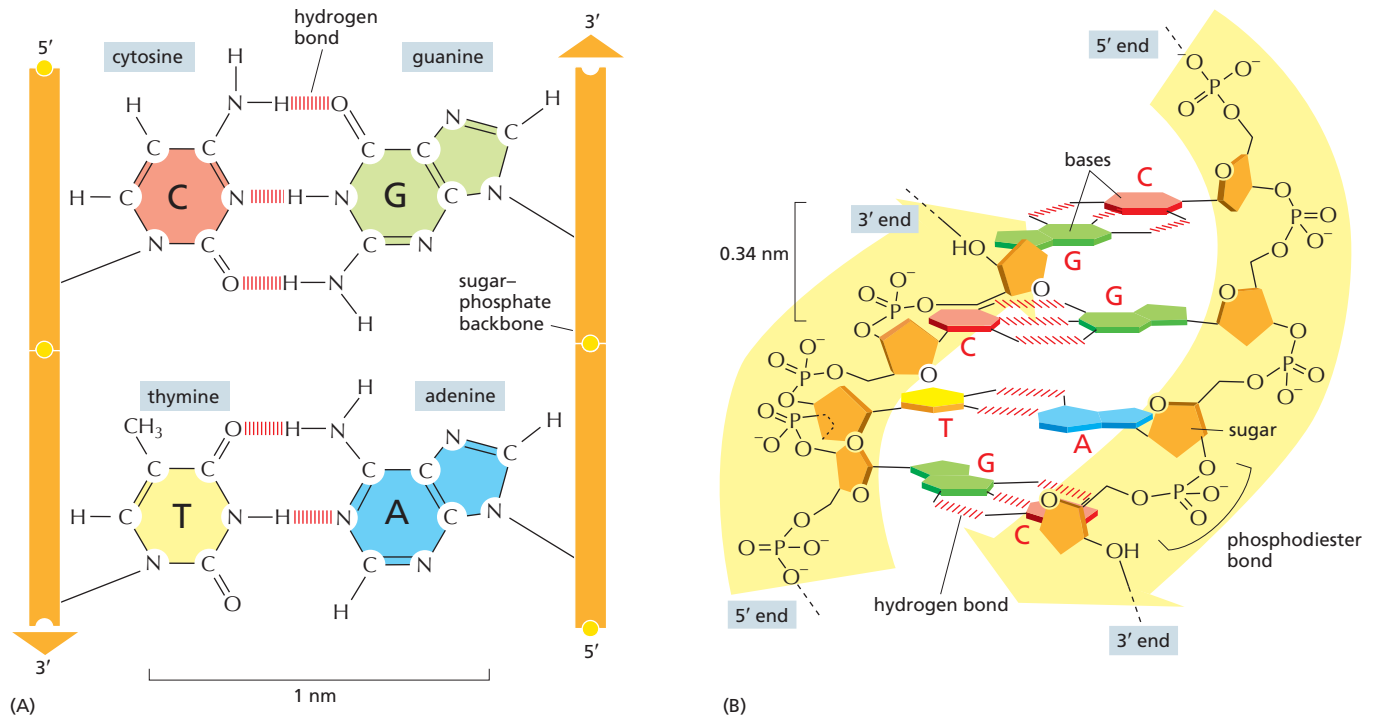


Figure 4-5 How the two strands of the DNA double helix are held together by hydrogen bonds between complementary base pairs.

(A) Schematic illustration showing how the shapes and chemical structures of the bases allow hydrogen bonds to form efficiently only between A and T and between G and C. As shown, two hydrogen bonds form between A and T, whereas three form between G and C. The bases can pair in this way only if the two polynucleotide chains that contain them are antiparallel; that is, oriented in opposite directions. (B) A short section of the double helix viewed from its side. Four base pairs are illustrated; note that they lie perpendicular to the axis of the helix, unlike those shown in the schematic of panel A. As shown in Figure 4-4, the nucleotides are linked together covalently by a *phosphodiester bond* that connects the 3'-hydroxyl ($-\text{OH}$) group of one sugar and the 5' phosphate ($-\text{PO}_3$) attached to the next (see Figure 4-4 to review how the carbon atoms in the sugar ring are numbered). This linkage gives each polynucleotide strand a chemical polarity; that is, its two ends are chemically different. The 3' end carries an unlinked $-\text{OH}$ group attached to the 3' position on the sugar ring; the 5' end carries a free phosphate group attached to the 5' position on the sugar ring.

DNA molecule contains a sequence of nucleotides that is exactly **complementary** to the nucleotide sequence of its partner strand.

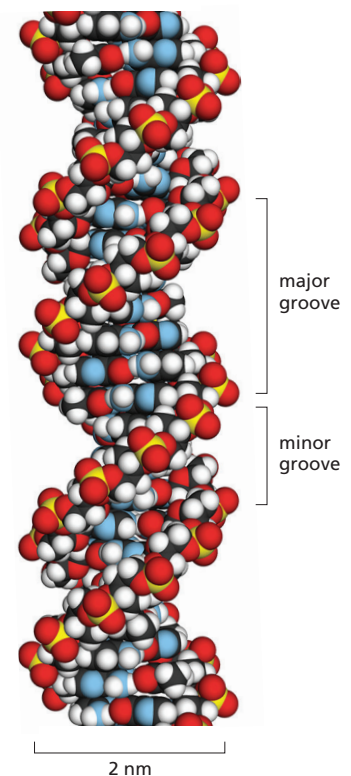
The Structure of DNA Provides a Mechanism for Heredity

The discovery of the structure of DNA immediately suggested answers to the two most fundamental questions about heredity. First, how could the information to specify an organism be carried in a chemical form? And second, how could this information be duplicated and copied from generation to generation?

The answer to the first question came from the realization that DNA is a linear polymer formed from four different kinds of monomers, strung out in a very long, defined sequence like the letters of a document written in an alphabetic script.

The answer to the second question came from the double-stranded nature of the structure: because each strand of DNA contains a sequence of nucleotides that is exactly complementary to the nucleotide sequence of its partner strand, each strand can act as a **template**, or mold, for the synthesis of a new complementary strand. In other words, if we designate the two DNA strands as S and S', strand S can serve as a template for making a new strand S', while strand S' can serve as a template for making a new strand S (**Figure 4-7**). Thus, the genetic

Figure 4-6 A space-filling model of the DNA double helix. The two strands wind around each other to form a right-handed helix (see Figure 3-23). Each turn of this helix contains 10.4 nucleotide pairs, and the center-to-center distance between adjacent nucleotide pairs is 0.34 nm. The coiling of the two strands around each other creates two grooves in the double helix: the wider groove is called the major groove, and the narrower groove is called the minor groove, as indicated. The colors of the atoms are N, blue; O, red; P, yellow; H, white; and C, black. (See **Movie 4.1**.)



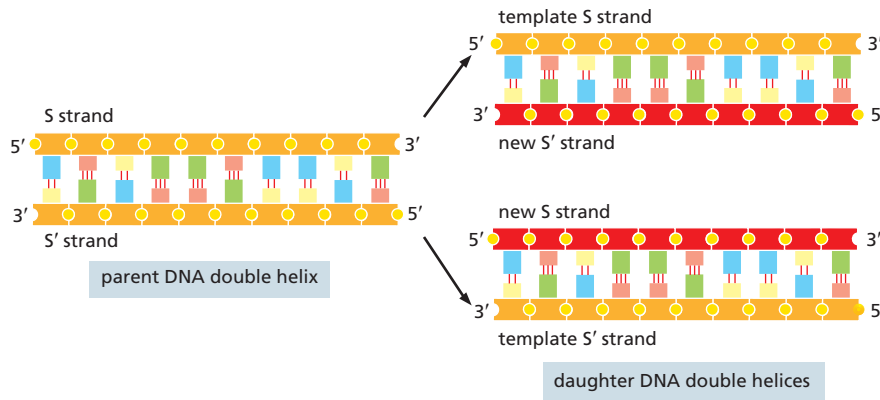


Figure 4–7 DNA as a template for its own duplication. Because the nucleotide A successfully pairs only with T, and G pairs with C, each strand of DNA can act as a template to specify the sequence of nucleotides in its complementary strand. In this way, double-helical DNA can be copied precisely, with each parent DNA helix producing two identical daughter DNA helices.

information in DNA can be accurately copied by the beautifully simple process in which strand S separates from strand S', and each separated strand then serves as a template for the production of a new complementary partner strand that is identical to its former partner.

The ability of each strand of a DNA molecule to act as a template for producing a complementary strand enables a cell to copy, or *replicate*, its genome before passing it on to its descendants. In Chapter 5, we shall describe the elegant machinery that the cell uses to perform this task.

Organisms differ from one another because their respective DNA molecules have different nucleotide sequences and, consequently, carry different biological messages. But how is the nucleotide alphabet used to make messages, and what do they spell out?

As discussed earlier, it was known well before the structure of DNA was determined that genes contain the instructions for producing proteins. If genes are made of DNA, the DNA must therefore somehow encode proteins. As discussed in Chapter 3, the properties of a protein, which are responsible for its biological function, are determined by its three-dimensional structure. This structure is determined in turn by the linear sequence of the amino acids of which it is composed. The linear sequence of nucleotides in a gene must therefore somehow spell out the linear sequence of amino acids in a protein. The exact correspondence between the four-letter nucleotide alphabet of DNA and the 20-letter amino acid alphabet of proteins—the *genetic code*—is not at all obvious from the DNA structure, and it took more than a decade after the discovery of the double helix before it was worked out. In Chapter 6, we will describe this code in detail in the course of elaborating the process of *gene expression*, through which a cell converts the nucleotide sequence of a gene first into the nucleotide sequence of an RNA molecule, and then into the amino acid sequence of a protein (**Figure 4–8**).

The complete store of information in an organism's DNA is called its **genome**, and it specifies all the RNA molecules and proteins that the organism will ever

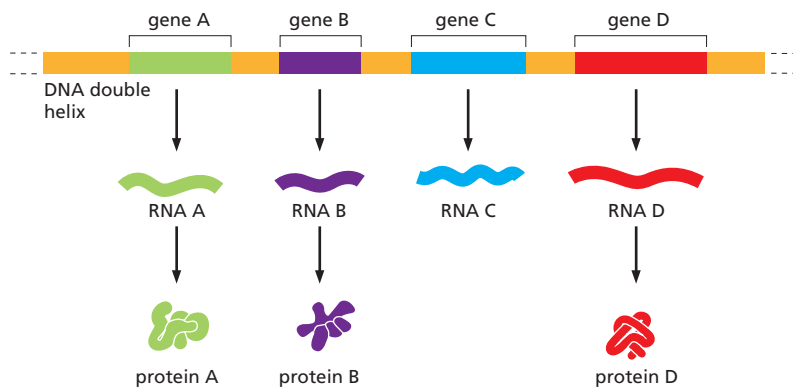


Figure 4–8 Most genes contain information to make proteins. As we discuss in Chapter 6, protein-coding genes each produce a set of RNA molecules (called mRNAs), which then direct the production of a specific protein molecule. Note that for a minority of genes, the final product is the RNA molecule itself, as shown here for gene C.

Figure 4–9 The nucleotide sequence of the human β -globin gene. By convention, a nucleotide sequence is written from its 5' end to its 3' end, and it should be read from left to right in successive lines down the page as though it were normal English text. This gene carries the information for the amino acid sequence of one of the two types of subunits of the hemoglobin molecule; a different gene, the α -globin gene, carries the information for the other. (Hemoglobin, the protein that carries oxygen in the blood, has four subunits, two of each type.) Only one of the two strands of the DNA double helix containing the β -globin gene is shown; the other strand has the exact complementary sequence.

The DNA sequences highlighted in *yellow* show the three regions of the gene that specify the amino acid sequence for the β -globin protein. We shall see in Chapter 6 how the cell splices these three sequences together at the RNA level to produce a messenger RNA molecule, and how this mRNA then guides the synthesis of a full-length β -globin protein.

synthesize. (The term “genome” is also used to describe the DNA that carries this information.) The amount of information contained in genomes is staggering. The nucleotide sequence of a very small human gene, written out in the four-letter nucleotide alphabet, occupies a quarter of a page of text (Figure 4–9), while the complete sequence of nucleotides in the human genome would fill more than a thousand books the size of this one. In addition to other critical information, the human genome contains roughly 20,000 protein-coding genes, which (through alternative splicing and other mechanisms described in Chapters 6 and 7) can give rise to a much greater number of distinct proteins.

In Eukaryotes, DNA Is Enclosed in a Cell Nucleus

As described in Chapter 1, nearly all the DNA in a eukaryotic cell is sequestered in a nucleus, which in many cells occupies about 10% of the total cell volume. This compartment is delimited by a *nuclear envelope* formed by two concentric lipid bilayer membranes (Figure 4–10). As illustrated, these two membranes are punctured at intervals by large nuclear pores, through which molecules move between the nucleus and the cytosol. The outer nuclear membrane is directly connected to the extensive system of intracellular membranes called the *endoplasmic reticulum*, which extend out from it into the cytoplasm. And the nuclear envelope is supported internally by a network of intermediate filaments called the *nuclear lamina*—a thin feltlike mesh just beneath the inner nuclear membrane (see Figure 4–10B).

The nuclear envelope allows the many proteins that act on DNA to be concentrated where they are needed in the cell, and, as we see in subsequent chapters, it also keeps nuclear and cytosolic proteins separate, a feature that is crucial for the proper functioning of eukaryotic cells.

Summary

Genetic information is carried in the linear sequence of nucleotides in DNA. Each molecule of DNA is a double helix formed from two complementary antiparallel strands of nucleotides held together by hydrogen bonds between G-C and A-T base pairs. Duplication of the genetic information occurs by the use of one DNA strand as a template for the formation of a complementary strand. The genetic information stored in an organism's DNA sequence contains the instructions for all the RNA molecules and proteins that the organism will ever synthesize, and it is said to be that organism's genome. In eukaryotes, DNA is contained in the cell nucleus, a large membrane-bound compartment.

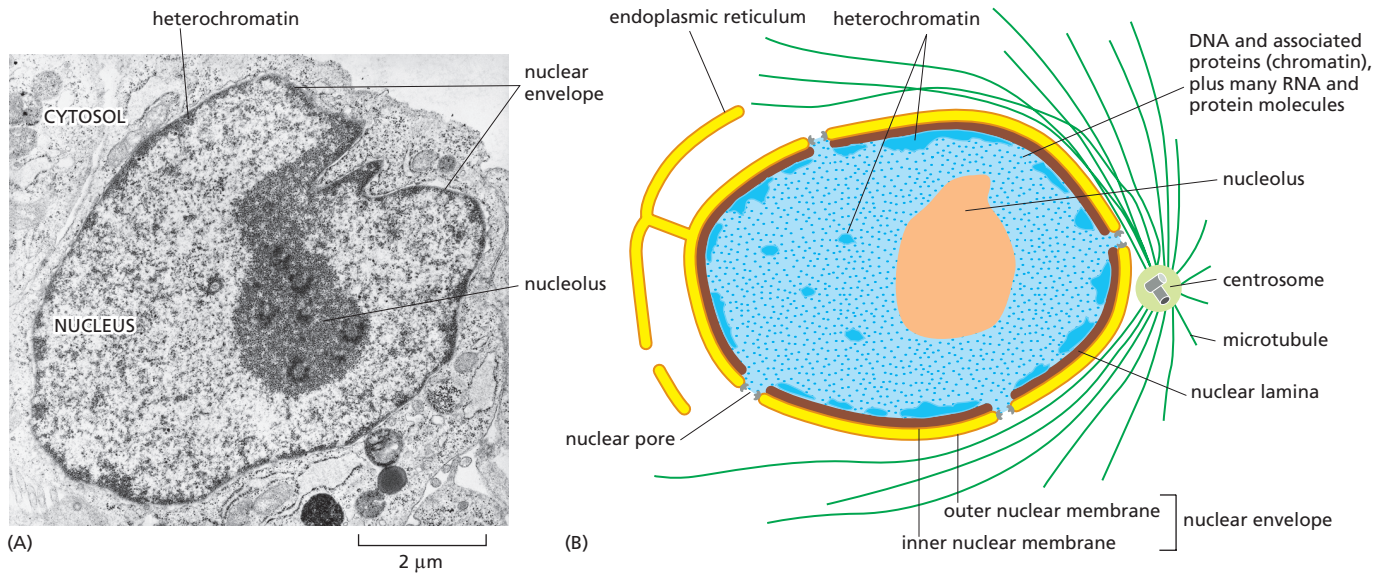
CHROMOSOMAL DNA AND ITS PACKAGING IN THE CHROMATIN FIBER

The most important function of DNA is to form genes, which are the information-carrying units that specify all the RNA molecules and proteins that make up an organism—including information about when, in what types of cells, and in what quantity each RNA molecule and protein is to be made. The nuclear DNA of eukaryotes is divided up into a set of chromosomes, each of which contains one

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5' CCCTGTGGAGCCACACCCTAGGGTTGGCCA
ATCTACTCCCAGGAGCAGGGAGGGCAGGAG
CCAGGGCTGGGCATAAAAGTCAGGGCAGAG
CCATCTATTGCTTACATTTGCTTCTGACAC
AACTGTGTTCCTAGCAACTCAACACAGACA
CCATGGTGCACCTGACTCCTGAGGAGAAGT
CTGCCGTTACTGCCCTGTGGGGCAAGGTGA
ACGTGGATGAAGTTGGTGGTGAGGCCCTGG
GCAGGTTGGTATCAAGGTTACAAGACAGGT
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GAGACAGAGAAGACTCTTGGGTTTCTGATA
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TTTCCCACCCITAGGCTGCTGGTGGTCTAC
CCTTGGACCCAGAGGTTCTTTGAGTCTTTT
GGGGATCTGTCCACTCCTGATGCTGTTATG
GGCAACCCTAAGGTGAAGGCTCATGGCAAG
AAAGTGCTCGGTGCCTTTAGTGATGGCCTG
GCTCACCTGGACAACCTCAAGGGCACCTTT
GCCACACTGAGTGAGCTGCACTGTGACAAG
CTGCACGTGGATCTCGAGAACTTCAGGCTG
AGTCTATGGGACCCCTGATGTTTTCTTTCC
CCTTCTTTTCTATGGTTAAGTTCATGTCAT
AGGAAGGGGAGAAGTAACAGGGTACAGTTT
AGAATGGGAAACAGACGAATGATTGCATCA
GTGTGGAAAGTCTCAGGATCGTTTAGTTTC
TTTTATTGCTGTTTATAACAATTGTTTTT
TTTTGTTTAATTCTGCTTTCTTTTTTTTT
CTTCTCCGCAATTTTACTATTATACCTTAA
TGCCTTAACATTGTGTATAACAAAAGGAAA
TATCTCTGAGATACATTAACTAATTAAAA
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GCTAATCATGTTTCATCTCTTATCTTCTCT
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CCCTAAGTCCAATACTAACTGGGGGATA
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CCTAATAAAAAACATTATTCTTCTGCAAT
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AAACCTTGGGAAAATACATATATCTTAAA
CTCCATGAAAGAAGGTGAGGCTGCAACCG
CTAATGCACATTGGCAACAGCCCTGATGC
CTATGCCCTTATTCATCCCTCAGAAAAGGAT
TCTGTAGAGGCTTGATTTGCAGGTTAAAG
TTTTGCTATGCTGATTTTACATTACTTAT
TGTTTTAGCTGTCTCATGAATGCTTTTC 3'

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enormously long DNA molecule. In this section we see how genes are arranged on chromosomes. In addition, we describe three specialized DNA sequences that are required for a chromosome to be accurately duplicated as a separate entity and passed on from one generation to the next.

We also confront the serious challenge of DNA packaging. If the double helices of all 46 chromosomes in a human cell could be laid end to end, they would reach approximately 2 meters; yet the nucleus, which contains the DNA, is only about 6 μm in diameter. This is geometrically equivalent to packing 40 km (24 miles) of extremely fine thread into a tennis ball. The complex task of packaging DNA is accomplished by specialized proteins that bind to the DNA and fold it, generating a series of organized coils and loops that prevent the DNA from becoming an unmanageable tangle. Amazingly, although the DNA is very tightly compacted, it nevertheless remains accessible to the many enzymes in the cell that replicate it, repair it, and use its genes to produce RNA molecules and proteins.

Eukaryotic DNA Is Packaged into a Set of Chromosomes

Each **chromosome** in a eukaryotic cell consists of a single, enormously long linear DNA molecule along with the proteins that fold the fine DNA thread into a more compact structure. In addition to proteins involved in packaging, chromosomes are also associated with many other proteins (as well as numerous RNA molecules). These are required for the processes of gene expression, DNA replication, and DNA repair. The complex of DNA and tightly bound protein is called *chromatin* (from the Greek *chroma*, “color,” because of its staining properties).

Bacteria lack a special nuclear compartment, and they generally carry their genes on a single DNA molecule, which is often circular (see Figure 1–38). This DNA is also associated with proteins that package and condense it, but they are different from the proteins that perform these functions in eukaryotes. Although the bacterial DNA with its attendant proteins is often called the bacterial “chromosome,” it does not have the same structure as the eukaryotic chromosomes that will be our main focus.

With the exception of the gametes (eggs and sperm) and a few highly specialized cell types that cannot multiply and either lack DNA altogether (for example, red blood cells) or have replicated their DNA without completing cell division (for example, megakaryocytes), each human cell nucleus contains two copies of each chromosome, one inherited from the mother and one from the father. The maternal and paternal chromosomes of a pair are called **homologous chromosomes (homologs)**. The only nonhomologous chromosome pairs are the sex chromosomes in males, where a *Y chromosome* is inherited from the father

Figure 4–10 A cross-sectional view of a typical cell nucleus. (A) Electron micrograph of a thin section through the nucleus of a human fibroblast. (B) Schematic drawing, showing that the nuclear envelope consists of two membranes, the outer one being continuous with the endoplasmic reticulum (ER) membrane (see also Figure 12–54). The space inside the endoplasmic reticulum (the ER lumen) is colored yellow; it is continuous with the space between the two nuclear membranes. The lipid bilayers of the inner and outer nuclear membranes are connected at each nuclear pore. A sheetlike network of filaments inside the nucleus forms the nuclear lamina (brown), providing mechanical support for the nuclear envelope (for details, see Chapter 12). Outside the nuclear envelope, a centrosome forms microtubules that help to organize the cytoplasm, as explained in Chapter 16. The dark-staining heterochromatin contains specially condensed regions of DNA that will be discussed later.

Prokaryotic cells (archaea and bacteria) have no nucleus. Lacking a nuclear envelope, their DNA is co-located with all other cell components in a single compartment, bounded by the cell's plasma membrane. (A, courtesy of E.G. Jordan and J. McGovern.)

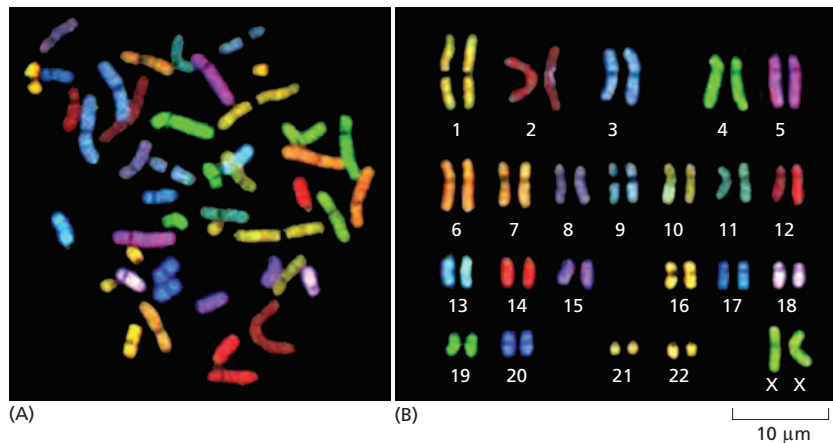


Figure 4-11 The complete set of human chromosomes. These chromosomes, from a female, were isolated from a cell undergoing nuclear division (mitosis) and are therefore highly compacted. Each chromosome has been “painted” a different color to permit its unambiguous identification under the fluorescence microscope, using a technique called “spectral karyotyping.” Chromosome painting can be performed by exposing the chromosomes to a large collection of DNA molecules (called DNA probes) whose sequences match known DNA sequences from the human genome. A set of sequences that matches a specific chromosome is coupled to a combination of fluorescent dyes. DNA molecules derived from chromosome 1 are labeled with one specific dye combination, those from chromosome 2 with another, and so on. Because the labeled DNA can form base pairs, or hybridize, only to the chromosome from which it was derived, each chromosome becomes labeled with a different combination of dyes. For such experiments, the chromosomes are subjected to treatments that separate the two strands of double-helical DNA in a way that permits base-pairing with the single-stranded, labeled DNA but keeps the overall chromosome structure relatively intact. (A) The chromosomes visualized as they originally spilled from the lysed cell. (B) The same chromosomes artificially lined up in their numerical order. This arrangement of the full chromosome set is called a karyotype. (Adapted from N. McNeil and T. Ried, *Expert Rev. Mol. Med.* 2:1–14, 2000. With permission from Cambridge University Press.)

and an *X chromosome* from the mother. Thus, each human cell contains a total of 46 chromosomes—22 pairs common to both males and females, plus two so-called sex chromosomes (X and Y in males, two X's in females). These human chromosomes can be readily distinguished by “painting” each one a different color using a technique that is based on *DNA hybridization* (Figure 4-11). In this method (see p. 505), short strands of nucleic acid tagged with fluorescent dyes serve as “probes” that pick out their complementary DNA sequence, lighting up the target chromosome at any site where they bind. Chromosome painting is most frequently done at the stage in the cell cycle called mitosis, when chromosomes are especially compacted and easy to visualize (discussed shortly).

Another more traditional way to distinguish one chromosome from another is to stain them with dyes that reveal a striking and reproducible pattern of bands along each mitotic chromosome (Figure 4-12). These banding patterns reflect variations in chromatin structure and/or base composition, although their basis is not well understood. Nevertheless, the pattern of these bands on each type of chromosome is unique, and it provided the initial means to identify and number each human chromosome reliably.

The display of the 46 human chromosomes at mitosis shown in Figure 4-11B is called the human **karyotype**. If parts of chromosomes are lost or are switched between chromosomes, these changes can be detected either by changes in the banding patterns or—with greater sensitivity—by changes in the pattern of chromosome painting (Figure 4-13). Cytogeneticists use these alterations to detect inherited chromosome abnormalities and to reveal the chromosome rearrangements that occur in cancer cells as they progress to malignancy (discussed in Chapter 20).

Chromosomes Contain Long Strings of Genes

Chromosomes carry genes—the functional units of heredity. A gene is often defined as a segment of DNA that contains the instructions for making a particular protein (or a set of closely related proteins), but this definition is too narrow. Genes that code for protein are indeed the majority, and most of the genes with clear-cut mutant phenotypes fall under this heading. In addition, however, there are many “RNA genes”—segments of DNA that generate a functionally significant RNA molecule, instead of a protein, as their final product. We shall say more about these RNA genes and their products later.

As might be expected, there can be a correlation between the complexity of an organism and the number of genes in its genome (see Table 1-2, p. 29). For example, some simple bacteria have only 500 genes, compared to about 25,000 for humans. Bacteria, archaea, and some single-celled eukaryotes, such as yeast, have concise genomes, consisting of little more than strings of closely packed genes. However, the genomes of multicellular plants and animals, as well as many other eukaryotes, contain, in addition to genes, a large quantity of interspersed DNA whose function

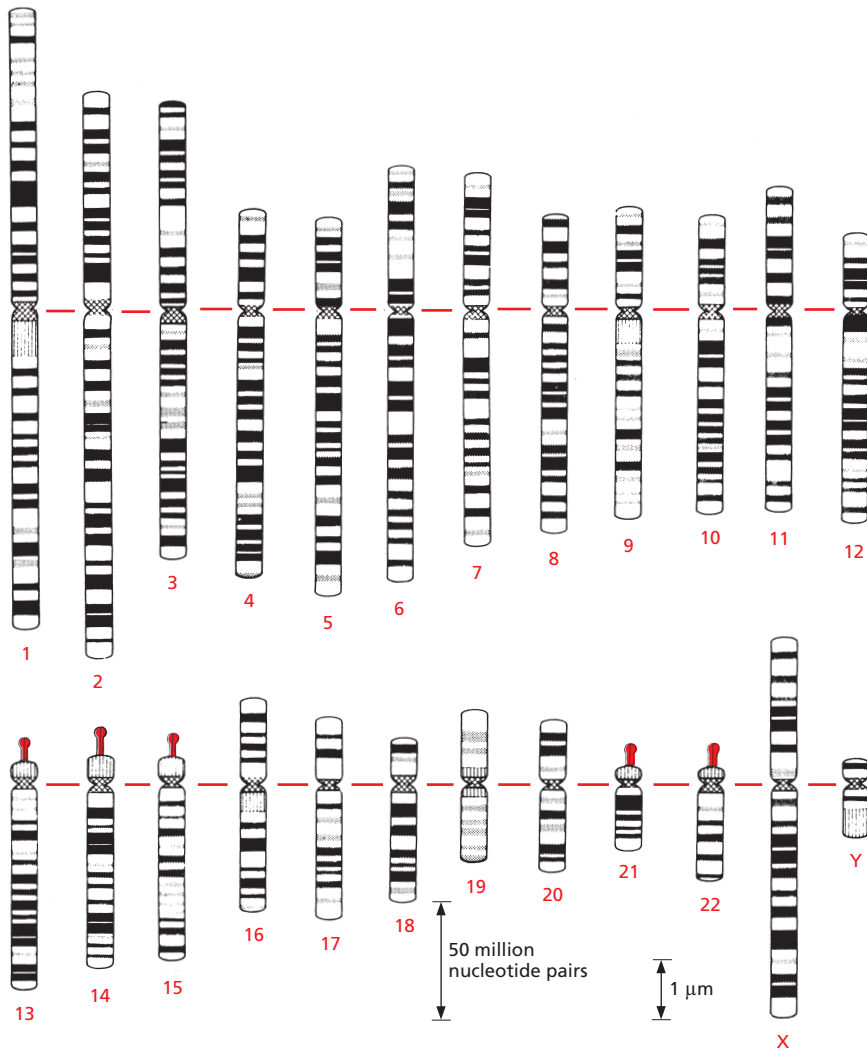
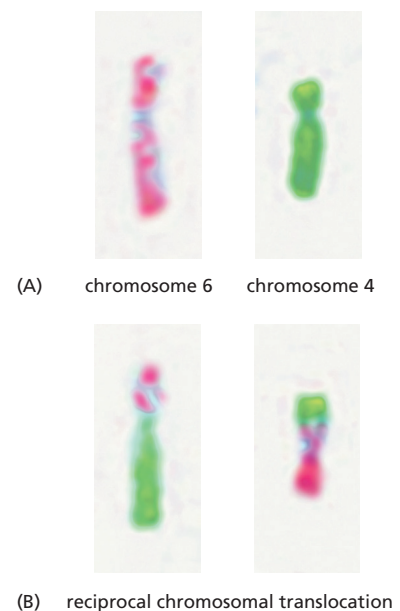


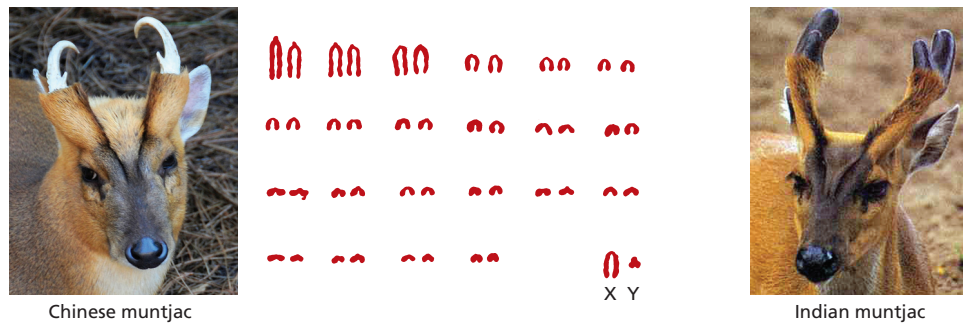
Figure 4-12 The banding patterns of human chromosomes. Chromosomes 1–22 are numbered in approximate order of size. A typical human cell contains two of each of these chromosomes, plus two sex chromosomes—two X chromosomes in a female, one X and one Y chromosome in a male. The chromosomes used to make these maps were stained at an early stage in mitosis, when the chromosomes are incompletely compacted. The horizontal red line represents the position of the centromere (see Figure 4-18), which appears as a constriction on mitotic chromosomes. The red knobs on chromosomes 13, 14, 15, 21, and 22 indicate the positions of genes that code for the large ribosomal RNAs and form the nucleolus (discussed in Chapter 6). These banding patterns are obtained by staining chromosomes with Giemsa stain, and they are observed under a light microscope. (Adapted from U. Francke, *Cytogenet. Cell Genet.* 31:24–32, 1981.)

is poorly understood. Some of this additional DNA is crucial for the proper control of gene expression, but this only partly explains why there is so much of it in multicellular organisms, whose genes need to be switched on and off according to complicated rules during development (discussed in Chapters 7 and 21).

Differences in the amount of noncoding DNA, most of it interspersed between genes—far more than differences in numbers of genes—account for the astonishing variations in genome size that we see when we compare one species with another (see Figure 1-30). For example, the human genome is 200 times larger than that of the yeast *Saccharomyces cerevisiae*, but 30 times smaller than that of some plants and amphibians and 200 times smaller than that of a species of amoeba. Moreover, because of differences in the amount of noncoding DNA, the genomes of closely related organisms (bony fish, for example) can vary several hundredfold in their DNA content, even though they contain roughly the same number of genes. Whatever the excess DNA may do, it seems clear that it is not a great handicap for a eukaryotic cell to carry a large amount of it.

Figure 4-13 Aberrant human chromosomes. (A) Two normal human chromosomes, 4 and 6. (B) In an individual carrying a reciprocal *chromosomal translocation*, the DNA double helix in one chromosome has crossed over with the DNA double helix in the other chromosome because of an abnormal recombination event. The chromosome painting technique used on the chromosomes in each of the sets allows the identification of short pieces of chromosomes that have become translocated, a frequent event in cancer cells. (Courtesy of Zhenya Tang and the NIGMS Human Genetic Cell Repository at the Coriell Institute for Medical Research.)





How the genome is divided into chromosomes also differs from one eukaryotic species to the next. For example, while the cells of humans have 46 chromosomes, those of some small deer have only 6, while those of the common carp contain more than 100. Even closely related species with similar genome sizes can have very different numbers and sizes of chromosomes (Figure 4-14). Thus, there is no simple relationship between chromosome number, complexity of the organism, and total genome size. Rather, the genomes and chromosomes of modern-day species have each been shaped by a unique history of seemingly random genetic events, acted on by poorly understood selection pressures over long evolutionary times.

Figure 4-14 Two closely related species of deer with very different chromosome numbers. In the evolution of the Indian muntjac, initially separate chromosomes in an ancestor fused, without having a major effect on the animal. These two species contain a similar number of genes. (Chinese muntjac photo courtesy of Deborah Carreno, Natural Wonders Photography; Indian muntjac photo courtesy of Beatrice Bourger.)

The Nucleotide Sequence of the Human Genome Shows How Our Genes Are Arranged

With the determination of the full DNA sequence of the human genome, it became possible to see in detail how the genes are arranged along each of our chromosomes (Figure 4-15). Although many decades will pass before the information contained in the human genome sequence is fully analyzed, it has already stimulated an enormous number of new experiments that have had major effects on the content of every chapter in this book.

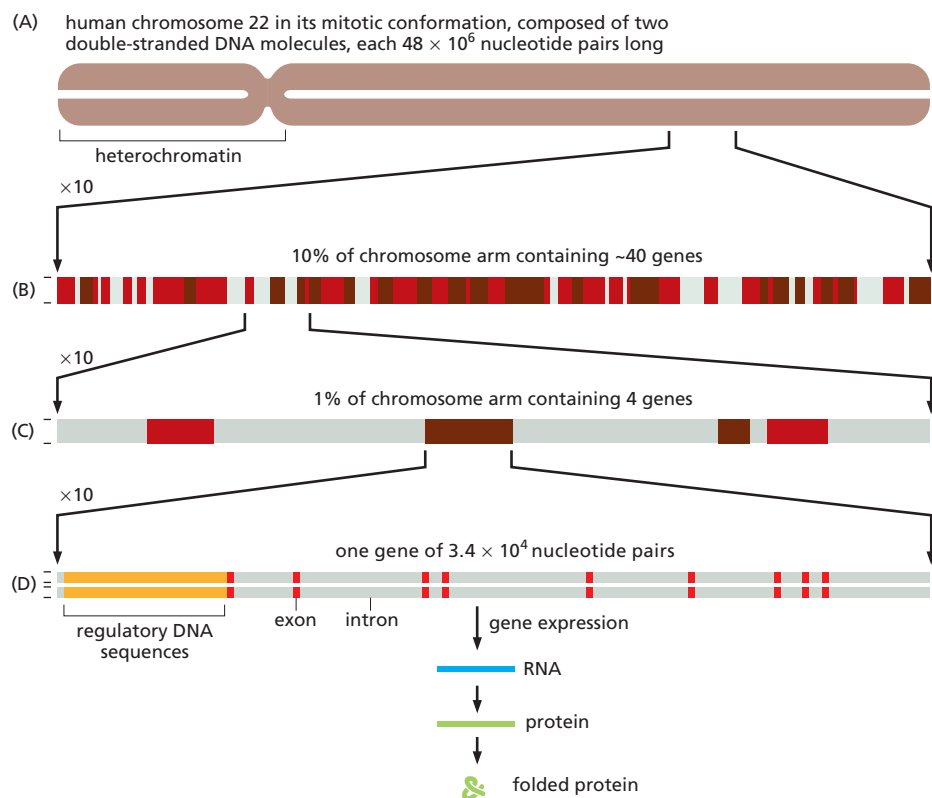


Figure 4-15 The organization of genes on a human chromosome, as determined from the initial sequencing of the human genome. (A) Chromosome 22, one of the smallest human chromosomes, contains 48×10^6 nucleotide pairs and makes up approximately 1.5% of the human genome. Most of the left arm of chromosome 22 consists of short repeated sequences of DNA that are packaged in a particularly compact form of chromatin (heterochromatin) discussed later in this chapter. (B) A tenfold expansion of a portion of chromosome 22, with about 40 genes indicated. Those in dark brown were known genes, and those in red were predicted genes. (C) An expanded portion of B showing four genes. (D) The intron-exon arrangement of a typical gene is shown after a further tenfold expansion. Each exon (red) codes for a portion of the protein, while the DNA sequence of the introns (gray) is relatively unimportant, as discussed in detail in Chapter 6. (Adapted from International Human Genome Sequencing Consortium, *Nature* 409:860–921, 2001.)

The human genome (3.1×10^9 nucleotide pairs) is the totality of genetic information belonging to our species. Almost all of this genome is distributed over the 22 different autosomes and 2 sex chromosomes (see Figures 4-11 and 4-12) found within the nucleus. A minute fraction of the human genome (16,569 nucleotide pairs—in multiple copies per cell) is found in the mitochondria (introduced in Chapter 1 and discussed in detail in Chapter 14). The term “human genome sequence” refers to the complete nucleotide sequence of DNA in the 24 nuclear chromosomes and the mitochondria. Being diploid, a human somatic cell nucleus contains twice the haploid amount of DNA, or 6.2×10^9 nucleotide pairs, before duplicating its chromosomes in preparation for division.

TABLE 4–1 Some Vital Statistics for the Human Genome

Human genome DNA length	3.1×10^9 nucleotide pairs*
Number of genes coding for proteins	About 20,000 (19,116, plus hundreds of genes encoding proteins of 50 amino acids or less)**
Largest gene coding for protein	2.5×10^6 nucleotide pairs
Median size for protein-coding genes	26,000 nucleotide pairs
Median size for messenger RNAs (including 5' and 3' untranslated regions)	2938 nucleotides
Median size for amino-acid coding in messenger RNA sequences	1290 nucleotides
Protein-coding gene features	
Median size of protein produced	430 amino acids
Smallest number of exons per gene	1 (1068 of these unspliced genes)
Largest number of exons per gene	363
Median number of exons per gene	9.0
Median exon size	131 nucleotide pairs
Median intron size	1747 nucleotide pairs
Number of noncoding RNA genes	About 5000, with 4849 annotated***
Number of pseudogenes	More than 20,000****
Percentage of protein-coding DNA sequence (in exons)	1%
Percentage of DNA in other highly conserved sequences	3.5%
Percentage of DNA transcribed (in protein-coding plus annotated noncoding RNA genes)	45%
Percentage of DNA in high-copy-number repetitive elements	Approximately 50%

*The sequence of 2.85 billion nucleotides is known precisely (error rate of only about 1 in 100,000 nucleotides). The remaining DNA primarily consists of short sequences that are tandemly repeated many times over, with repeat numbers differing from one individual to the next. These highly repetitive blocks are difficult to sequence accurately.

**RefSeq genes, with at least one reviewed/validated mRNA (A. Piovesan et al., *BMC Res. Notes* 12:315–319, 2019). For genes encoding tiny proteins, see T.F. Martinez et al., *Nat. Chem. Biol.* 16:458–468, 2020).

***A considerable number of potential ncRNA genes are still uncharacterized and not annotated.

****A pseudogene is a DNA sequence closely resembling that of a functional gene but containing numerous mutations that prevent its proper expression or function; most are still uncharacterized and not annotated in databases.

Courtesy of Allison Piovesan, University of Bologna, Bologna, Italy.

The first striking feature of the human genome is how little of the DNA sequence (only about 1%) codes for proteins (Table 4–1 and Figure 4–16). It is also notable that nearly half of the chromosomal DNA is made up of mobile pieces of DNA that have gradually inserted themselves in the chromosomes over evolutionary time, multiplying like parasites in the genome and appearing as genome-wide repeat sequences (see Figure 4–63). We discuss these *transposable elements* in detail in later chapters.

A second notable feature of the human genome is the large average gene size—about 26,000 nucleotide pairs. As discussed earlier, a typical gene carries in

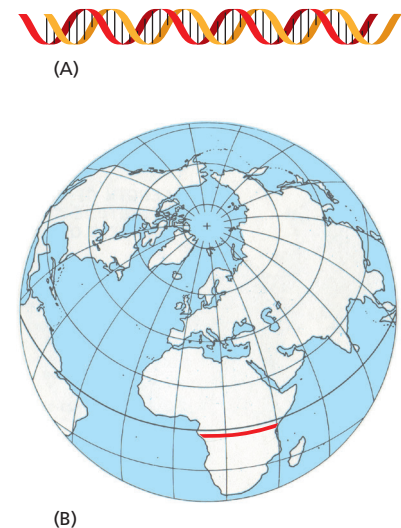


Figure 4–16 Scale of some human genome features. If drawn with a 1-mm space between each nucleotide pair (as in A), the human genome would extend 3100 km (nearly 2000 miles), far enough to stretch across the center of Africa, the site of our human origins (red line in B). At this scale, there would be, on average, a protein-coding gene every 160 meters. An average gene would extend for 26 meters, but the coding sequences in this gene would add up to only just over a meter.

its linear sequence of nucleotides the information for the linear sequence of the amino acids of a protein. Only about 1300 nucleotides are required to encode a protein of average size (about 430 amino acids in humans). Most of the remaining sequence in a gene consists of long stretches of noncoding DNA that interrupt the relatively short segments of DNA that code for protein. As will be discussed in detail in Chapter 6, the coding sequences are included in the **exons**; the intervening sequences in genes are noncoding and called **introns** (see Figure 4–15 and Table 4–1). The majority of human genes thus consist of a long string of alternating exons and introns, with most of the gene consisting of introns. In contrast, the majority of genes from organisms with concise genomes either lack introns (as in prokaryotes) or contain relatively short ones. This helps to account for the much smaller size of their genes, as well as for the much higher fraction of coding DNA in their chromosomes.

In addition to introns and exons, each gene is associated with *regulatory DNA sequences*, which are responsible for ensuring that the gene is turned on or off at the proper time, expressed at the appropriate level, and only in the proper type of cell. In humans, the regulatory sequences for a typical gene are spread out over hundreds of thousands of nucleotide pairs. As would be expected, these regulatory sequences are much more compact and less numerous in organisms with concise genomes. In Chapter 7 we discuss how regulatory DNA sequences work.

Detailed studies of genome sequences have surprised biologists with the discovery that, in addition to 20,000 protein-coding genes, the human genome contains thousands of genes that encode RNA molecules that do not produce proteins, but instead have a variety of other important functions. Although the roles of some of these non-coding RNA genes have been known for decades, many more remain mysterious, as will be discussed in Chapters 6 and 7.

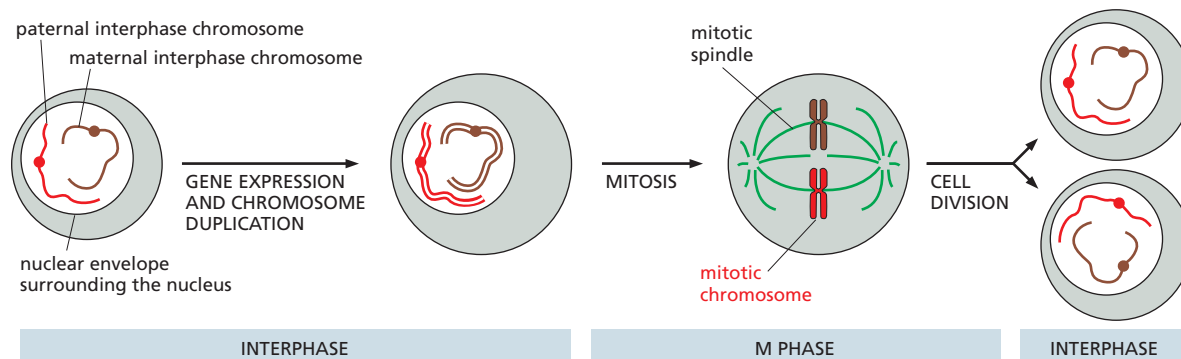
Last, but not least, the nucleotide sequence of the human genome has revealed that the archive of information needed to produce a human seems to be in an alarming state of chaos. As one commentator described our genome, “In some ways it may resemble your garage/bedroom/refrigerator/life: highly individualistic, but unkempt; little evidence of organization; much accumulated clutter (referred to by the uninitiated as ‘junk’); virtually nothing ever discarded; and the few patently valuable items indiscriminately, apparently carelessly, scattered throughout.” We shall discuss how this is thought to have come about in the final section of this chapter, which is titled “How Genomes Evolve.”

Each DNA Molecule That Forms a Linear Chromosome Must Contain a Centromere, Two Telomeres, and Replication Origins

To form a functional chromosome, a DNA molecule must be able to do more than simply carry genes: it must be able to replicate, and the replicated copies must be separated and reliably partitioned into daughter cells at each cell division. This process occurs through an ordered series of stages, collectively known as the **cell cycle**, which provides for a temporal separation between the duplication of chromosomes and their segregation into two daughter cells. The eukaryotic cell cycle is briefly summarized in **Figure 4–17**, and it is discussed in detail in Chapter 17.

Figure 4–17 A simplified view of the eukaryotic cell cycle. During interphase, the cell is actively expressing its genes and is therefore synthesizing proteins. Also, during interphase and before cell division, the DNA is replicated, and each chromosome is duplicated to produce two closely paired sister DNA molecules (called sister chromatids). A cell with only one type of chromosome, present in maternal and paternal copies, is illustrated here.

Once DNA replication is complete, the cell can enter *M phase*, when mitosis occurs and the nucleus is divided into two daughter nuclei. During this stage, the chromosomes condense, the nuclear envelope breaks down, and the mitotic spindle forms from microtubules and other proteins. The condensed mitotic chromosomes are captured by the mitotic spindle, and one complete set of chromosomes is then pulled to each end of the cell by separating the members of each sister-chromatid pair. A nuclear envelope re-forms around each chromosome set, and in the final step of *M phase*, the cell divides to produce two daughter cells. Most of the time in the cell cycle is spent in interphase; *M phase* is brief in comparison, occupying only about an hour in many mammalian cells.



Briefly, during a long *interphase*, genes are expressed and chromosomes are replicated, with the two replicas remaining together as a pair of *sister chromatids*. Throughout this time the chromosomes are extended, and much of their chromatin exists as long threads in the nucleus so that individual chromosomes cannot be easily distinguished. It is only during a much briefer period of *mitosis* that each chromosome condenses so that its two sister chromatids can be separated and distributed to the two daughter nuclei. The highly condensed chromosomes in a dividing cell are known as *mitotic chromosomes* (Figure 4-18). This is the form in which chromosomes are most easily visualized; in fact, the images of chromosomes shown so far in the chapter are of chromosomes in mitosis.

Each chromosome operates as a distinct structural unit. For a faithful copy of a chromosome to be passed on to each daughter cell at division, it must have been completely replicated during interphase to produce two identical DNA molecules. In addition, these newly replicated, enormously long DNA copies must then be separated and partitioned correctly into the two daughter cells. These basic functions are controlled by three types of specialized sites on the DNA, each of which binds specific proteins that guide the machinery that replicates and segregates chromosomes (Figure 4-19).

Experiments in budding yeast, whose chromosomes are relatively small and easy to manipulate, have identified the minimal DNA sequence elements responsible for each of these functions. One type of nucleotide sequence acts as a **DNA replication origin**, the location at which duplication of the DNA begins. Eukaryotic chromosomes contain many origins of replication to ensure that the entire chromosome can be replicated rapidly, as discussed in detail in Chapter 5.

After DNA replication, the two sister chromatids that form each chromosome remain attached to one another and, as the cell cycle proceeds, they are condensed further to produce mitotic chromosomes. In budding yeasts, the presence of a second specialized DNA sequence, called a **centromere**, allows one copy of each duplicated and condensed chromosome to be pulled into each daughter cell when a cell divides. A protein complex called a *kinetochore* forms at the centromere and attaches the duplicated chromosomes to the mitotic spindle, in a manner that causes the two sister chromatids to be pulled apart (discussed in Chapter 17).

The third specialized DNA sequence forms **telomeres**, the ends of a chromosome. Telomeres contain repeated nucleotide sequences that enable the ends of chromosomes to be efficiently replicated. Telomeres also perform another function: the repeated telomere DNA sequences, together with the regions adjoining them, form structures that protect the end of the chromosome from

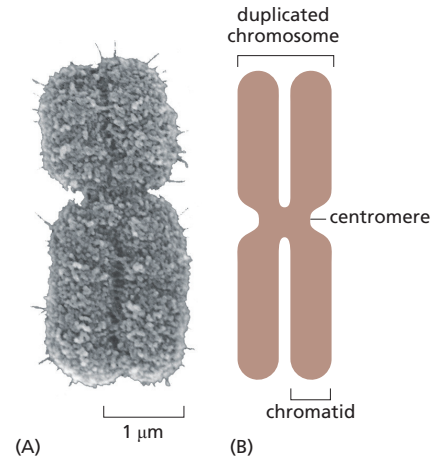


Figure 4-18 A mitotic chromosome. At the metaphase stage of M phase, each chromosome exists as a condensed duplicated chromosome, in which the two replicated chromosomes, called sister chromatids, are still linked together (see Figure 4-17). The constricted region indicates the position of the centromere. (A) Electron micrograph; (B) schematic drawing. (A, courtesy of Terry D. Allen.)

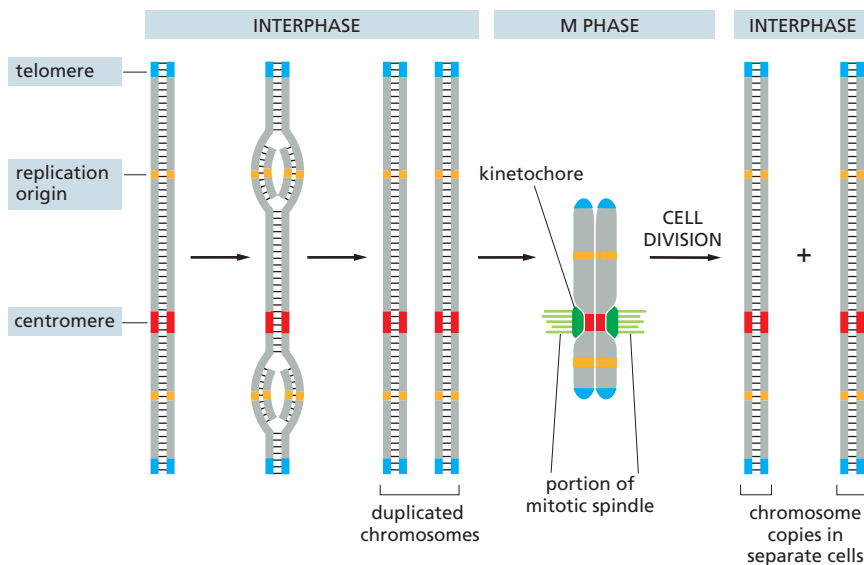


Figure 4-19 The three specialized sites on the DNA required to produce a eukaryotic chromosome that can be replicated and then segregated accurately at mitosis. Each chromosome has multiple origins of replication, one centromere, and two telomeres. Shown here is the sequence of events that a typical chromosome follows during the cell cycle. The DNA replicates during the portion of interphase known as S phase, beginning at the origins of replication and proceeding bidirectionally from the origins across the chromosome. In M phase, the centromere attaches the duplicated chromosomes to the mitotic spindle so that a copy of the entire genome is distributed to each daughter cell during mitosis; the special structure that attaches the centromere to the spindle is a protein complex called the kinetochore (dark green). The centromere also helps to hold the duplicated chromosomes together until they are ready to be moved apart. As explained in the text, the telomeres form special caps at each chromosome end.

being mistaken by the cell for a broken DNA molecule in need of repair. We discuss both this type of repair and the structure and function of telomeres in Chapter 5.

In budding yeast cells, the three types of sequences required to propagate a chromosome are relatively short (typically fewer than 1000 base pairs each) and therefore use only a tiny fraction of the information-carrying capacity of a chromosome. Although telomere sequences are fairly simple and short in all eukaryotes, the DNA sequences that form centromeres and replication origins in more complex organisms are much longer than their yeast counterparts. For example, experiments suggest that a human centromere can contain up to a million nucleotide pairs and that it may not require a stretch of DNA with a defined nucleotide sequence. Instead, as we shall discuss later in this chapter, a human centromere is thought to consist of a large, regularly repeating protein–nucleic acid structure that can be inherited when a chromosome replicates.

DNA Molecules Are Highly Condensed in Chromosomes

All eukaryotic organisms have special ways of packaging DNA into chromosomes. Thus, if the 48 million nucleotide pairs of DNA in human chromosome 22 could be laid out as one long double helix stretched out end to end, the molecule would extend for about 1.5 cm. But chromosome 22 measures only about 2 μm in length in mitosis (see Figure 4–11), representing an end-to-end compaction ratio of more than 7000-fold. This remarkable feat of compression is performed by proteins that successively coil and fold the DNA into higher levels of organization.

Although much less condensed than mitotic chromosomes, the DNA of human interphase chromosomes is still tightly packed. But interphase chromosomes have a dynamic structure. Specific regions of interphase chromosomes decondense to allow access to specific DNA sequences for gene expression, DNA repair, and replication—and then recondense when these processes are completed. The packaging of DNA in chromosomes is therefore accomplished in a way that allows rapid localized, on-demand access to the DNA. In the next sections, we discuss the specialized proteins that make this type of packaging possible.

Nucleosomes Are a Basic Unit of Eukaryotic Chromosome Structure

The proteins that bind to the DNA to form eukaryotic chromosomes are traditionally divided into two classes: the **histones** and the *non-histone chromosomal proteins*, each contributing about the same mass to a chromosome. The complex of both classes of protein with the nuclear DNA of eukaryotic cells is known as **chromatin** (Figure 4–20; Movie 4.2).

Histones are responsible for the first and most basic level of chromosome packing, a protein–DNA complex called the **nucleosome**. When interphase nuclei are broken open very gently and their contents examined under the electron microscope, most of the chromatin appears to be in the form of a fiber

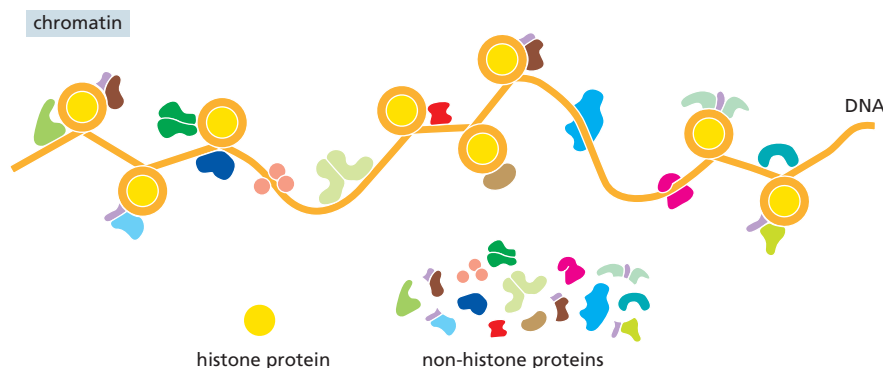


Figure 4–20 Chromatin. As illustrated, chromatin consists of DNA bound to both histone and non-histone proteins. The mass of histone protein present is about equal to the total mass of non-histone protein, but—as schematically indicated here—the latter class is composed of an enormous number of different species. In total, a chromosome is about one-third DNA and two-thirds protein by mass.

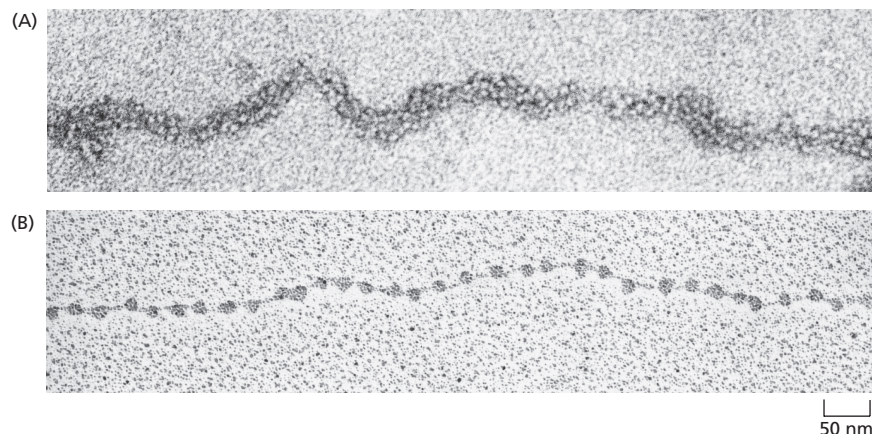


Figure 4-21 Nucleosomes as seen in the electron microscope. (A) Chromatin isolated directly from an interphase nucleus appears in the electron microscope as a thread of associated nucleosomes. (B) This electron micrograph shows a length of chromatin that has been experimentally decondensed after isolation to show the nucleosomes. (A, courtesy of Barbara Hamkalo; B, courtesy of Victoria Foe.)

(Figure 4-21A). If this chromatin is subjected to treatments that cause it to unfold partially, it can be seen under the electron microscope as a series of “beads on a string” (Figure 4-21B). The string is DNA, and each bead is a *nucleosome core particle* that consists of DNA wound around a histone core.

The nucleosome core particles can be isolated by digesting chromatin with particular enzymes (called nucleases) that cut DNA. After digestion for a short period, the exposed DNA between the nucleosome core particles, the *linker DNA*, is degraded. Each individual nucleosome core particle is found to consist of a complex of eight histone proteins—two molecules each of histones H2A, H2B, H3, and H4—and double-stranded DNA that is 147 nucleotide pairs long. This *histone octamer* forms a protein core around which the double-stranded DNA is wound (Figure 4-22).

The region of linker DNA that separates each nucleosome core particle from the next can vary in length from a few nucleotide pairs up to about 80. (The term “nucleosome” technically refers to a nucleosome core particle plus one of its adjacent DNA linkers, but it is often used synonymously with “nucleosome core particle.”) On average, therefore, nucleosomes repeat at intervals of about 200 nucleotide pairs. For example, a diploid human cell with 6.2×10^9 nucleotide pairs contains approximately 30 million nucleosomes. The formation of nucleosomes converts a DNA molecule into a chromatin thread that is about one-third of its initial length.

The Structure of the Nucleosome Core Particle Reveals How DNA Is Packaged

The high-resolution structure of a nucleosome core particle reveals a disc-shaped histone core around which the DNA is tightly wrapped in a left-handed coil of 1.7 turns (Figure 4-23). All four of the histones that make up the core of the nucleosome are relatively small proteins (102–135 amino acids), and they share a structural motif, known as the *histone fold*, formed from three α helices connected by two loops (Figure 4-24). In assembling a nucleosome, the histone folds first bind to each other to form H3–H4 and H2A–H2B dimers, and the H3–H4 dimers combine to form tetramers. An H3–H4 tetramer then further combines with two H2A–H2B dimers to form a compact *histone octamer*, the core around which the DNA is wound.

The interface between DNA and histone is extensive: 142 hydrogen bonds are formed between DNA and the histone core in each nucleosome. Nearly half

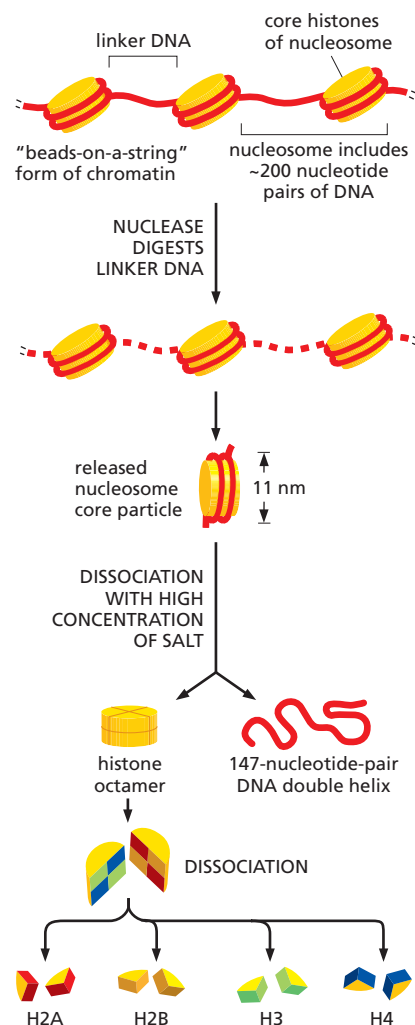
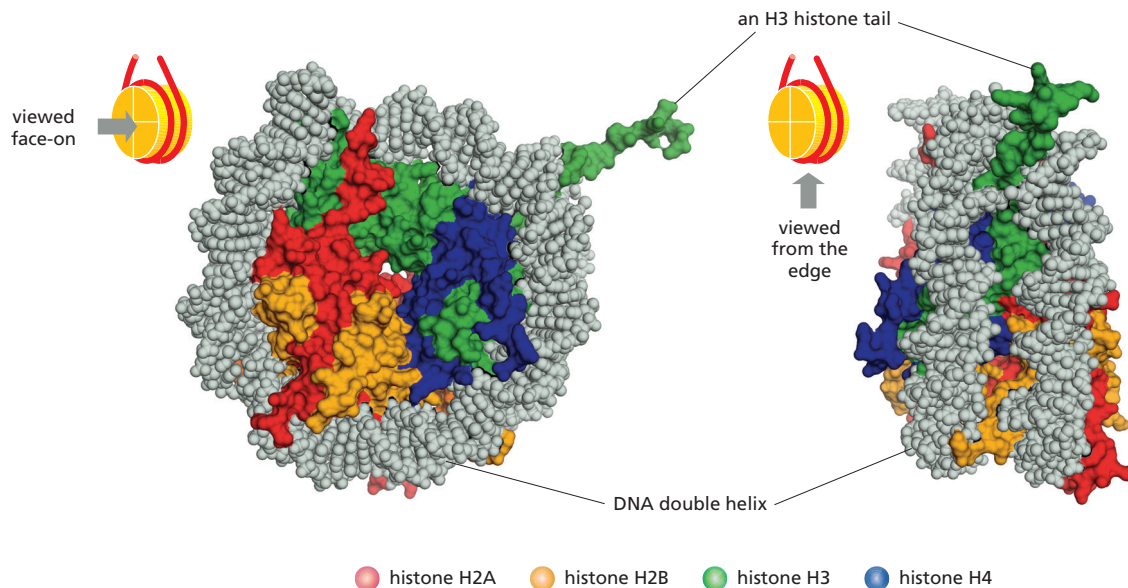


Figure 4-22 Structural organization of the nucleosome. A nucleosome contains a protein core made of eight histone molecules. In biochemical experiments, the nucleosome core particle can be released from isolated chromatin by digestion of the linker DNA with a nuclease, an enzyme that hydrolyzes the phosphodiester bonds that connect the nucleotides in DNA. (The nuclease can degrade the exposed linker DNA but cannot attack the DNA wound tightly around the nucleosome core.) After dissociation of the isolated nucleosome into its protein core and DNA, the length of the DNA that was wound around the core can be determined. This length of 147 nucleotide pairs is sufficient to wrap 1.7 times around the histone core.



of these bonds form between the amino acid backbone of the histones and the sugar-phosphate backbone of the DNA. Numerous hydrophobic interactions and salt linkages also hold DNA and protein together in the nucleosome. More than one-fifth of the amino acids in each of the core histones are either lysine or arginine (two amino acids with basic side chains), and their positive charges can effectively neutralize the negatively charged DNA backbone. These numerous interactions explain in part why DNA of virtually any sequence can be bound on a histone octamer core. The path of the DNA around the histone core is not smooth; rather, several kinks are seen in the DNA, as expected from the nonuniform surface of the core.

The bending of the DNA requires a substantial compression of the minor groove of the DNA helix. Certain dinucleotides in the minor groove are especially

Figure 4-23 The structure of a nucleosome core particle, as determined by x-ray diffraction analyses of crystals. Each histone is colored according to the scheme in Figure 4-22, with the DNA double helix in light gray. One H3 N-terminal tail (green) can be seen extending from the nucleosome core; the positions of the tails of the other histones could not be determined, due to their disorder. (Adapted from K. Luger et al., *Nature* 389:251–260, 1997.)

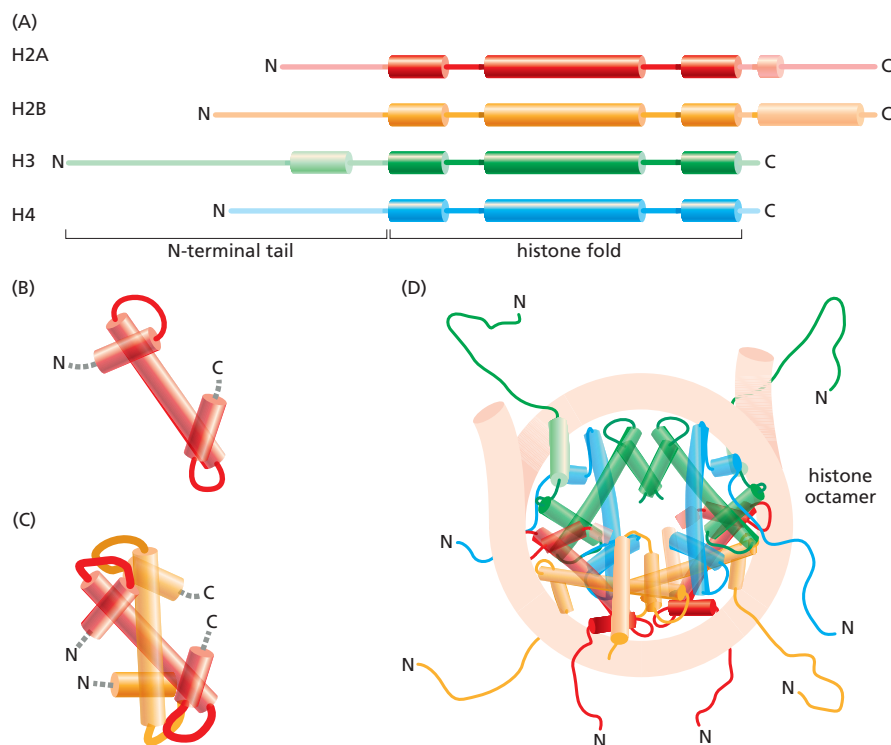


Figure 4-24 The overall structural organization of the core histones. (A) Each of the core histones contains an N-terminal tail, which is subject to several forms of covalent modification, and a histone fold region, as indicated. (B) The structure of the histone fold, which is formed by all four of the core histones. (C) Histones H2A and H2B form a dimer through an interaction known as the “handshake.” Histones H3 and H4 form a dimer through the same type of interaction. (D) The final histone octamer around which the DNA is wound. Note that all eight N-terminal tails of the histones protrude from the disc-shaped core structure. Their conformations are highly flexible, and they serve as binding sites for sets of other proteins.

easy to compress, and some nucleotide sequences bind the histone core more tightly than others (Figure 4-25). This probably explains some striking, but unusual, cases of very precise positioning of nucleosomes along a stretch of DNA. However, the sequence preference of nucleosomes must be weak enough to allow other factors to dominate, inasmuch as nucleosomes can occupy any one of a number of positions relative to the DNA sequence in most chromosomal regions.

In addition to its histone fold, each of the core histones has a largely unstructured N-terminal amino acid “tail,” which extends out from the DNA-histone core (see Figure 4-24D). These histone tails are “hot spots” for different types of covalent modifications that control critical aspects of chromatin structure and function, as we shall discuss shortly.

As a reflection of their fundamental role in DNA function through controlling chromatin structure, the histones are among the most highly conserved eukaryotic proteins. For example, the amino acid sequence of histone H4 from a pea differs from that of a human at only two of the 102 positions. Although histones H2A and H2B have been somewhat less constrained in their evolution than histones H3 and H4, such a strong evolutionary conservation suggests that the functions of histones involve nearly all of their amino acids, so that a change in any position is deleterious to the cell.

In addition to this remarkable conservation, eukaryotic organisms also produce smaller amounts of specialized variant core histones that differ in amino acid sequence from the main ones. As discussed later, these variants, combined with the surprisingly large number of covalent modifications that can be added to the histones in nucleosomes, give rise to a variety of chromatin structures in cells.

Nucleosomes Have a Dynamic Structure and Are Frequently Subjected to Changes Catalyzed by ATP-dependent Chromatin-remodeling Complexes

For many years biologists thought that, once formed in a particular position on DNA, a nucleosome would remain fixed in place because of the very tight association between its core histones and DNA. If true, this would pose problems for genetic readout mechanisms, which require easy access to many specific DNA sequences. It would also hinder the rapid passage of the DNA transcription and replication machinery through chromatin. But kinetic experiments show that the DNA in an isolated nucleosome unwraps from each end at a rate of about four times per second, remaining exposed for 10–50 milliseconds before the partially unwrapped structure recloses. Thus, most of the DNA in an isolated nucleosome is in principle available for binding other proteins (see Figure 7-12).

Inside a cell, a further loosening of DNA-histone contacts is clearly required, because eukaryotic cells contain a variety of ATP-dependent *chromatin-remodeling complexes*. These abundant proteins include a subunit that hydrolyzes ATP (an ATPase evolutionarily related to the DNA helicases discussed in Chapter 5). This “motor subunit” binds both to the protein core of the nucleosome and to the double-stranded DNA that winds around it. By using the energy of ATP hydrolysis to move this DNA relative to the core, the remodeling complex changes the structure of a nucleosome temporarily, making the DNA less tightly bound to the histone core. Through repeated cycles of ATP hydrolysis that pull the DNA helix along the nucleosome core, a remodeling complex can catalyze *nucleosome sliding* (Figure 4-26). Because nucleosomes are frequently repositioned in this way, all of the DNA sequences in chromatin are potentially available for binding to other proteins in the cell.

In addition, some types of remodeling complexes are able to remove either all or part of the nucleosome core from a nucleosome—catalyzing either an exchange of its H2A–H2B histones or the complete removal of the octameric core from the DNA (Figure 4-27). As a result, a typical nucleosome is replaced on the DNA every 1 or 2 hours inside the cell.

Cells contain dozens of different ATP-dependent chromatin-remodeling complexes that are specialized for different roles, with roughly one complex present per five nucleosomes. Most are large protein complexes that can contain 10 or

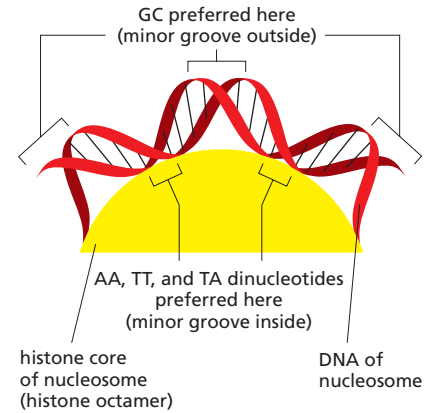


Figure 4-25 The bending of DNA in a nucleosome. The DNA helix makes 1.7 tight turns around the histone octamer. This diagram illustrates how the minor groove is compressed on the inside of the turn. Owing to structural features of the DNA molecule, the indicated dinucleotides are preferentially accommodated in such a narrow minor groove, which helps to explain why certain DNA sequences will bind more tightly than others to the nucleosome core.

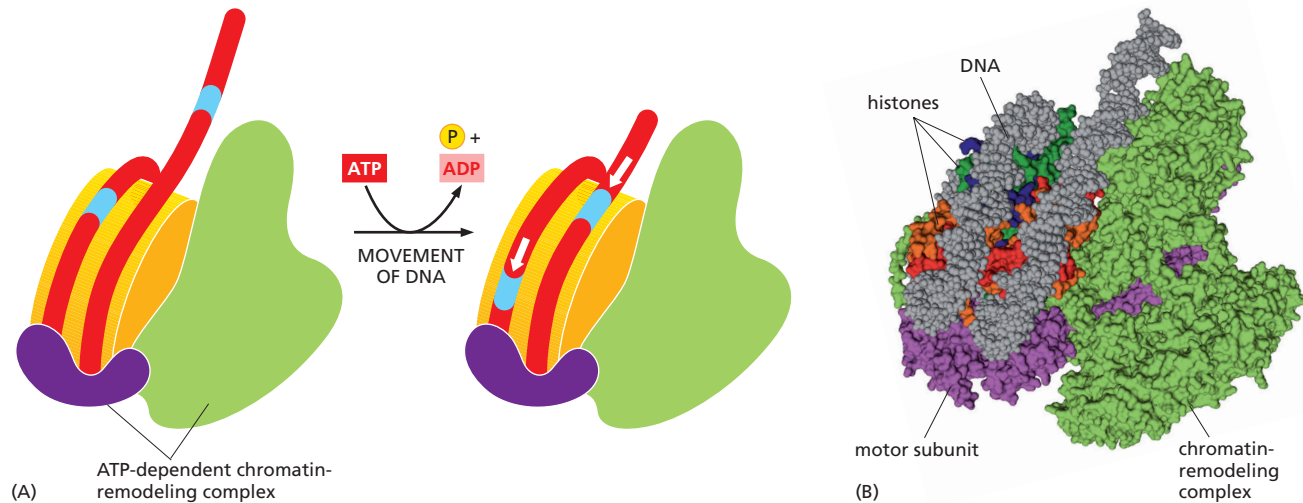


Figure 4-26 The nucleosome movements catalyzed by ATP-dependent chromatin-remodeling complexes. (A) Using the energy of ATP hydrolysis, a remodeling complex that catalyzes nucleosome sliding pulls on the DNA of its bound nucleosome and loosens its attachment to the histone octamer. Each cycle of ATP binding, ATP hydrolysis, and release of the ADP and phosphate products thereby moves the DNA in the direction of the arrows in this diagram. Because each cycle moves the DNA by one base pair, it requires many such cycles to produce the nucleosome sliding shown. (B) The structure of the yeast SWR1 chromatin-remodeling complex. This complex catalyzes an exchange of an H2A–H2B dimer for a dimer that contains an H2A histone variant. Its ATP-driven motor subunit is colored *purple*. (B, PDB code: 6GEJ.)

more subunits, some of which bind to specific modifications on histones. The activity of these complexes is controlled by the cell. As genes are turned on and off, chromatin-remodeling complexes are brought to specific regions of DNA where they act locally to influence chromatin structure (discussed in Chapter 7).

Although some DNA sequences bind more tightly than others to the nucleosome core (see Figure 4-25), the most important influence on nucleosome positioning appears to be the presence of other tightly bound proteins on the DNA. Some bound proteins favor the formation of a nucleosome adjacent to them. Others create obstacles that force the nucleosomes to move elsewhere. The exact positions of nucleosomes along a stretch of DNA therefore depend mainly

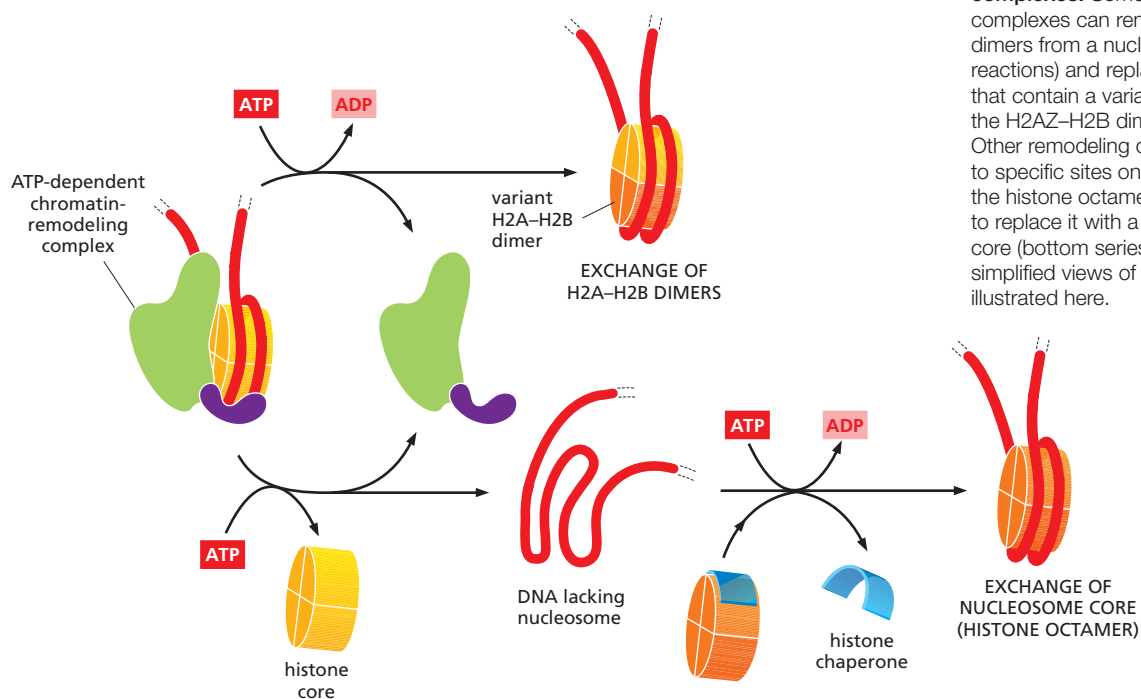


Figure 4-27 Nucleosome removal and histone exchange catalyzed by ATP-dependent chromatin-remodeling complexes. Some chromatin-remodeling complexes can remove the H2A–H2B dimers from a nucleosome (top series of reactions) and replace them with dimers that contain a variant histone, such as the H2AZ–H2B dimer (see Figure 4-36). Other remodeling complexes are attracted to specific sites on chromatin to remove the histone octamer completely and/or to replace it with a different nucleosome core (bottom series of reactions). Highly simplified views of the processes are illustrated here.

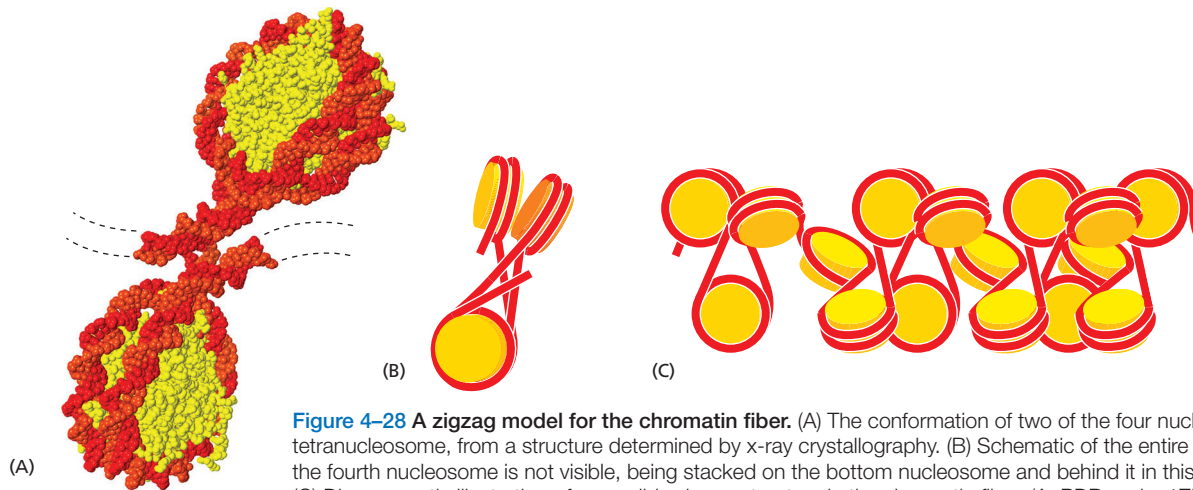


Figure 4-28 A zigzag model for the chromatin fiber. (A) The conformation of two of the four nucleosomes in a tetranucleosome, from a structure determined by x-ray crystallography. (B) Schematic of the entire tetranucleosome; the fourth nucleosome is not visible, being stacked on the bottom nucleosome and behind it in this diagram. (C) Diagrammatic illustration of a possible zigzag structure in the chromatin fiber. (A, PDB code: 1ZBB; C, adapted from C.L. Woodcock, *Nat. Struct. Mol. Biol.* 12:639–640, 2005.)

on the presence and nature of other proteins bound to the DNA. And due to the presence of ATP-dependent chromatin-remodeling complexes, the arrangement of nucleosomes on DNA can be highly dynamic, changing rapidly according to the needs of the cell.

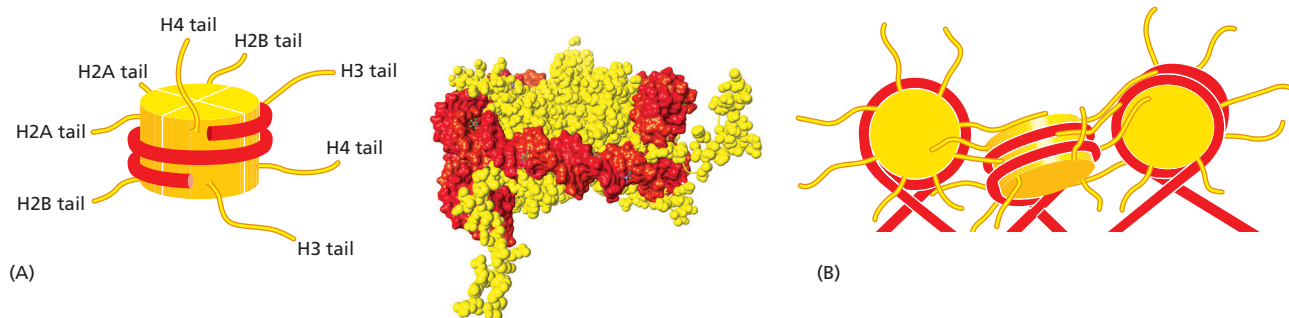
Attractions Between Nucleosomes Compact the Chromatin Fiber

Although enormously long strings of nucleosomes form on the chromosomal DNA, chromatin in a living cell probably rarely adopts the extended “beads-on-a-string” form. Instead, the nucleosomes are packed on top of one another, generating arrays in which the DNA is more highly condensed. Thus, when nuclei are very gently lysed onto an electron microscope grid, much of the chromatin is seen to be in the form of a fiber that is considerably wider than an individual nucleosome (see Figure 4-21A).

How nucleosomes are organized into condensed arrays is unclear. The structure of a tetranucleosome (a complex of four nucleosomes) obtained by x-ray crystallography and high-resolution electron microscopy of reconstituted chromatin has been used to support a zigzag model for the stacking of nucleosomes (Figure 4-28). But cryo-electron microscopy of carefully prepared nuclei suggests that most regions of chromatin are less regularly structured.

What causes nucleosomes to stack on each other? Nucleosome-to-nucleosome attractions that involve histone tails, most notably the H4 tail, constitute one important factor (Figure 4-29). In addition, an additional histone is often present in a 1-to-1 ratio with nucleosome cores, known as **histone H1**. This so-called *linker histone* is larger than the individual core histones, and it has been considerably less well conserved during evolution. A nucleosome can bind a single histone H1 molecule; this H1 molecule contacts both the DNA and the histone octamer and changes the path of the DNA as it exits from the nucleosome

Figure 4-29 A model for the role played by histone tails in the compaction of chromatin. (A) A schematic diagram shows the approximate exit points of the eight N-terminal histone tails, one from each histone protein, that extend from each nucleosome. The actual structure is shown to its right. In the high-resolution structure of the nucleosome, most of the histone tails are not visible, suggesting that they are relatively unstructured and highly flexible. (B) As indicated, the histone tails are thought to be involved in interactions between nucleosomes that help to hold them together. (A, PDB code: 1KX5.)



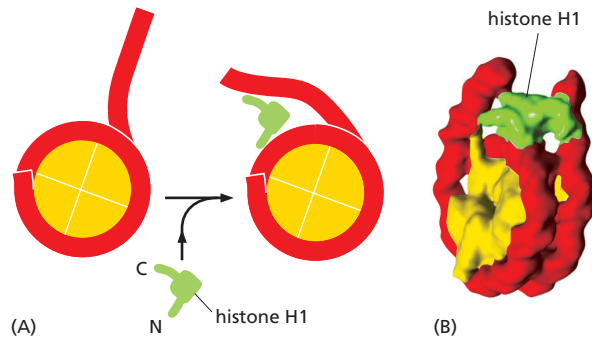


Figure 4-30 How the linker histone binds to the nucleosome. The position and structure of histone H1 are shown. H1 constrains the DNA where it exits from the nucleosome and thereby compacts chromatin. (A) Schematic, and (B) structure inferred for a single nucleosome from a structure determined by high-resolution electron microscopy of a reconstituted chromatin fiber. (B, adapted from F. Song et al., *Science* 344:376–380, 2014.)

(**Figure 4-30**). The change in the exit path of DNA is thought to help compact nucleosomal DNA. The presence of many other DNA-binding proteins, as well as proteins that bind directly to histones, is certain to add important additional features to any array of nucleosomes.

Summary

A gene is a nucleotide sequence in a DNA molecule that acts as a functional unit for the production of a protein, a structural RNA, or a catalytic or regulatory RNA molecule. In eukaryotes, protein-coding genes are usually composed of a string of alternating introns and exons associated with regulatory regions of DNA. A chromosome is formed from a single, enormously long DNA molecule that contains a linear array of many genes, bound to a large set of proteins. The human genome contains 3.1×10^9 DNA nucleotide pairs, divided between 22 different autosomes (present in two copies each) and 2 sex chromosomes. Only a small percentage of this DNA codes for proteins or functional RNA molecules. A chromosomal DNA molecule also contains three other types of important nucleotide sequences: replication origins and telomeres allow the DNA molecule to be efficiently replicated, while a centromere attaches the sister DNA molecules to the mitotic spindle, ensuring their accurate segregation to daughter cells during the M phase of the cell cycle.

The DNA in eukaryotes is tightly bound to an equal mass of histones, which form repeated arrays of DNA–protein particles called nucleosomes. The nucleosome is composed of an octameric core of histone proteins around which the DNA double helix is wrapped. Nucleosomes are spaced at intervals of about 200 nucleotide pairs, and they associate with neighboring nucleosomes to create a more compact chromatin fiber. Chromatin structure must be highly dynamic to allow access to the DNA. Some spontaneous DNA unwrapping and rewinding occurs in the nucleosome itself; however, the general strategy for reversibly changing local chromatin structure in a cell features ATP-driven chromatin-remodeling complexes. Cells contain a large set of such complexes, which are targeted to specific regions of chromatin at appropriate times. The remodeling complexes allow nucleosome cores to be repositioned, reconstituted with different histones, or completely removed to expose the underlying DNA.

THE EFFECT OF CHROMATIN STRUCTURE ON DNA FUNCTION

Having described how DNA is packaged into nucleosomes to create a chromatin fiber, we next describe the mechanisms that create different chromatin structures in different regions of a cell's genome and how this affects DNA function in cells. Most strikingly, we shall see that some types of chromatin structure can be inherited; that is, the structure can be directly passed down from a cell to its descendants. Because this creates a cell memory that is not based on an inherited change in DNA sequence, it creates a form of **epigenetic inheritance**. The prefix *epi* is Greek for “on”; this is appropriate, because epigenetics represents a form of inheritance that is superimposed on the DNA-based genetic inheritance.

Chapter 7 explains the many different ways in which the expression of genes is regulated; there we will discuss epigenetic inheritance in detail and present several different mechanisms that can produce it. Here, we are concerned only with those epigenetic mechanisms based on chromatin structure. We shall emphasize some of the chemistry that makes this possible—the covalent modification of histones in nucleosomes. These modifications serve as recognition sites for protein domains that bind different non-histone protein complexes to different regions of chromatin. In total, these complexes are so abundant that, roughly speaking, chromatin consists of one-third DNA, one-third histones, and one-third non-histone proteins by mass (see Figure 4-20). The varieties of chromatin structures produced have critical effects not only on gene expression but also on many other DNA-dependent processes—playing an important role in the development, growth, and maintenance of all eukaryotic organisms, including ourselves.

Different Regions of the Human Genome Are Packaged Very Differently in Chromatin

Light-microscope studies in the 1930s distinguished two types of chromatin in the interphase nuclei of higher eukaryotic cells: a highly condensed form, called **heterochromatin**, and all the rest, called **euchromatin**, which is less condensed. Powerful new types of molecular analyses have now allowed scientists to classify these two chromatin types more precisely. Chromatin is now considered to be either “open and active” or “closed and inactive.” As illustrated in **Figure 4-31**, roughly 80% of the human genome is in the closed form, half of which is compactly packaged in heterochromatin, with the other, less condensed half (part of the euchromatin) designated as “quiescent.” Only about 20% of the genome is packaged in that portion of the euchromatin associated with the actively expressed genes that will be the focus of Chapters 6 and 7.

Heterochromatin Is Highly Condensed and Restricts Gene Expression

Heterochromatin represents an especially compact form of chromatin (see Figure 4-10), and we are finally beginning to understand its molecular properties. Some heterochromatin is concentrated in specialized chromosomal regions, most notably at the centromeres and telomeres introduced previously (see Figure 4-19). But heterochromatin is also present at many other locations along chromosomes—locations that can vary according to the developmental state of the cell.

The packaging of DNA in heterochromatin typically prevents gene expression. However, we know now that the term *heterochromatin* encompasses a number of

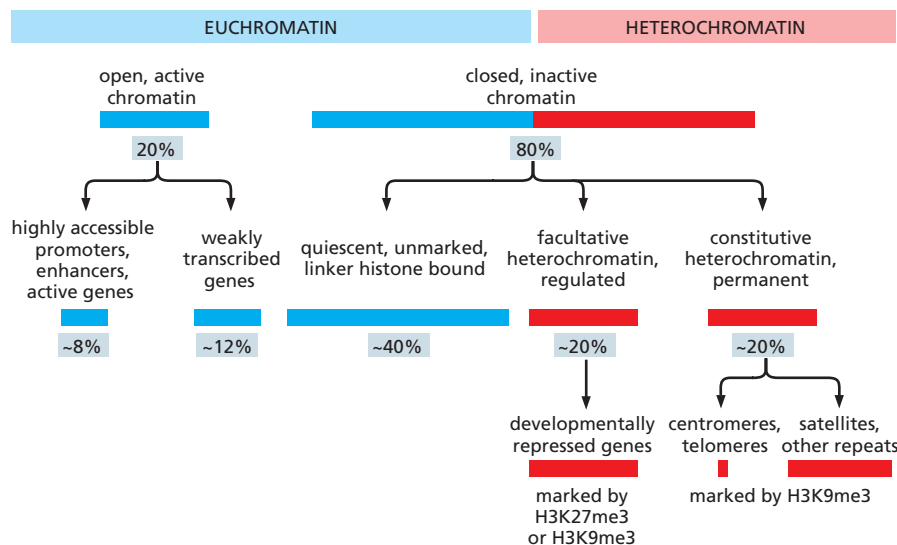
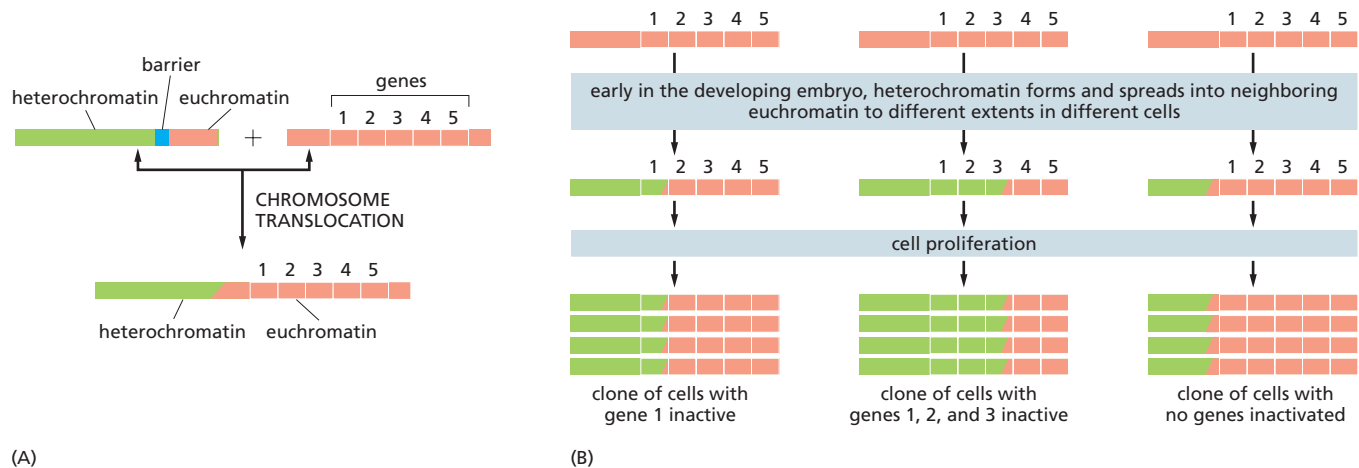


Figure 4-31 Some distinct ways that DNA is packaged in chromatin in a mammalian cell. Although all of these types of chromatin are based on long strings of nucleosomes, the nucleosomes are differently organized through association with different non-histone proteins. As we discuss shortly, these associations can depend on the particular ways that the histones in each nucleosome have been covalently “marked.” For example, as indicated here, the histone H3 molecules in heterochromatin are marked either with trimethylated lysine 9 (H3K9me3) or with trimethylated lysine 27 (H3K27me3), depending on the particular structure that is formed (see Figure 4-40).



distinct modes of chromatin compaction that have different functions. There are two broad classes, distinguished as *constitutive heterochromatin* and *facultative heterochromatin* (see Figure 4-31). The constitutive class permanently condenses many regions of the genome (and hence its name), whereas the facultative class of heterochromatin can be regulated to control gene expression.

As described next, heterochromatin has the critical property of being able to self-propagate.

The Heterochromatic State Can Spread Along a Chromosome and Be Inherited from One Cell Generation to the Next

Through errors that can occur in chromosome rejoining when DNA double-strand breaks are repaired, a piece of chromosome that is normally euchromatic can be accidentally translocated into the neighborhood of heterochromatin. Remarkably, this often causes the *silencing*—inactivation—of normally active genes, a phenomenon referred to as a *position effect*. First recognized in the fruit fly *Drosophila*, such position effects have now been observed in many eukaryotes, including yeasts, plants, and humans. Position effects are caused by a spreading of the heterochromatic state into an originally euchromatic region, and detailed studies of this phenomenon have provided important clues to the mechanisms that create and maintain heterochromatin.

After the above type of chromosome breakage-and-rejoining event, the zone of silencing, where euchromatin is converted to a heterochromatic state, is found to spread for different distances in different cells of the early fly embryo. Remarkably, these differences then are perpetuated for the rest of the animal's life: in each cell, once the heterochromatic condition is established on a piece of chromatin, it tends to be stably inherited by all of that cell's progeny (Figure 4-32). This remarkable phenomenon, called *position effect variegation*, was first recognized as a mottled loss of red pigment in the fly eye (Figure 4-33).

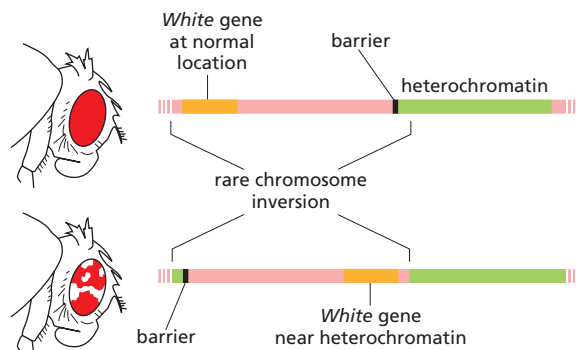
These observations, taken together, point to a fundamental feature of heterochromatin formation: heterochromatin begets more heterochromatin. This

Figure 4-32 The cause of position effect variegation in *Drosophila*.

(A) Heterochromatin (green) is normally prevented from spreading into adjacent regions of euchromatin (red) by barrier DNA sequences, which we shall discuss shortly. In flies that inherit certain chromosomal rearrangements, however, this barrier is no longer present. (B) During the early development of such flies, heterochromatin can spread into neighboring chromosomal DNA, proceeding for different distances in different cells. This spreading soon stops, but the established pattern of heterochromatin is subsequently inherited, so that large clones of progeny cells are produced that have the same neighboring genes condensed into heterochromatin and thereby inactivated (hence the “variegated” appearance of some of these flies; see Figure 4-33).

Figure 4-33 The discovery of position effects on gene expression.

The *White* gene in the fruit fly *Drosophila* controls eye pigment production and is named after the mutation that first identified it. Wild-type flies with a normal *White* gene (*White*⁺) have normal pigment production, which gives them red eyes, but if the *White* gene is mutated and inactivated, the mutant flies (*White*[−]) make no pigment and have white eyes. In flies in which a normal *White* gene has been moved near a region of heterochromatin, the eyes are mottled, with both red and white patches. The white patches represent cell lineages in which the *White* gene has been silenced by the effects of the heterochromatin. In contrast, the red patches represent cell lineages in which the *White* gene is expressed. Early in development, when the heterochromatin is first formed, it spreads into neighboring euchromatin to different extents in different embryonic cells (see Figure 4-32). The presence of large patches of red and white cells reveals that the state of transcriptional activity, as determined by the packaging of this gene into chromatin in those ancestor cells, is inherited by all daughter cells.



(A) LYSINE ACETYLATION AND METHYLATION ARE COMPETING REACTIONS

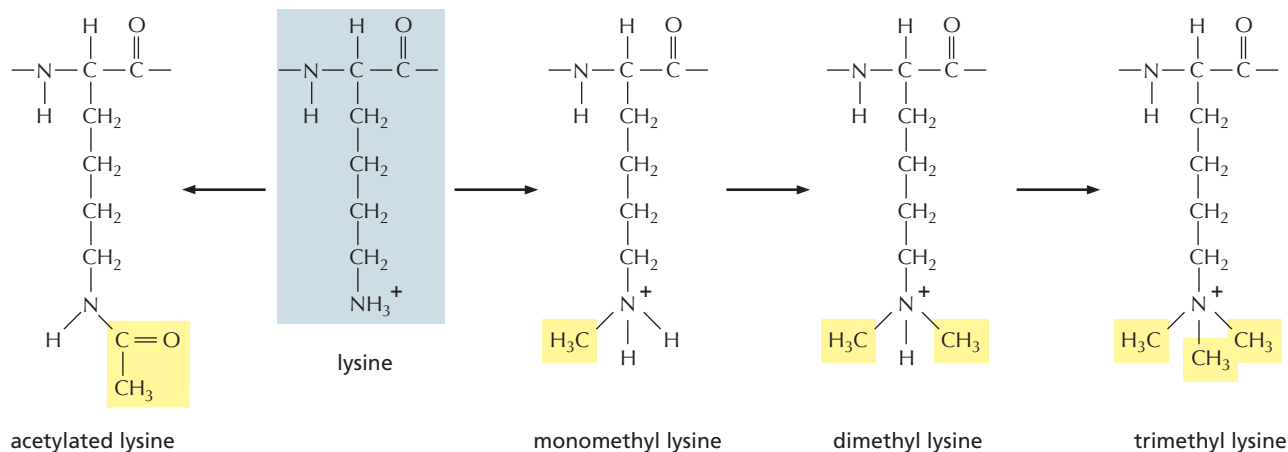
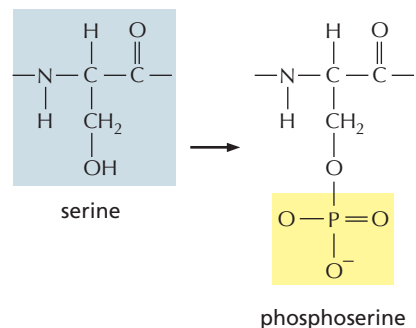


Figure 4-34 Some prominent types of covalent amino acid side-chain modifications found on nucleosomal histones. (A) Three different levels of lysine methylation are shown; each can be recognized by a different binding protein, and thus each can have a different significance for the cell. Note that acetylation removes the plus charge on lysine, and that, perhaps most important, an acetylated lysine cannot be methylated, and vice versa. (B) Serine phosphorylation adds a negative charge to a histone. Modifications of histones not shown here include the mono- or dimethylation of an arginine, the phosphorylation of a threonine, the addition of ADP-ribose to a glutamic acid, and the addition of a ubiquitin, sumoyl, or biotin group to a lysine.

(B) SERINE PHOSPHORYLATION



positive feedback can operate both in space, causing the heterochromatic state to spread along the chromosome, and in time, across cell generations, propagating the heterochromatic state of the parent cell to its daughters. The challenge is to explain the molecular mechanisms that underlie this remarkable behavior.

As a first step, one can carry out a search for the molecules that are involved. This has been done by means of *genetic screens*, in which large numbers of mutants are generated, after which one picks out those that show an abnormality of the process in question. Extensive genetic screens in *Drosophila*, fungi, and mice have identified more than 100 genes whose products either enhance or suppress the spread of heterochromatin and its stable inheritance—in other words, genes that serve as either *enhancers* or *suppressors* of position effect variegation. Many of these genes turn out to code for non-histone chromosomal proteins that interact with histones and are involved in modifying or maintaining chromatin structure. These include genes that encode some of the enzymes that add or remove covalent modifications to histone side chains, as we discuss next.

The Core Histones Are Covalently Modified at Many Different Sites

The amino acid side chains of the four histones in the nucleosome core are subjected to a remarkable variety of covalent modifications, including the acetylation of lysines, the mono-, di-, and trimethylation of lysines, and the phosphorylation of serines (Figure 4-34). A large number of these side-chain modifications occur on the eight relatively unstructured N-terminal “histone tails” that protrude from the nucleosome (Figure 4-35). However, there are also more than 20 specific side-chain modifications on the nucleosome’s globular core.

All of the above types of modifications are reversible, with one enzyme serving to create a particular type of modification and a different enzyme serving to remove it. These enzymes are highly specific. Thus, for example, acetyl groups are added to specific lysines by a set of different *histone acetyl transferases* (HATs) and removed by a set of *histone deacetylase complexes* (HDACs). Likewise, methyl groups are added to lysine side chains by a set of different *histone methyl transferases* and removed by a set of *histone demethylases*. Each enzyme is recruited to specific sites on the chromatin at defined times in each cell’s life history. The initial recruitment can depend on *transcription regulatory proteins* (sometimes

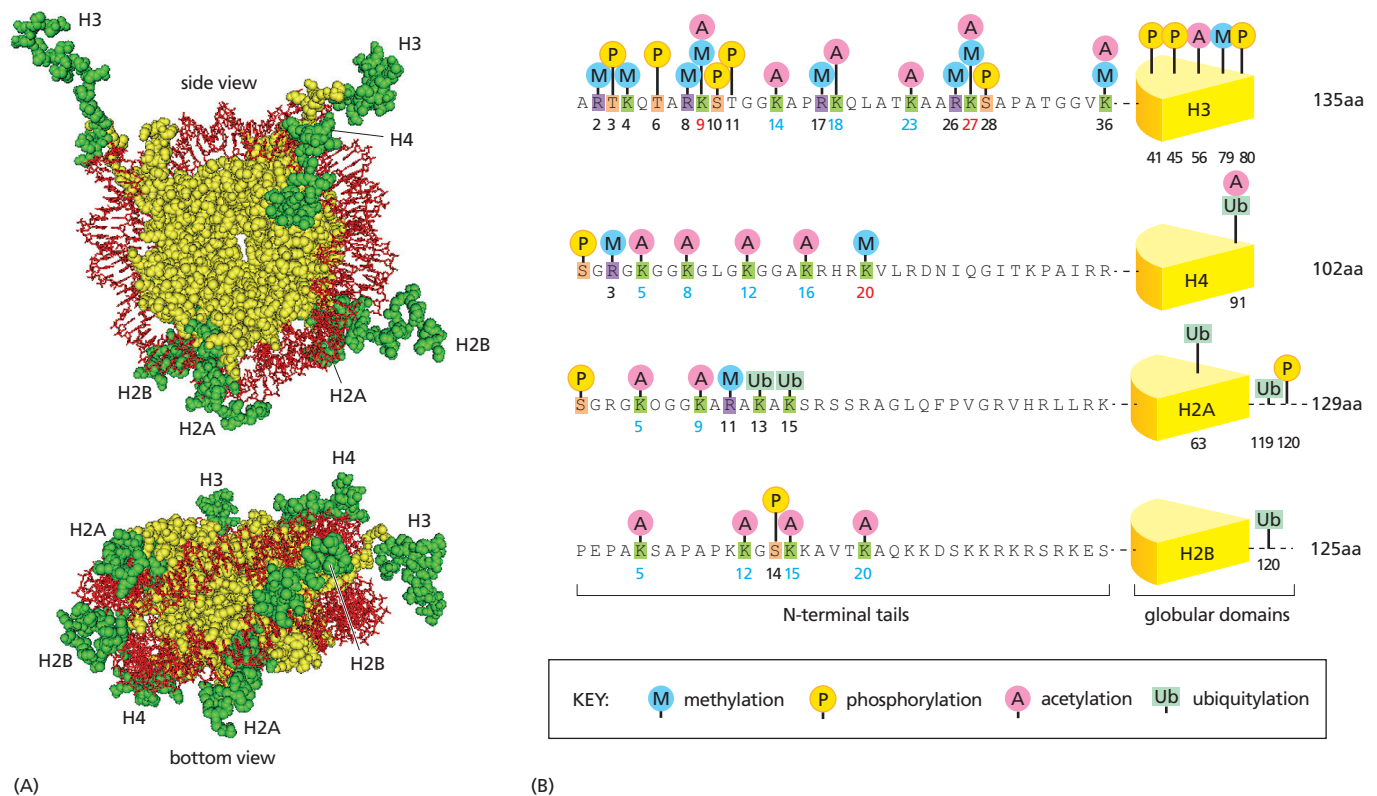


Figure 4-35 The covalent modification of core histones. (A) The structure of the nucleosome highlighting the location of the first 30 or so amino acids in each of its eight N-terminal histone tails (green). These tails are unstructured and highly mobile, and thus will change their conformation depending on other bound proteins. (B) Some well-documented modifications of the four histone core proteins are indicated. Notably, although only a single symbol is used here for methylation (M), each lysine (K) or arginine (R) can be methylated in several different ways. Thus, for example, mono-, di-, or trimethyl lysine groups can have very different effects. Note also that some positions (for example, lysine 9 of H3) can be modified either by methylation or by acetylation, but not both. Most of the modifications shown add a relatively small molecule onto a histone; the exception is ubiquitin, a 76-amino-acid protein also used for other cell processes (see Figure 3-65). Note that while most of the modifications occur in unstructured regions, including on short tails at the C-terminus of histone H2A and histone H2B, there are also important modifications in structured histone folds. (A, PDB code: 1KX5; B, based on D. Allis et al., *Epigenetics*, 2nd ed., Overview and concepts. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press, 2015.)

called “transcription factors”). As we shall explain in Chapter 7, these proteins recognize and bind to specific DNA sequences in the chromosomes. They are produced at different times and places in the life of an organism, thereby determining where and when the chromatin-modifying enzymes will act. In this way, the DNA sequence ultimately determines how histones are modified. But as we shall discuss shortly, the covalent modifications on nucleosomes in heterochromatin can persist long after the transcription regulatory proteins that first induced them have disappeared, thereby providing the cell with a memory of its developmental history. Most remarkably, as in the related phenomenon of position effect variegation discussed earlier, this “heterochromatin memory” can be transmitted from one cell generation to the next.

Very different patterns of covalent modification are found on different groups of nucleosomes, depending both on their exact position in the genome and on the history of the cell. The modifications of the histones are carefully controlled, and they can have important consequences. The acetylation of lysines on the N-terminal tails loosens chromatin structure, in part because adding an acetyl group to lysine removes its positive charge. However, the most profound effects of the histone modifications lie in their ability to recruit specific other proteins to the modified stretch of chromatin. For example, as we discuss shortly, the trimethylation of one specific lysine on the histone H3 tail attracts the protein HP1 and promotes the establishment and spread of one type of heterochromatin,

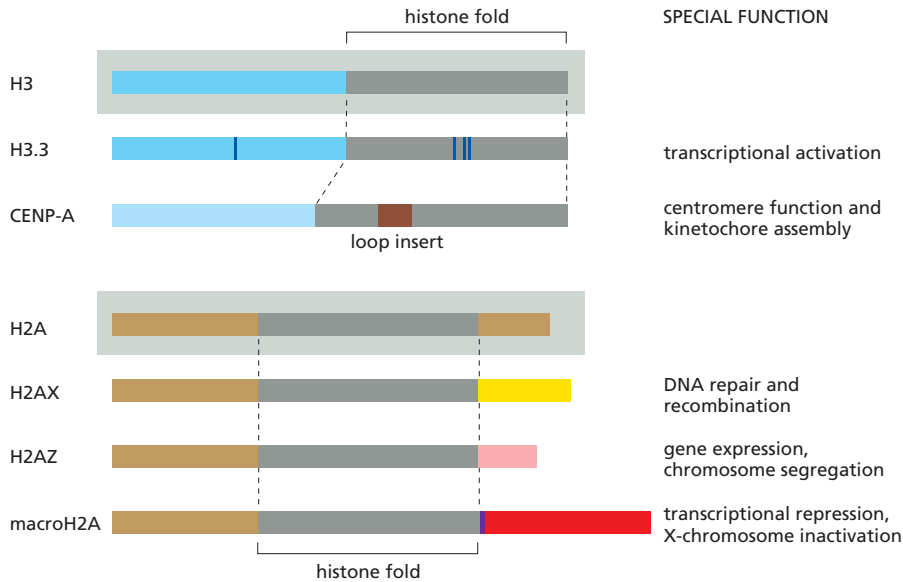


Figure 4-36 The structure of some histone variants compared with the major histone that they replace in a histone octamer. The histone variants are inserted into nucleosomes at specific sites on chromosomes by ATP-dependent chromatin-remodeling complexes (see Figure 4-26). The CENP-A (centromere protein-A) variant of histone H3 is discussed later in this chapter (see Figure 4-42); other variants are discussed in Chapter 7. The sequences in each variant that are colored differently (compared to the major histone above it) denote regions with an amino acid sequence different from this major histone. (Adapted from K. Sarma and D. Reinberg, *Nat. Rev. Mol. Cell Biol.* 6:139–149, 2005.)

while the trimethylation of a different lysine attracts a different protein to form a second heterochromatin type (see Figure 4-40). In this way and many others, the recruited proteins act with the modified histones to determine how and when genes will be expressed, as well as other critical chromosome functions.

Chromatin Acquires Additional Variety Through the Site-specific Insertion of a Small Set of Histone Variants

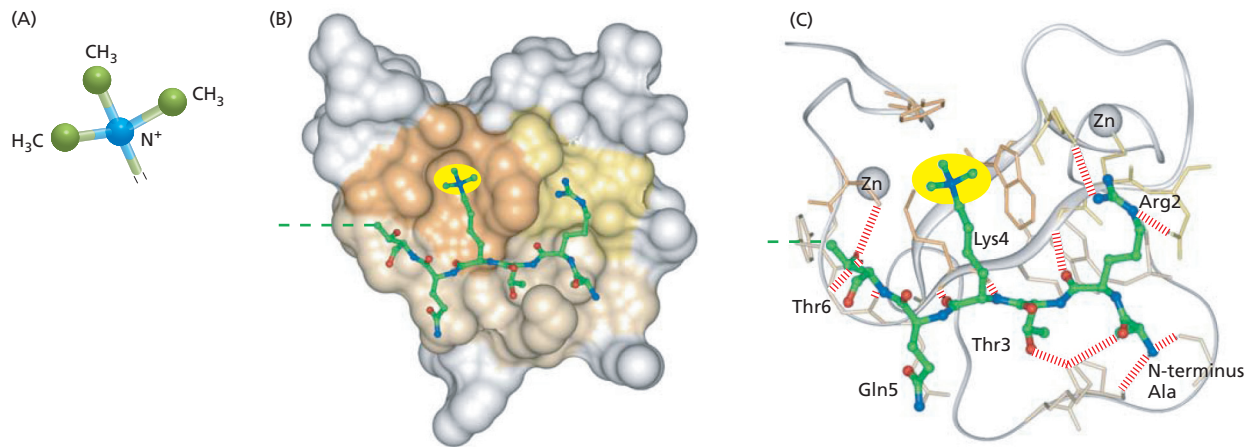
In addition to the four highly conserved standard core histones, eukaryotes also contain a few variant histones that can also assemble into nucleosomes. These histones are present in much smaller amounts than the major histones, and they have been less well conserved over long evolutionary times. Variants are known for each of the core histones with the exception of H4; some examples are shown in **Figure 4-36**.

The major histones are synthesized primarily during the S phase of the cell cycle and assembled into nucleosomes on the daughter DNA helices just behind the replication fork (see Figure 5-32). In contrast, most histone variants are synthesized throughout interphase. They are often inserted into already-formed chromatin. This requires a histone-exchange process catalyzed by a special ATP-dependent chromatin-remodeling complex that binds both the variant histone and the nucleosome containing the histone to be exchanged (see Figure 4-27). These remodeling complexes contain subunits that cause them to bind to specific sites on chromatin. As a result, each histone variant is inserted into chromatin in a highly selective manner.

Covalent Modifications and Histone Variants Can Act in Concert to Control Chromosome Functions

The number of possible distinct markings on an individual nucleosome is in principle enormous, and this potential for diversity becomes still greater when one considers the nucleosomes that contain histone variants. The histone modifications are known to occur in coordinated sets, and more than 15 such sets can be identified in mammalian cells. However, it is not yet clear how many different types of chromatin are functionally important.

Some combinations are known to have a specific meaning for the cell in the sense that they determine how and when the DNA packaged in the nucleosomes is to be accessed or manipulated—a fact that led to the idea of a “histone code.” For example, one type of marking signals that a stretch of chromatin has been newly replicated, another signals that the DNA in that chromatin has been



damaged and needs repair, while others signal when and how gene expression should take place. Various regulatory proteins contain small domains that bind to specific marks, recognizing, for example, a trimethylated lysine at a specific position in one of the histones (Figure 4-37; see also Figure 9-52). These domains are often linked together as modules in a large protein complex, such as in an ATP-dependent chromatin remodeler, allowing that complex to recognize a specific *combination* of histone modifications. These so-called *reader protein complexes* (Figure 4-38) allow particular combinations of markings on chromatin to attract a set of additional proteins, so as to execute an appropriate biological function at the right time.

Figure 4-37 How a mark on a nucleosome is read. The figure shows the structure of a protein module (called an ING PHD domain) that specifically recognizes histone H3 trimethylated on lysine 4. (A) A trimethyl group. (B) Space-filling model of an ING PHD domain bound to a histone tail (green, with the trimethyl group highlighted in yellow). (C) A ribbon model showing how the N-terminal six amino acids in the H3 tail are recognized. The *hatched red lines* represent hydrogen bonds. This is one of a family of PHD domains that recognize methylated lysines on histones; different members of the family bind tightly to lysines located at different positions, and they can discriminate between a mono-, di-, and trimethylated lysine. In a similar way, other small protein modules recognize specific histone side chains that have been marked with acetyl groups, phosphate groups, and so on. (Adapted from P.V. Peña et al., *Nature* 442:100–103, published 2006 by Nature Publishing Group. Reproduced with permission of SNCSC.)

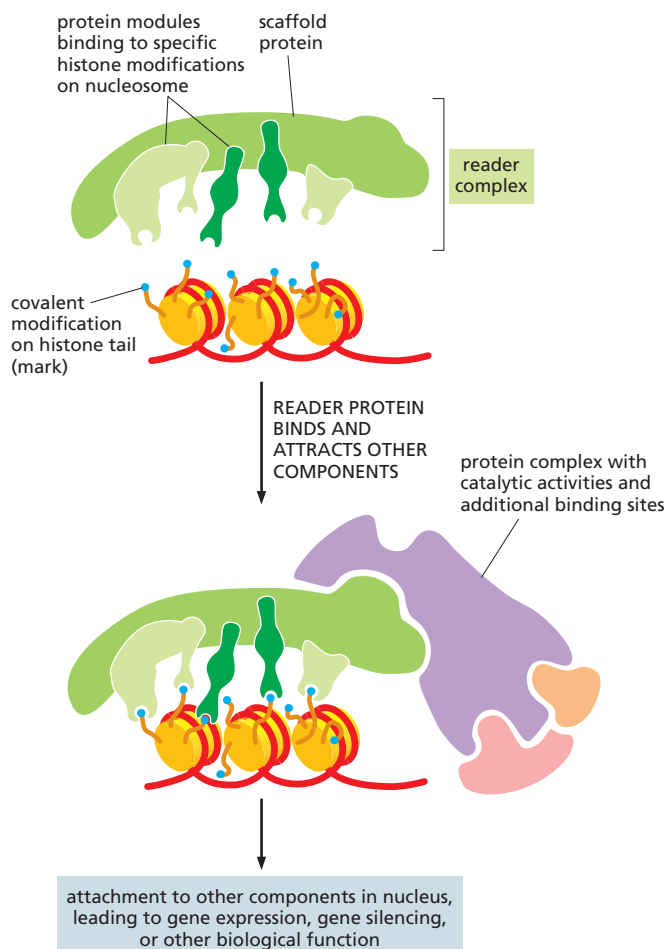


Figure 4-38 Schematic diagram showing how a particular combination of histone modifications can be recognized by a reader protein complex. A large protein complex that contains a series of protein modules, each of which recognizes a specific histone mark, is schematically illustrated (green). This *reader complex* will bind tightly only to a region of chromatin that contains several of the different histone marks that it recognizes. Therefore, only a specific combination of marks will cause the complex to bind to chromatin and attract the additional protein complexes (purple) needed to catalyze a biological function.

TABLE 4–2 Some Effects of Histone Modifications

Histone modification	Association with chromatin type	Gene expression	Abundance (%)
H4K4me3	Highly accessible, open chromatin	ON	1
H3K9ac	Highly accessible, open chromatin	ON	1
H3K9me3	Heterochromatin (either constitutive or facultative)	OFF	25
H3K27me3	Facultative heterochromatin	OFF	13

In general, sets of modifications act in combination, but only a small number of their meanings are clear. For how the two different kinds of heterochromatin listed here form, see Figure 4–40.

Each of the marks on nucleosomes is denoted as follows: listed first is an abbreviation for the particular histone involved, written as H3, H4, H2A, or H2B. The marked amino acid side chain then follows, using its one-letter abbreviation followed by its distance from that histone’s amino-terminus—thus, for example, H3K9 or H4K4 (see Figure 4–35). Listed last is the type of modification on that amino acid side chain, as in H3K9ac (acetylated), H3K9me2 (dimethylated), or H3K9me3 (trimethylated).

All of the covalent additions to histones are dynamic, being constantly removed and added at rates that depend both on their chromosomal locations and on specific states of the cell. Because the histone tails extend outward from the nucleosome core and are likely to be accessible even when chromatin is condensed, they would seem to provide an especially suitable format for creating marks that can be readily altered as a cell’s needs change. Although much remains to be learned about the meaning of the different histone modifications, a few well-studied examples of the information that can be encoded in histone modifications are listed in [Table 4–2](#).

A Complex of Reader and Writer Proteins Can Spread Specific Chromatin Modifications Along a Chromosome

The phenomenon of position effect variegation described previously requires that some modified forms of chromatin have the ability to spread for substantial distances along a chromosomal DNA molecule (see Figure 4–32). How is this possible?

The enzymes that add or remove modifications to histones in nucleosomes are part of multisubunit complexes. They can initially be brought to a particular region of chromatin by one of the sequence-specific DNA-binding proteins (transcription regulators) discussed in Chapters 6 and 7 (for a specific example, see Figure 7–23). But after a modifying enzyme “writes” its mark on one or a few neighboring nucleosomes, events that resemble a chain reaction can ensue. In such a case, the *writer enzyme* works in concert with a *reader protein* located in the same protein complex. The reader protein contains a module that recognizes the mark and binds tightly to the newly modified nucleosome (see Figure 4–37), allosterically activating an attached writer enzyme and positioning it near an adjacent nucleosome. Through many such read–write cycles, the reader protein can carry the writer enzyme along the DNA—spreading the mark in a hand-over-hand manner along the chromosome ([Figure 4–39A](#); Movie 4.2).

As important examples, there are two major classes of heterochromatin in mammalian cells, one centered on the trimethylation of H3K9 and the other on the trimethylation of H3K27 (see Figure 4–31). The H3K27me3 class

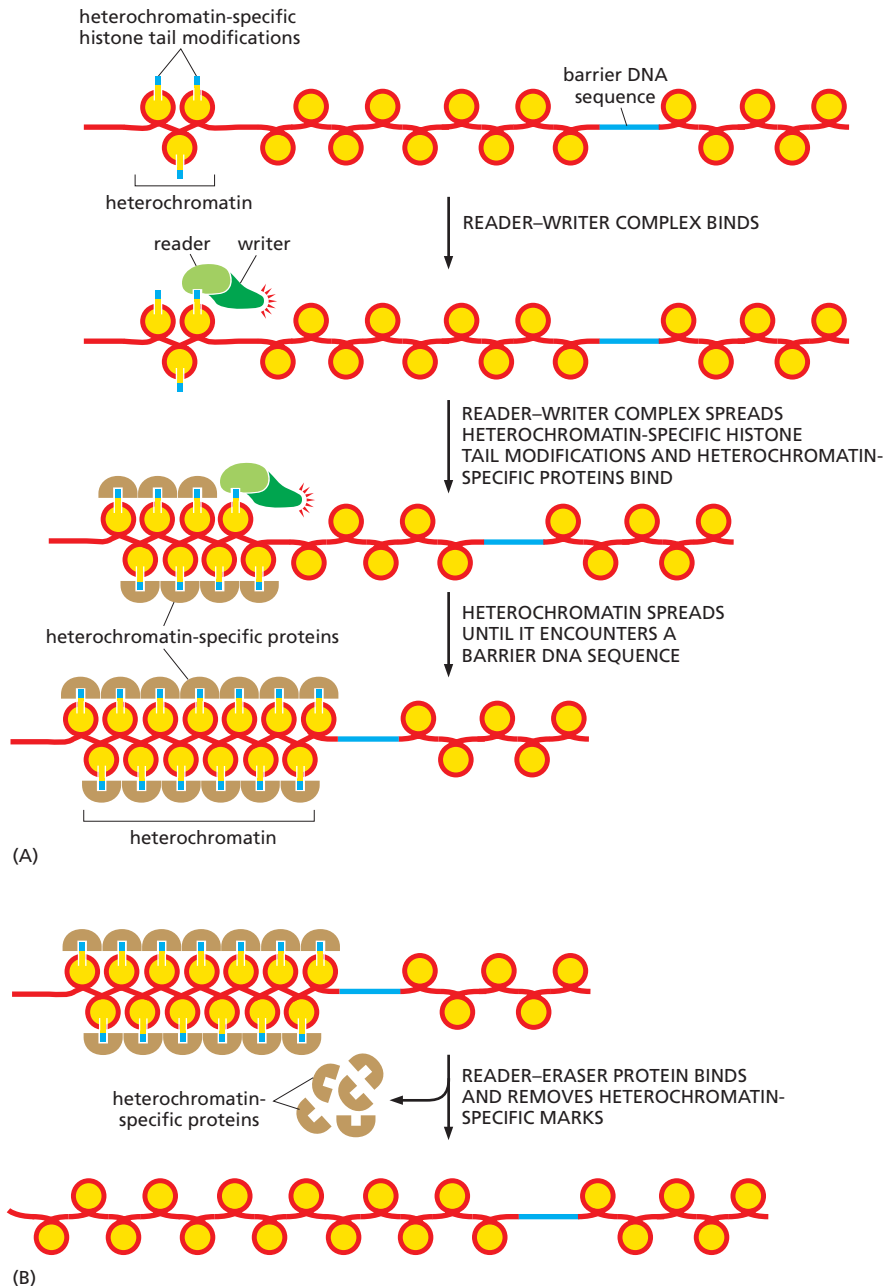


Figure 4-39 How the recruitment of reader-writer and reader-eraser complexes can spread chromatin changes along a chromosome.

(A) The writer is an enzyme that creates a particular modification on one of the four nucleosomal histones. After a writer enzyme modifies a few nucleosomes following its recruitment to a specific site on a chromosome—for example, by a transcription regulatory protein—the writer collaborates with a reader protein to spread its mark from nucleosome to nucleosome by means of the indicated reader-writer complex. For this mechanism to work, the reader must recognize the same histone modification mark that the writer produces, and its binding to that mark should activate the writer. In this schematic example, a spreading wave of chromatin condensation is thereby induced, forming heterochromatin. (B) A reader-eraser complex (not shown) reverses the chromatin change illustrated in A. As described in the text, additional proteins are involved in these two types of spreading events, including ATP-dependent chromatin-remodeling complexes.

of heterochromatin is generated by the extensively studied *polycomb repressive complex (PRC)*. Because this class is highly regulated, it is designated as facultative heterochromatin. The H3K9me3 class of heterochromatin plays an important structural role in forming the centromere (to be described shortly) and in silencing a variety of “selfish DNA” elements (see p. 465). It was once thought to remain unchanged in all cells of a multicellular organism (“constitutive heterochromatin”). However, recent data reveal that some of its forms are reversible, and that it can be used to regulate gene expression. For example, it is used to tightly repress genes that are active early in embryonic development, once they are no longer needed (see Chapter 7).

The H3K9me3 class of heterochromatin also serves to block the frequent genetic recombination that would otherwise occur between the highly repeated DNA sequences in genomes. Roughly speaking, 40% of the human genome is packaged into heterochromatin, with the ratio of the H3K9me3 class to the H3K27me3 class of heterochromatin being about 2:1 (see Figure 4-31).

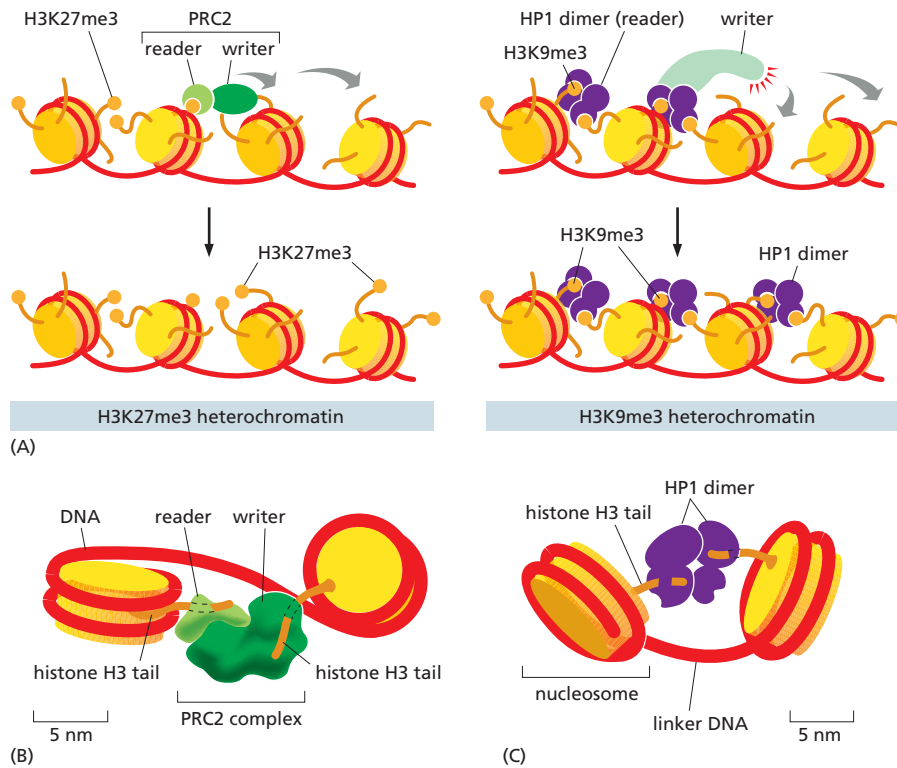


Figure 4-40 Some of the proteins required for the formation of two classes of heterochromatin in mammalian cells.

(A) Schematic diagrams that compare the reader–writer complexes for two classes of heterochromatin: one that catalyzes the spread of H3K27me3 marks along chromatin and the other that catalyzes the spread of H3K9me3 marks.

(B) The H3K27me3 class of heterochromatin is produced by the *polycomb repressive complex (PRC)*. This complex, first discovered in *Drosophila*, is composed of a PRC1 protein complex, which creates the initial mark, plus a PRC2 complex that spreads it. Shown here is the three-dimensional structure of a PRC2 reader–writer complex bridging two adjacent nucleosomes. (C) The reader for the H3K9me3 class of heterochromatin, the HP1 protein, is present in very large amounts compared to its writer enzyme. In addition to binding the writer it exists as a dimer whose two H3K9me3 binding sites enable it to bridge two adjacent nucleosomes as shown, thereby helping to package the marked nucleosomes (see also panel A, above). (B, based on S. Poepsel et al., *Nat. Struct. Mol. Biol.* 25:154–162, 2018; C, based on S. Machida et al., *Mol. Cell* 69:385–397, 2018.)

Figure 4-40 outlines what is known about the two different processes that spread and maintain these abundant heterochromatin forms.

A related type of process is used to remove specific histone modifications from a region of the DNA. In this case, an *eraser enzyme*, such as a histone demethylase or a histone deacetylase, is recruited to the complex, producing a read–erase cycle that spreads along a chromosome (**Figure 4-39B**).

In reality, the process is more complicated than the schemes just described. Both readers and writers are part of a protein complex that may contain multiple readers and writers, as well as erasers, and require multiple marks on the nucleosome to spread. Moreover, many of these reader–writer complexes also contain an ATP-dependent chromatin-remodeling protein (see **Figure 4-27**), with the reader, writer, and remodeling proteins working in concert to alter long stretches of chromatin as the reader moves progressively along the nucleosome-packaged DNA. Some idea of the complexity can be derived from the results of genetic screens for genes that either enhance or suppress the spreading and stability of heterochromatin, as manifest for example in effects on position effect variegation in *Drosophila* (see **Figure 4-33**). As pointed out previously, more than 100 such genes are known, and many of them are likely to code for subunits in one or more reader–writer–remodeling protein complexes. However, nearly all of the details remain to be deciphered by future research.

Barrier DNA–Protein Complexes Block the Spread of Reader–Writer Complexes and Thereby Separate Neighboring Chromatin Domains

The above mechanisms for spreading chromatin structures raise a potential problem. Inasmuch as each chromosome contains one continuous, very long DNA molecule, what prevents a cacophony of confusing cross-talk between adjacent chromatin domains of different structure and function? Early studies of position effect variegation had suggested an answer: certain DNA sequences mark the boundaries of chromatin domains and separate one such domain from another (see **Figure 4-32**). Several such *barrier DNA sequences* have now

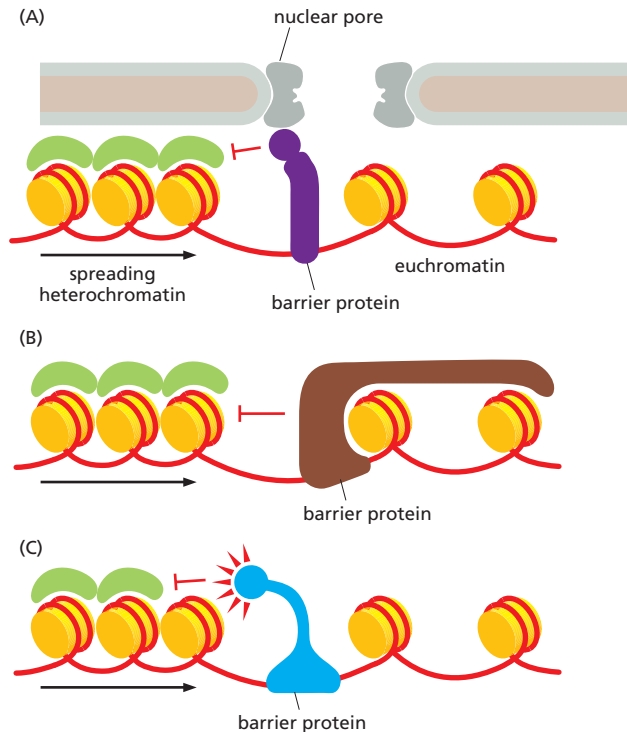


Figure 4-41 Some mechanisms of barrier action. These models are derived from experimental analyses of barrier action, and a combination of several of them may function at any one site. (A) The tethering of a region of chromatin to a large fixed site, such as the nuclear pore complex illustrated here, can form a barrier that stops the spread of heterochromatin. (B) The tight binding of barrier proteins to a group of nucleosomes can make this chromatin resistant to heterochromatin spreading. (C) By recruiting a group of highly active histone-modifying enzymes, barriers can erase the histone marks that are required for heterochromatin to spread. For example, a potent acetylation of lysine 9 on histone H3 will compete with lysine 9 methylation, thereby preventing the binding of the HP1 protein needed to form a major form of heterochromatin. (Based on A.G. West and P. Fraser, *Hum. Mol. Genet.* 14:R101–R111, 2005.)

been identified and characterized through the use of genetic engineering techniques that allow specific DNA segments to be deleted from or inserted into chromosomes.

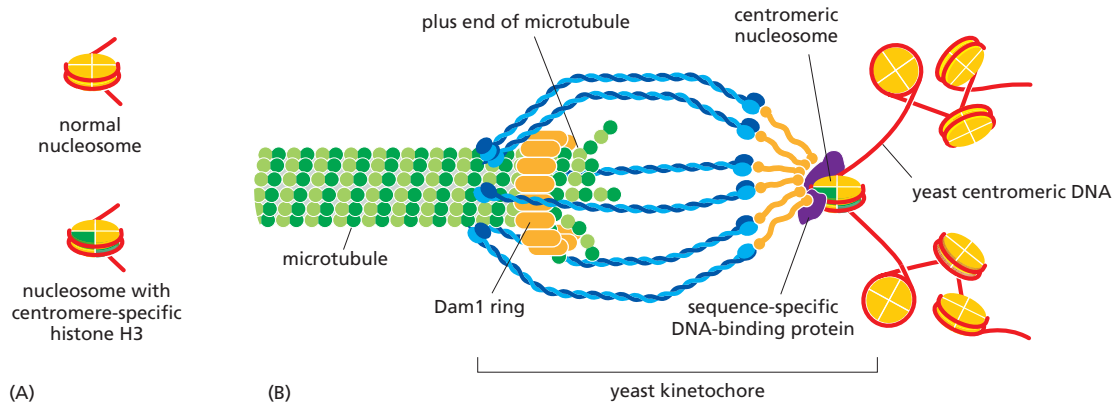
Analysis of one type of barrier sequence reveals that it contains a cluster of binding sites for histone acetylase enzymes. The acetylation of a lysine side chain is incompatible with the methylation of the same side chain, and specific lysine methylations are required to spread heterochromatin (see Figure 4-40). Histone acetylases are therefore logical candidates for the formation of the barriers that stop heterochromatin spreading, as are the histone-demethylating enzymes that erase the marks on histones specific to heterochromatin. Other types of barrier mechanisms are also known (Figure 4-41).

Centromeres Have a Special, Inherited Chromatin Structure

There is a specialized chromatin structure at the centromere, the region of each chromosome required for its orderly attachment to the mitotic spindle. In many complex organisms, including humans, each centromere is embedded in a stretch of special *centromeric chromatin* that persists throughout interphase, even though the centromere-mediated attachment to the spindle and movement of DNA occur only during mitosis. This chromatin contains a centromere-specific variant H3 histone, known as CENP-A (centromere protein-A; see Figure 4-36), plus additional proteins that pack the nucleosomes into particularly dense arrangements and form the kinetochore, the structure required for mitotic spindle attachment (see Figure 4-19).

In the yeast *Saccharomyces cerevisiae*, a specific DNA sequence of approximately 125 nucleotide pairs is sufficient to serve as a centromere. Despite its small size, more than a dozen different proteins assemble on this DNA sequence; the proteins include the CENP-A histone H3 variant that, along with the three other core histones, forms a centromere-specific nucleosome. The additional proteins at the yeast centromere form a kinetochore, which links this nucleosome to a single microtubule from the yeast mitotic spindle (Figure 4-42).

The centromeres in more complex organisms are considerably larger than those in budding yeasts. For example, human centromeres extend over several million nucleotide pairs, and, while they contain multiple copies of CENP-A and bind



about 20 microtubules, they do not seem to contain a centromere-specific DNA sequence. These centromeres largely consist of short, repeated DNA sequences, known as *alpha satellite DNA*. But the same repeat sequences are also found at other (non-centromeric) positions on chromosomes, indicating that they are not sufficient to direct centromere formation. And most striking, in some unusual cases, new human centromeres (called neocentromeres) have been observed to form spontaneously on fragmented chromosomes at positions that were originally euchromatic and lack alpha satellite DNA altogether (Figure 4-43).

It seems that centromeres in complex organisms are defined by an assembly of proteins rather than by a specific DNA sequence. Essential for this assembly is a set of CENP-A nucleosomes. And once a special region of centromeric chromatin forms, this assembly is thereafter faithfully inherited when a chromosome replicates, despite the fact that no special DNA sequence need be involved.

The inactivation of some centromeres and the genesis of others *de novo* occur as organisms evolve. Different species, even when quite closely related, often have different numbers of chromosomes (see Figure 4-14 for an extreme example). And, as we discuss later, detailed genome comparisons show that many changes in chromosomes have arisen through chromosome breakage-and-rejoining events; these create novel chromosomes, some of which must initially have contained abnormal numbers of centromeres—either more than one or

Figure 4-42 The structure of a simple centromere. (A) In the yeast *Saccharomyces cerevisiae*, a special centromeric DNA sequence assembles a single nucleosome in which two copies of an H3 variant histone (called CENP-A in most organisms) replace the normal H3. (B) How amino acid sequences unique to this variant histone (see Figure 4-36) help to assemble additional proteins, some of which form a kinetochore. The yeast kinetochore is unusual in capturing only a single microtubule; humans have much larger centromeres and form kinetochores that can capture 20 or more microtubules. The kinetochore is discussed in detail in Chapter 17.

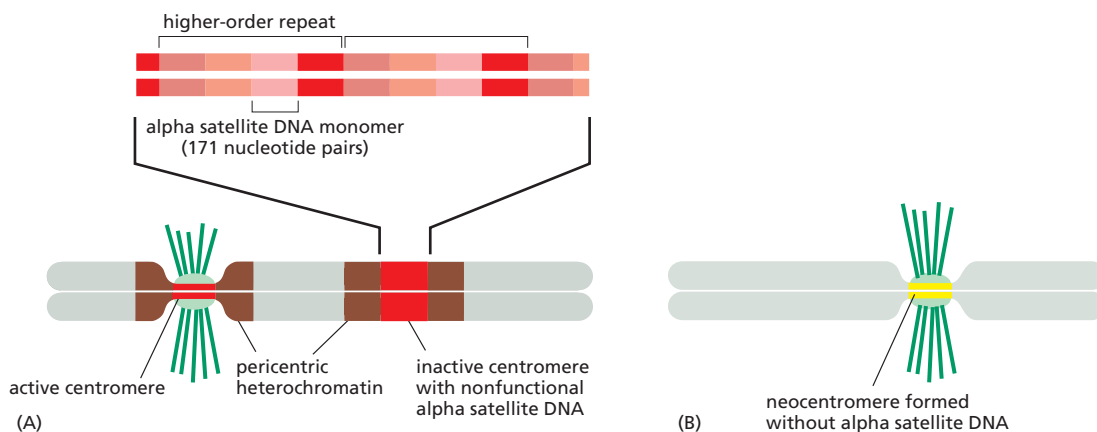


Figure 4-43 Evidence for the plasticity of human centromere formation. (A) A series of A-T-rich alpha satellite DNA sequences is repeated many thousands of times at each human centromere (red) and is surrounded by *pericentric heterochromatin* (brown). The pericentric heterochromatin contains H3K9me3, along with HP1 protein, and it is an example of "classical" *constitutive heterochromatin* (see Figure 4-31). As indicated, some human chromosomes contain two blocks of alpha satellite DNA, each of which presumably functioned as a centromere in its original chromosome. (B) In a small fraction (1/2000) of human births, extra chromosomes are observed in cells of the offspring. Some of these extra chromosomes, which have formed from a breakage event, contain alpha satellite DNA sequences that have been co-opted to form new centromeres (neocentromeres); other neocentromeres lack alpha satellite DNA altogether and have arisen from what was originally euchromatic DNA.

none at all. Yet stable inheritance requires that each chromosome should contain one centromere, and one only. It seems that surplus centromeres must have been inactivated and/or new centromeres created, so as to allow the rearranged chromosome sets to be stably maintained.

Some Forms of Chromatin Can Be Directly Inherited

The changes in centromere activity just discussed, once established, need to be perpetuated through subsequent cell generations. What could be the mechanism of this type of epigenetic inheritance?

It has been proposed that *de novo* centromere formation requires an initial seeding event, which is followed by the formation of a specialized DNA-protein structure that contains nucleosomes formed with the CENP-A variant of histone H3. In humans, this seeding event happens more readily on arrays of alpha satellite DNA than on other DNA sequences. The entire centromere then forms as an all-or-none entity, suggesting that the creation of centromeric chromatin is a highly cooperative process, spreading out from an initial seed. And once established, this special form of chromatin is passed on to each daughter cell when a cell divides.

Both the spreading and the inheritance of centromeric chromatin mimic the phenomenon of position effect variegation that we discussed earlier (see Figure 4-32). The spreading of a particular chromatin structure can be explained by the action of reader-writer complexes (see Figure 4-39A). But how can we explain the inheritance of centromeric chromatin from one cell generation to the next?

Notably, experiments have revealed that the H3-H4 tetramers from each nucleosome on the parent DNA helix are directly inherited by both of the sister DNA helices at a replication fork, being equally partitioned between them. This is quickly followed by the addition of two H2A-H2B dimers to complete each “half-old” nucleosome, and by the deposition of the new histone octamers needed to restore the normal complement of nucleosomes (see Figure 5-32). Therefore, once a set of CENP-A-containing nucleosomes has been assembled on a stretch of DNA, it is easy to understand how a new centromere could be generated in the same place on both daughter chromosomes after each round of cell division. One need only assume that the presence of the CENP-A histone in an inherited nucleosome selectively recruits more CENP-A histone to its newly formed neighbors.

An analogous scheme is thought to explain the observation that both the H3K9me3 and H3K27me3 forms of heterochromatin, once formed, are directly inherited after each round of chromosome replication. In those cases, the H3-H4 tetramers that contain a particular modified histone will be passed to each daughter DNA helix, followed by the rapid addition of H2A-H2B to re-form specifically marked nucleosomes. The action of reader-writer complexes can then spread the identical marks to the new neighboring nucleosomes—either H3K9me3 or H3K27me3 (see Figure 4-40). Thus, the activities of reader-writer complexes can explain not only the spreading of specific forms of chromatin along a chromosome, but also the propagation of heterochromatin across cell generations—from parent cell to daughter cell (Figure 4-44).

In vertebrates, but not in flies, this inheritance of heterochromatin is often reinforced by a second process that silences the same genes. As described in Chapter 7, this process is based on a DNA methylation system that generates an inherited pattern of methylated C nucleotides (see Figures 7-47 and 7-48).

The Abnormal Perturbations of Heterochromatin That Arise During Tumor Progression Contribute to Many Cancers

As described in Chapter 20, cancer arises from a series of accidental changes in a cell's control systems, each of which is inherited and accumulates progressively in a clone of daughter cells. Many of these sequential changes are the result of mutations that alter the sequence of the DNA encoding an important regulatory protein, such as a protein kinase involved in cell signaling. Other

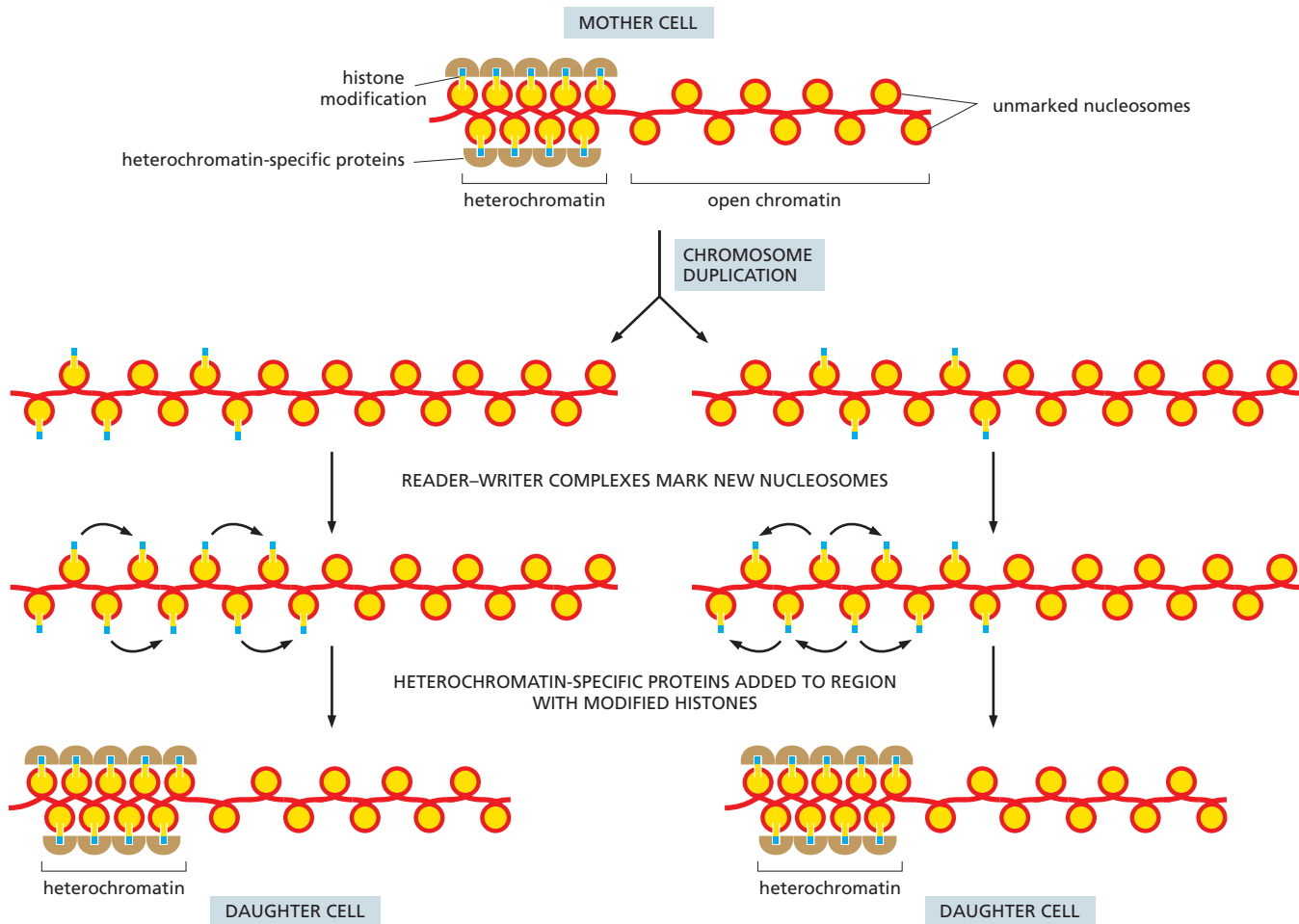


Figure 4–44 The propagation of heterochromatin across generations. After DNA replication, all of the old H3–H4 molecules in the parent chromatin will be passed directly to the daughter DNA helices, half to each daughter (see Figure 5–32). This causes the specially marked nucleosomes in heterochromatin to be inherited, as indicated. Reader–writer complexes can then add the same pattern of histone modification to their new, unmarked nucleosome neighbors, as previously illustrated in Figure 4–40. As the original pattern of histone marks is reestablished, binding sites are created for the heterochromatin-specific non-histone proteins that assemble to reproduce the parent chromatin structure. This type of process is thought to occur for both H3K9me3 and H3K27me3 classes of heterochromatin, as well as for the chromatin at centromeres; it is thought not to occur for the open structure of chromatin at active genes.

mutations that are known to drive tumor progression alter chromatin structure, for example by affecting the readers, writers, or erasers of histone marks—or by altering a chromatin-remodeling complex. Because changes in chromatin can alter gene expression, as we explore further in Chapter 7, this finding is not surprising.

Much more surprising is the discovery that a change in a single amino acid in a histone can cause cancer. As very abundant proteins, each histone is encoded by multiple copies of its histone gene. As a result, any mutation that changes a histone should alter less than 10% of the molecules, meaning that any effect observed must be dominant—overriding the presence of a large excess of the normal histone. Such *oncohistone* mutations appear to be present in about 4% of all tumors, and for a few special cancers they are a predominant cancer driver. In particular, a mutation that changes the lysine at position 27 of histone H3 to a methionine (denoted as H3K27M) is almost universally present in a lethal type of pediatric brain tumor (diffuse intrinsic pontine glioma; DIPG), while also being occasionally found in acute myeloid leukemias and melanomas in adults. This mutation has been shown to exert its dominant effect by binding abnormally

tightly to the PCR2 protein complex, thereby altering the overall pattern and level of H3K27me3 modifications across the human genome (see Figure 4–40), a finding that illustrates the important role that heterochromatin plays in controlling genes.

Summary

In the chromosomes of eukaryotes, DNA is assembled into long strings of nucleosomes, but a variety of different chromatin structures is possible. This variety is based on a large set of reversible covalent modifications of the four histones in the nucleosome core. The modifications include the mono-, di-, and trimethylation of many different lysine side chains, an important reaction that is incompatible with the acetylation that can occur on the same lysines. Specific combinations of the modifications mark many nucleosomes, governing their interactions with other proteins. These marks are read when protein modules that are part of a larger protein complex bind to the modified nucleosomes in a region of chromatin. These reader proteins then attract additional proteins that perform various functions.

Some reader protein complexes contain a histone-modifying enzyme, such as a histone lysine methylase, that “writes” the same mark that the reader recognizes. A reader–writer–remodeling complex of this type can spread a specific form of chromatin along a chromosome. In particular, large regions of condensed heterochromatin are thought to be formed in this way. Heterochromatin is commonly found around centromeres and near telomeres, but it is also present at many other positions in chromosomes. The tight packaging of DNA into heterochromatin usually silences the genes within it.

The phenomenon of position effect variegation provides strong evidence for the inheritance of condensed states of chromatin from one cell generation to the next. A similar mechanism appears to be responsible for maintaining the specialized chromatin at centromeres. More generally, the ability to propagate specific chromatin structures across cell generations makes possible an epigenetic cell memory process that plays a role in maintaining the set of different cell states required by complex multicellular organisms.

THE GLOBAL STRUCTURE OF CHROMOSOMES

Having discussed the DNA and protein molecules from which the chromatin fiber is made, we now turn to the organization of the chromosome on a more global scale and the way in which its various domains are arranged in space. Packaged into nucleosomes, a typical human chromosome would be able to span the nucleus thousands of times. Thus, a higher level of folding is required, even in interphase chromosomes. As we shall see, this higher-order packaging involves the folding of each chromosome into a series of large loops through a process catalyzed by ring-shaped SMC (structural maintenance of chromosomes) protein complexes.

We begin this section by describing some unusual chromosomes that can be easily visualized. Exceptional though they are, these special cases reveal features that are relevant for all eukaryotic chromosomes. Next, we describe how the interphase chromosomes are arranged in the mammalian cell nucleus. Finally, we discuss the mechanisms that cause a special compaction of chromosomes during their passage from interphase to mitosis.

Chromosomes Are Folded into Large Loops of Chromatin

An early insight into the structure of the chromosomes in interphase cells came from studies of the stiff and enormously extended chromosomes in growing amphibian oocytes (immature eggs). These very unusual **lampbrush chromosomes** (the largest chromosomes known), paired in preparation for meiosis, are clearly visible in the light microscope, where they are seen to contain a

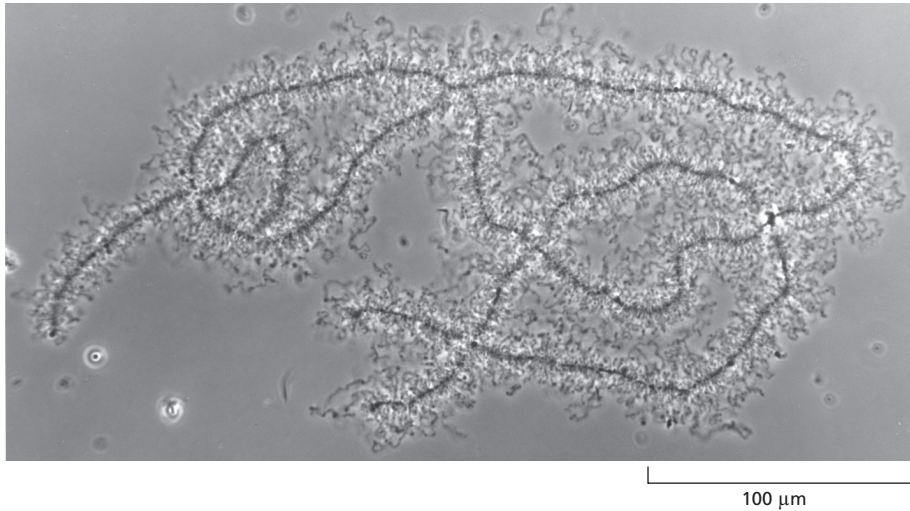


Figure 4-45 A light micrograph of lampbrush chromosomes in an amphibian oocyte. Early in oocyte differentiation, each chromosome replicates to begin meiosis, and the homologous replicated chromosomes pair to form this highly extended structure. Each lampbrush chromosome consists of two aligned sets of paired *sister chromatids*, with large chromatin loops as a prominent feature. The chromosome set contains thousands of loops, each containing a particular DNA sequence that remains extended in the same manner as the oocyte grows, producing huge amounts of RNA for storage in the oocyte. The lampbrush chromosome stage persists for months or years, while the oocyte builds up a supply of materials required for its ultimate development into a new individual. These chromosomes were first described in 1878. (Courtesy of Joseph G. Gall.)

series of large chromatin loops emanating from a linear chromosomal axis (**Figure 4-45**).

Figure 4-46 summarizes lampbrush chromosome structure. Each chromosome is formed by a very long DNA helix packed into chromatin, and it is joined to a sister DNA helix, called a *sister chromatid*. Most of the DNA is located near the junction of the two chromatids, and it is highly condensed and transcriptionally inert; this DNA serves to organize each chromatid along a linear chromosome axis. In contrast, the highly transcribed regions of the DNA extend from the axis as large loops, which range in length from tens of thousands to hundreds of thousands of nucleotide pairs.

Meiotic chromosomes are found to adopt the lampbrush chromosome state in the growing oocytes of all vertebrates, except for mammals. However, if incubated in amphibian oocyte cytoplasm, human sperm chromosomes will form lampbrush chromosomes. This reveals that chromosomes are highly dynamic structures that can restructure in different environments.

Polytene Chromosomes Are Uniquely Useful for Visualizing Chromatin Structures

Further early insight into the structure of interphase chromosomes came from a second unusual type of cell—the *polytene cells* of flies, such as the fruit fly *Drosophila*. Some special cells, in many organisms, grow abnormally large through

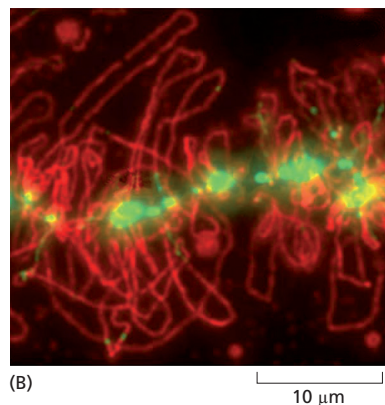
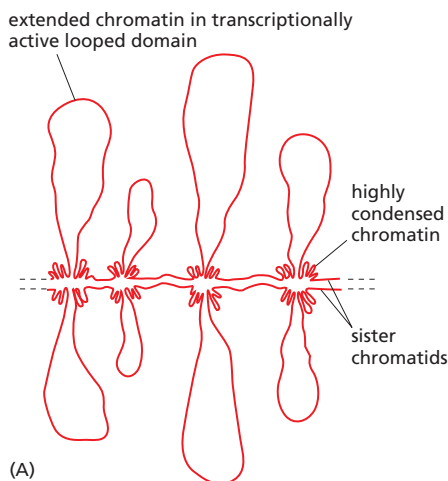
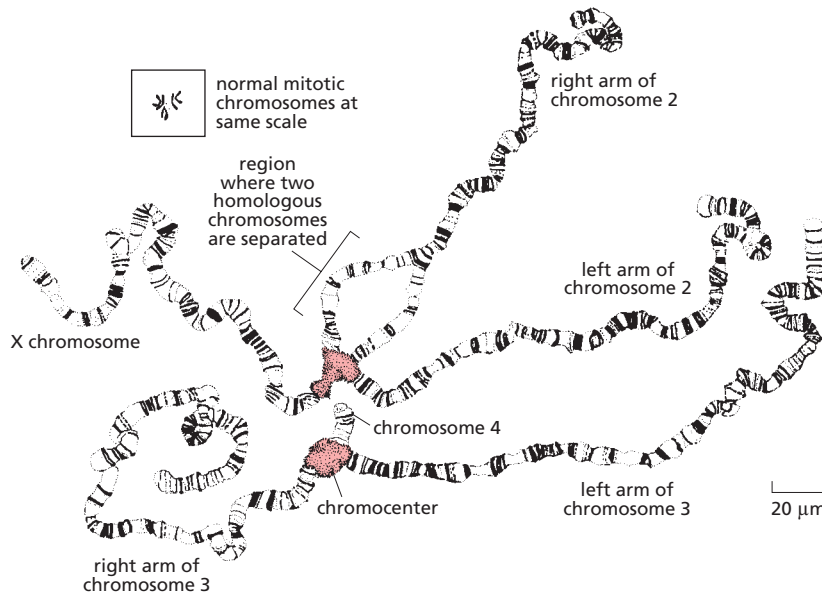


Figure 4-46 The structure of lampbrush chromosomes. (A) Model for a small portion of one pair of sister chromatids. Two identical DNA double helices are aligned side by side, packaged into different types of chromatin (see Figure 4-45). (B) Merged light micrographs of the lampbrush chromosomes in an axolotl (a type of salamander), stained for transcriptionally active (RNA polymerase, red) and inactive (DNA containing 5-methyl C, green) chromatin regions. The loops are stiff and extended because they are being unusually highly transcribed, with RNA polymerases spaced only about 100 nucleotide pairs apart. Most of the rest of the DNA in each chromosome (the great majority) remains condensed and is located close to the chromatid axis. (B, from G.T. Morgan et al., *Chromosome Res.* 20:925–942, 2012. Reproduced with permission of SNCSC.)



multiple cycles of DNA synthesis without cell division. Such cells, containing increased numbers of standard chromosomes, are said to be *polyploid*. In the salivary glands of fly larvae, this process is taken to an extreme degree, creating huge cells that contain hundreds or thousands of copies of the genome. Moreover, in this case, all the copies of each chromosome are aligned side by side in exact register, like drinking straws in a box, to create giant **polytene chromosomes**. These chromosomes allow features to be detected that are thought to be shared with ordinary interphase chromosomes but are normally hard to see.

When polytene chromosomes from a fly's salivary glands are viewed in the light microscope, distinct alternating dark *bands* and light *interbands* are visible (Figure 4-47), each formed from a thousand identical DNA sequences arranged side by side in register. About 95% of the DNA in polytene chromosomes is in bands, and 5% is in interbands. A very thin band can contain 3000 nucleotide pairs, while a thick band may contain 200,000 nucleotide pairs in each of its chromatin strands. The chromatin in each band appears dark because the DNA is more condensed than the DNA in interbands; it may also contain a higher concentration of proteins (Figure 4-48).

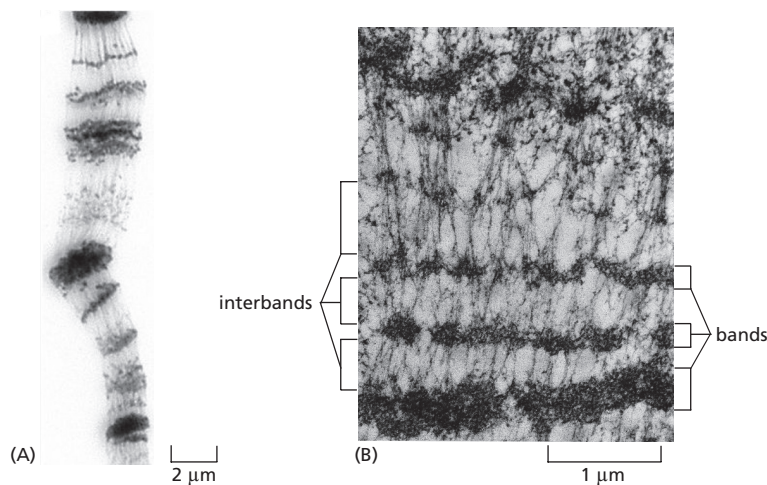


Figure 4-47 The entire set of polytene chromosomes in one *Drosophila* salivary cell. In this drawing of a light micrograph, the giant chromosomes have been spread out for viewing by squashing them against a microscope slide. *Drosophila* has four chromosomes, and there are four different chromosome pairs present. But each chromosome is tightly paired with its homolog (so that each pair appears as a single structure), which is not true in most nuclei (except in meiosis). Each chromosome has undergone multiple rounds of replication, and the homologs and all their duplicates have remained in exact register with each other, resulting in huge chromatin cables many DNA strands thick.

The four polytene chromosomes are normally linked together by heterochromatic regions near their centromeres that aggregate to create a single large chromocenter (*pink region*). In this preparation, however, the chromocenter has been split into two halves by the squashing procedure used. (Adapted from T.S. Painter, *J. Hered.* 25:465–476, 1934. With permission from Oxford University Press.)

Figure 4-48 Micrographs of polytene chromosomes from *Drosophila* salivary glands. (A) Light micrograph of a portion of a chromosome. The DNA has been stained with a fluorescent dye, but a reverse image is presented here that renders the DNA *black* rather than *white*; the bands are clearly seen to be regions of increased DNA concentration. This chromosome has been processed by a high-pressure treatment so as to show its distinct pattern of bands and interbands more clearly. (B) An electron micrograph of a small section of a *Drosophila* polytene chromosome seen in thin section. Bands of very different thickness can be readily distinguished, separated by interbands, which contain less condensed chromatin. (A, adapted from D.V. Novikov et al., *Nat. Methods* 4:483–485, published 2007 by Nature Publishing Group. Reproduced with permission of SNCSC; B, courtesy of Veikko Sorsa.)

There are approximately 3700 bands and 3700 interbands in the complete set of *Drosophila* polytene chromosomes, a number that should be compared to the 14,000 genes in this fruit fly (see Table 1–2, p. 29). The bands can be recognized by their different thicknesses and spacings, and each one has been given a number to generate a chromosome “map” that has been indexed to the finished genome sequence of *Drosophila*.

The *Drosophila* polytene chromosomes provide a good starting point for examining how chromatin is organized on a large scale. In the previous section, we saw that there can be many forms of chromatin, each of which contains nucleosomes with a different combination of modified histones. Specific sets of non-histone proteins assemble on these nucleosomes to affect biological function in different ways. Recruitment of some of these non-histone proteins can spread for long distances along the DNA, imparting a similar chromatin structure to broad tracts of the genome (see Figure 4–39A). At low resolution, the interphase chromosome can therefore be considered as a mosaic of chromatin structures, each containing particular nucleosome modifications associated with a particular set of non-histone proteins.

Polytene chromosomes allow us to see details of this mosaic of domains in the light microscope and to observe some of the changes associated with gene expression. For example, by staining *Drosophila* polytene chromosomes with antibodies and using ChIP (chromatin immunoprecipitation) analysis (see Chapter 8), the locations of specific histone modifications and non-histone proteins in chromatin can be mapped across the entire *Drosophila* DNA sequence. These results suggest that three types of repressive chromatin predominate in this organism, along with two types of chromatin on actively transcribed genes, and that each type is associated with a different complex of non-histone proteins. In addition to these five major chromatin types, other more minor forms of chromatin appear to be present, each of which may be differently regulated and have distinct roles in the cell. Much remains to be learned about the molecular structures that underlie these findings.

Chromosome Loops Decondense When the Genes Within Them Are Expressed

When an organism containing polytene chromosomes progresses from one developmental stage to another, distinctive *chromosome puffs* arise and old puffs recede in its polytene chromosomes as new genes become expressed and old ones are turned off (Figure 4–49). From inspection of each puff when it is relatively small and the banding pattern is still discernible, it seems that most puffs arise from the decondensation of a single chromosome band.

The individual chromatin fibers that make up a puff can be visualized with an electron microscope. In favorable cases, loops are seen. When genes in the loop are not expressed, the loop assumes a thickened, condensed structure, but when gene expression is occurring, the loop becomes more extended—resembling the loops seen in lampbrush chromosomes. This again demonstrates that chromosome structures are dynamic, as we had concluded from lampbrush experiments.

Mammalian Interphase Chromosomes Occupy Discrete Territories in the Nucleus, with Their Heterochromatin and Euchromatin Distributed Differently

Light microscopy after chromosome painting reveals that each of the 46 interphase chromosomes in a human cell occupies its own discrete territory within the nucleus; that is, the chromosomes are not extensively entangled with one another (Figure 4–50). However, pictures such as these present only an average view of the DNA in each chromosome. By staining a heterochromatic region of a chromosome, one finds that it is often closely associated with the nuclear envelope,

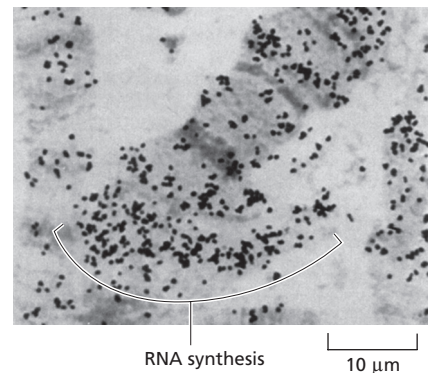


Figure 4–49 RNA synthesis in polytene chromosome puffs. An autoradiograph of a single puff in a polytene chromosome from the salivary glands of the freshwater midge *Chironomus tentans*, the fly in which polytene chromosomes were first discovered in 1881. As outlined in Chapter 1 and described in detail in Chapter 6, the first step in gene expression is the synthesis of an RNA molecule using the DNA as a template. The decondensed portion of the chromosome is undergoing RNA synthesis and has become labeled with ^3H -uridine, an RNA precursor molecule that is incorporated into growing RNA chains. The black dots are produced where the radioactivity emitted interacts with an overlying photographic emulsion. (From J.J. Bonner and M.L. Pardue, *Cell* 12:227–234, 1977. With permission from Elsevier.)

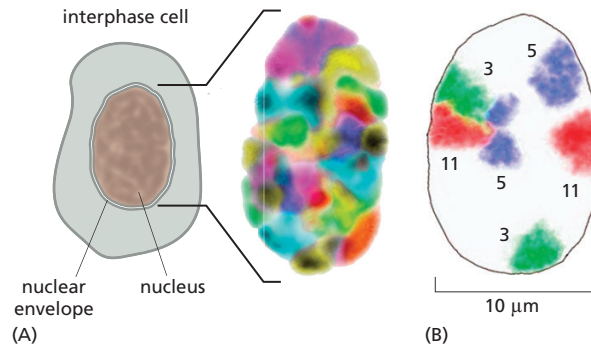


Figure 4-50 Simultaneous visualization of the chromosome territories for all of the human chromosomes in a single interphase nucleus. Here, the DNA probes for a specific chromosome are labeled so as to fluoresce at specific wavelengths, and a different combination of dyes is used to label the probes for each chromosome so that each chromosome fluoresces differently. This “chromosome painting” technique allows DNA–DNA hybridization to be used to detect each chromosome, as in Figure 4-11. Three-dimensional reconstructions were then produced. (A) Viewed in a fluorescence microscope, the nucleus is seen to be filled with a patchwork of discrete colors. (B) To highlight their distinct locations, three sets of chromosomes are singled out: chromosomes 3, 5, and 11. Note that pairs of homologous chromosomes, such as the two copies of chromosome 3, are not generally located in the same position. (Adapted from M.R. Hübner and D.L. Spector, *Annu. Rev. Biophys.* 39:471–489, 2010. With permission from Annual Reviews.)

regardless of the chromosome examined. And probes that preferentially stain chromosomal regions containing a high density of active genes reveal that most of these regions extend out from the territory of each chromosome into the nucleoplasm and away from the nuclear envelope (**Figure 4-51**).

Studies of mammalian cells also show that highly folded loops of chromatin expand to occupy an increased volume when a gene within them is expressed. For example, **Figure 4-52** demonstrates how a large, highly transcribed gene in a human cell, which appears as a dot in an interphase cell when inactive, is extended just like a lampbrush loop by its extensive transcription.

As we shall discuss in Chapter 6, the interior of the nucleus is very heterogeneous, with functionally different regions that are produced by *biomolecular condensates* specialized to speed different biochemical processes—such as RNA synthesis, RNA splicing, and DNA replication. The fact that a section of a chromosome can move away from its chromosome territory, as we have just seen, helps cells to use nuclear condensates to accelerate various reactions. These condensates include the nucleolus specialized for producing ribosomes and the nuclear speckles involved in RNA production (see pp. 353–357).

Although much is known about DNA transcription and how genes are turned on and off at the molecular level, we defer that discussion to Chapters 6 and 7. Here we instead concentrate on the mechanism that packages the long linear DNA molecule in each mammalian chromosome into a series of loops in the cell nucleus.

A Biochemical Technique Called Hi-C Reveals Details of Chromosome Organization

As explained in Chapter 8, new automated technologies allow scientists to determine massive amounts of DNA sequence at low cost. A powerful *chromosome conformation capture* method that exploits this ability, *Hi-C*, has made it possible

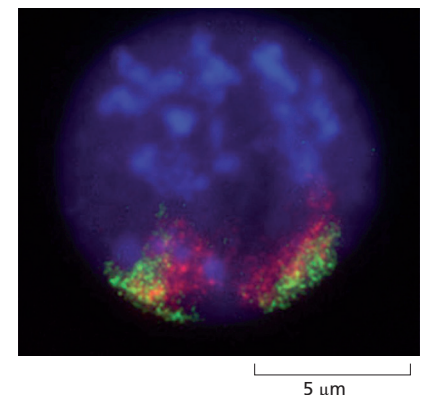


Figure 4-51 The looping out of gene-rich regions of the genome from chromosome territories. In this light micrograph of a cell nucleus, DNA has been fluorescently labeled with a dye (blue). In addition, by using a chromosome painting technique, all of the DNA of one particular mouse chromosome has been stained green, and the gene-rich regions of that same chromosome have been stained red. (From W.A. Bickmore and B. van Steensel, *Cell* 152:1270–1284, 2013. With permission from Elsevier.)

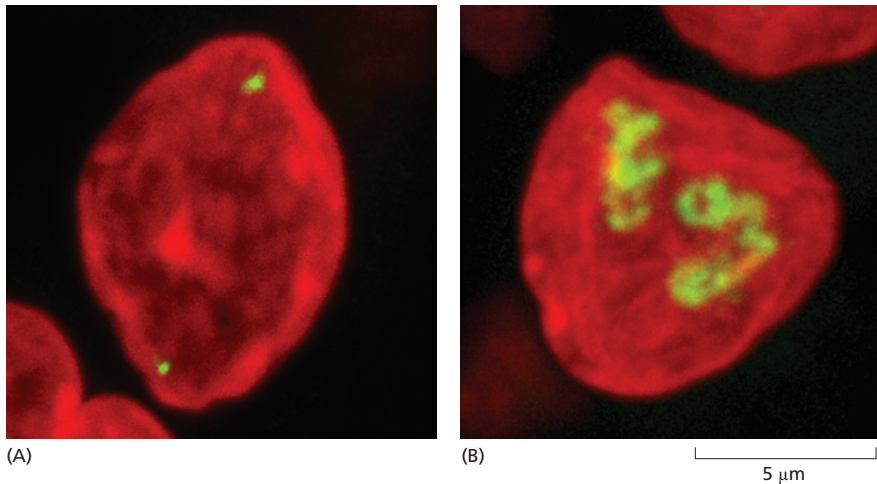


Figure 4-52 A large gene forms a highly extended chromosome loop when highly expressed in a human interphase cell. The thyroglobulin gene codes for the extracellular protein thyroglobulin, which is secreted by thyroid gland cells. This gene is both unusually large (270 kb) and unusually highly transcribed. DNA and RNA hybridization techniques reveal that it forms a large stiff loop that extends into the interior of the nucleus from each of the two homologous chromosomes that express it. (A) An image of a cell that does not make thyroglobulin reveals the thyroglobulin locus as a small dot. (B) An image of a thyroid gland cell expressing the gene. Experiments demonstrate that these loops retract back to a dot if RNA synthesis is blocked by drug treatment. (Courtesy of Irina Solovei, Ludwig-Maximilians University, Munich.)

to assess the frequency with which any two genomic loci—whether along a single chromosome or between chromosomes—are held together (**Figure 4-53**).

Application of the Hi-C technique has revealed some fundamental features of chromosome organization. One is that each interphase chromosome occupies its own discrete territory within the nucleus. That is, the genomic loci within a chromosome contact each other more often than they do the loci on different chromosomes, showing that the chromosomes are not extensively entangled. This was expected from the earlier findings of discrete chromosome territories by light microscopy, as just described. In addition, Hi-C and related chromosome capture methods have revealed that the interphase chromosome is folded into a long series of *topologically associated domains*, or *TADs*, in which any two

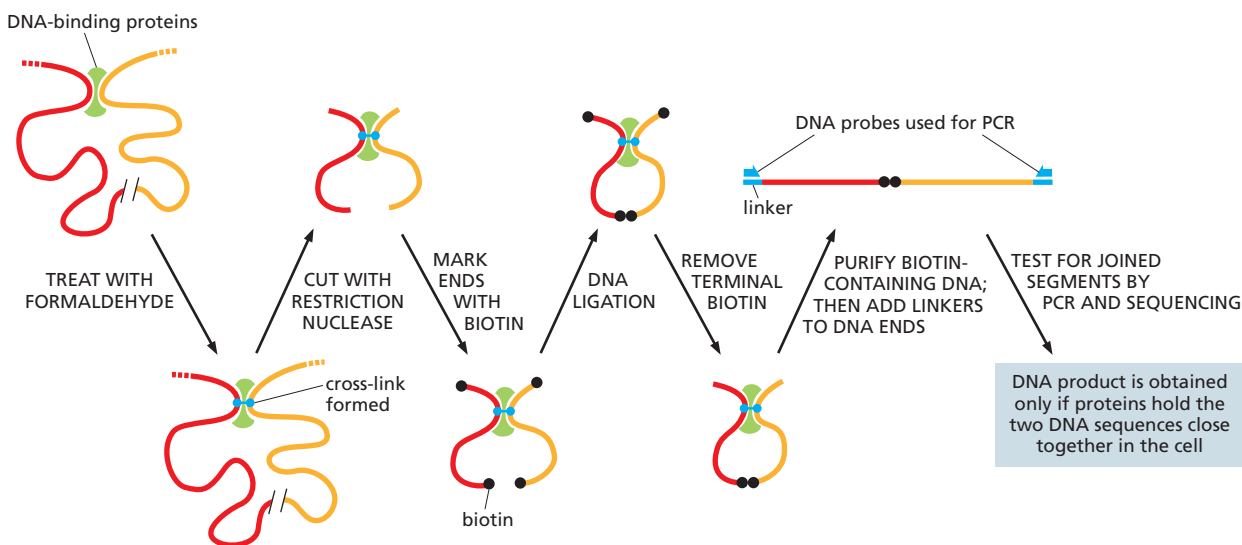


Figure 4-53 Use of Hi-C to determine the frequency with which any two DNA sequences are adjacent to each other in chromosomes. In the Hi-C technique, cells are treated with cross-linking agents such as formaldehyde to create covalent DNA–protein–DNA cross-links, as indicated. The DNA is then treated with an enzyme (called a restriction nuclease) that chops the DNA into many pieces, cutting at strictly defined nucleotide sequences and forming sets of identical “cohesive ends” (see Figure 8-23). These ends are then marked by the incorporation of biotinylated nucleotides, so as to enable their selective purification later. Any two DNA ends can become covalently joined if incubated with a DNA ligase enzyme. But importantly, prior to the DNA ligation step shown, the DNA is diluted so that only the fragments that have been kept in close proximity to each other (through cross-linking) are likely to join. After the ligation step, the cross-links are reversed, and all the biotin at unligated DNA ends is removed. This allows the newly ligated fragments of DNA to be selectively purified through their binding to streptavidin beads, amplified by PCR (polymerase chain reaction), and then sequenced (by methods described in Chapter 8). The results, combined with knowledge of the complete DNA sequence of each chromosome, generate detailed models for the conformation of chromosomes.

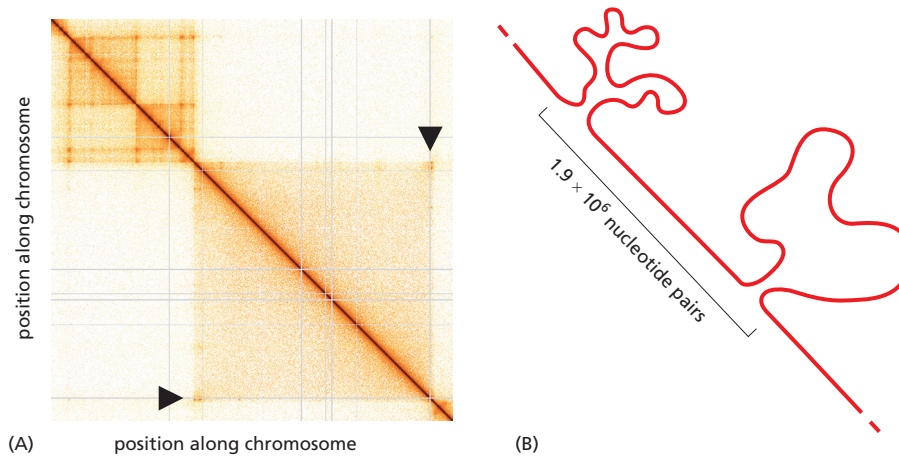


Figure 4-54 Chromosome capture techniques reveal topologically associated, looped domains in interphase chromosomes. (A) A plot of results for a small portion of human chromosome 8 in human fibroblasts. The intensity of the *orange* color denotes the number of times that any DNA sequence from a 2-kilobase (kb) region whose chromosome position is indicated on the X axis was found to be ligated to any DNA sequence from a 2-kb region whose chromosome position is indicated on the Y axis (see Figure 4-53). The darker the color, the more often these two regions were located in close proximity to each other inside the cell. The strong stripe along the diagonal results from the fact that DNA sequences that are very close to each other along the linear chromosome have a high probability of interacting with each other.

The largest square reveals a topologically associated domain (TAD), with the dots in the diagram (highlighted here by *black arrowheads*) showing the unusually persistent contacts between the DNA sequences located at the base of a simple looped domain that contains about a million nucleotide pairs. The box at the *upper left* shows a more complex “nested” TAD structure. Here, the multiple dots are interpreted as a “bouquet” of smaller loop subdomains gathered together to form a larger folding domain, as indicated in (B), which schematically compares the structures inferred for the two different TADs. [A, Courtesy of Oliver Rando, University of Massachusetts Medical School. The data for human chromosome 8 (125,034,652 to 126,909,657) are from N. Krietenstein et al., *Mol. Cell* 78:554–565, 2020.]

DNA sequences are much more likely to encounter each other than they are to encounter DNA sequences outside of that domain. Their locations along each chromosome can be derived from the series of squares observed in plots of results like those in [Figure 4-54](#). These reveal that DNA segments in a contiguous set (those within a TAD) have a relatively high probability (*dark color* in the figure) of becoming ligated together. And because any one DNA segment in a TAD can become ligated to many others in that TAD, one concludes that the chromatin inside each TAD must be folded in a way that allows its DNA sequences to frequently encounter any other DNA sequence within it.

Extensive data of this type have led to the conclusion that all interphase chromosomes are organized as a long linear series of folded *looped domains* of chromatin, with the DNA in each loop being compacted, but highly mobile. Although a typical loop in a human chromosome might contain between 50,000 and 200,000 nucleotide pairs of DNA, loops of a million nucleotide pairs also exist, and there appear to be roughly 10,000 loops in the human genome. How these loops are formed is the subject that we discuss next.

Chromosomal DNA Is Organized into Loops by Large Protein Rings

The enormously long DNA molecules that form chromosomes must be organized if they are to be effective as the carriers of the genetic information that every cell requires to survive and multiply. A mechanism for creating this organization that appears to be universal involves the folding of the DNA into loops by an **SMC protein complex**, a large protein ring that both binds to and encircles the DNA double helix. These rings—which function similarly in archaea, bacteria, and eukaryotes—have the structure illustrated in [Figure 4-55A](#). The subunits that give the complex its name are the pair of long, coiled-coil SMC (structural maintenance of chromosomes) proteins. Each long SMC protein chain folds upon itself to form a globular ATPase domain, and two of these chains join together to create a ring that is large enough for DNA that is packaged in chromatin to readily pass through it (compare ring size to nucleosomes in [Figure 4-55B](#)).

By associating with additional proteins, the two ATPase domains in an SMC protein complex allow the ring to motor rapidly along DNA. Why this type of activity is useful to the cell is perhaps most easily understood by considering the role of these rings in bacteria. The bacterial chromosome is a large circular DNA molecule that is duplicated from a single DNA replication origin to produce two identical DNA circles. Each of these two daughter chromosomes must be transferred to a different daughter cell when the parent cell divides. The required segregation process begins when a series of SMC protein complexes is loaded onto the DNA near the DNA replication origin; these protein rings then create a loop on each new DNA double helix and proceed to move continuously along the

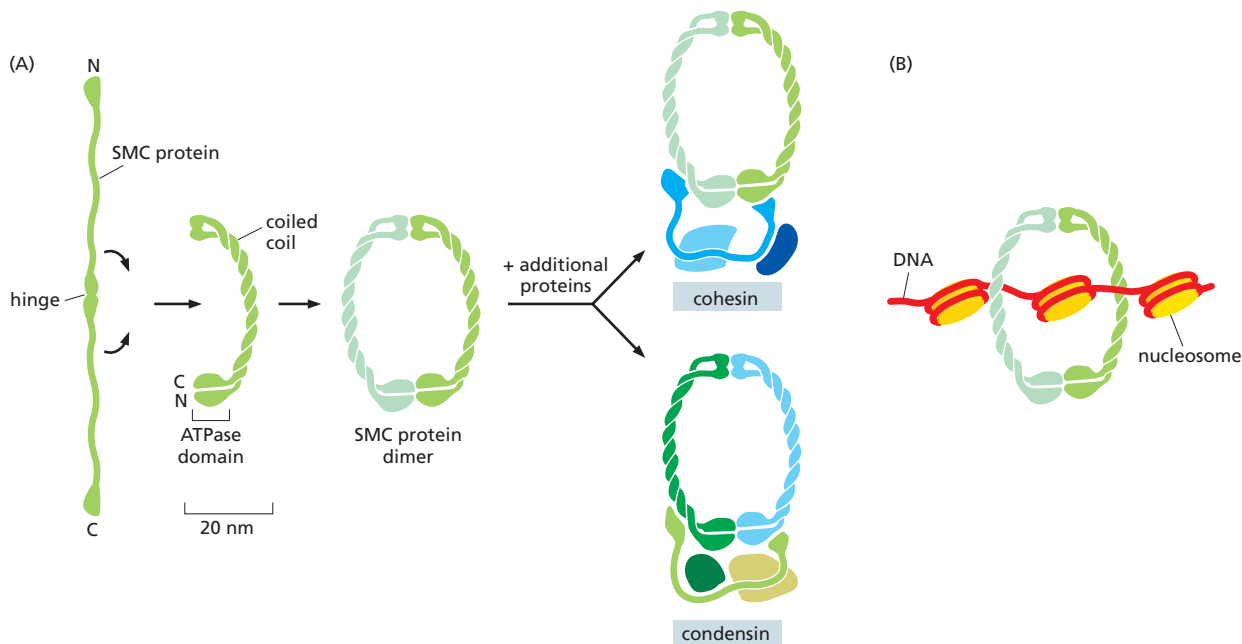


Figure 4-55 SMC protein complex formation. (A) These structures have important roles in bacteria, archaea, and eukaryotes. As indicated, their core forms from a large protein of 1000 to 1500 amino acids that folds on itself to form a antiparallel coiled-coil with a globular head; two of these molecules then pair to produce a ring structure with a flexible hinge at one end and an ATP-binding domain at the other end. Additional subunits are then added to form either cohesin or condensin, as indicated. (B) An SMC complex compared to the size of a nucleosome.

DNA, cleanly separating the two daughter chromosomes from each other in the manner illustrated in [Figure 4-56A](#). A model proposed to explain this movement is likewise shown ([Figure 4-56B](#)).

The DNA molecules that form eukaryotic chromosomes are generally much longer than those that form bacterial chromosomes, and many different DNA replication origins are therefore needed to copy each chromosome (see Chapter 5). Moreover, the accurate separation of two daughter chromosomes involves a process that is considerably more elaborate than the process in bacteria (described in Chapter 17). The SMC protein complexes have therefore diversified during the evolution of eukaryotes to play two different roles, each of which is critical for the function of chromosomes.

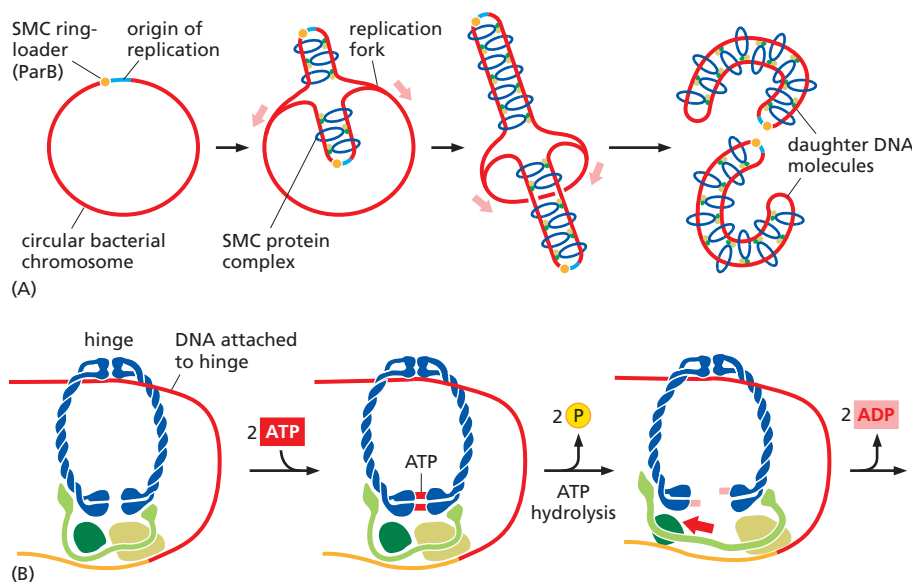


Figure 4-56 How moving SMC-containing protein rings function to separate bacterial chromosomes. (A) A DNA sequence adjacent to the origin of replication on the circular *Bacillus subtilis* chromosome binds a protein (ParB) that repeatedly loads bacterial SMC protein complexes. These complexes then use the energy released by repeated cycles of ATP hydrolysis to travel along the entire chromosome to the site where the two DNA replication forks that were formed at the origin meet, thereby partitioning the two very large daughter DNA molecules in a way that allows them to be easily segregated into two different daughter cells. (B) One model proposed to explain how SMC complexes move. These complexes function as sophisticated protein machines (see Figure 3-72), and it is still uncertain how they drive loop formation. In this "inchworm" model, the two ATPase domains separate and rejoin in sequence to motor along the DNA; for an alternative, see Figure 17-25B. (A, adapted from X. Wang et al., *Science* 355:524–527, 2017; B, adapted from M.H. Nichols and V.G. Corces, *Nat. Struct. Mol. Biol.* 25:906–910, 2018.)

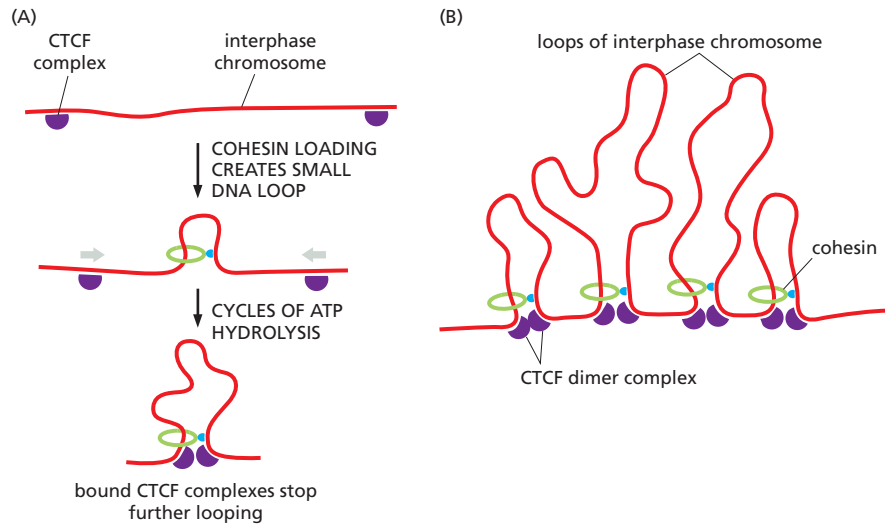


Figure 4-57 Moving cohesin rings divide eukaryotic interphase chromosomes into a long series of looped domains.

(A) The SMC protein complex *cohesin*, aided by accessory proteins that are not shown, is loaded onto chromatin to form a small DNA loop. Then, propelled by the energy of ATP hydrolysis (see Figure 4-56B), moving cohesin rings continually enlarge the loop, stopping when they encounter a DNA-bound CTCF complex on each side. A protein complex containing the CTCF dimer then becomes the loop base. (B) Schematic illustration of a portion of an interphase chromosome, organized as a series of looped domains.

During interphase, it is the SMC protein complex named **cohesin** that is critical (we shall see that its relative, *condensin*, functions in mitosis). Cohesin is thought to function in a manner similar to that of its bacterial counterpart, except that the cohesin rings are loaded at multiple sites along a chromosome and serve to fold the linear chromosomal DNA molecule into a long series of loops. As illustrated in [Figure 4-57A](#), because the traveling cohesin rings tend to stop at specific sites on the DNA, a loop is formed at a specific place on a chromosome. In vertebrates, these favored stop sites are usually marked by the sequence-specific DNA-binding protein *CTCF*, which forms a protein complex that not only stalls or stops the moving cohesin ring, but also holds the two ends of each DNA loop together.

As will be described in Chapter 7, CTCF is one of the *insulator proteins* that helps to maintain discrete domains of chromatin function (see Figure 7-28). It can also serve as part of a barrier that prevents the spreading of chromatin structures by reader-writer remodeling complexes (see Figure 4-39A). Because this protein commonly demarks the base of each looped domain in a chromosome, one can propose a general model for the structure of eukaryotic chromosomes that accounts for observations made in a diverse range of organisms. In this view, not only is the very long, linear DNA molecule that forms the core of each chromosome folded into a series of looped domains, but also the chromatin in each domain will often differ with respect to both its non-histone proteins and its covalent histone modifications—just as it does in the insect polytene chromosomes discussed earlier ([Figure 4-57B](#)). These differences are important because they help to control the selective expression of genes.

It is satisfying to find that a model recently derived for vertebrate interphase chromosomes is consistent with the observations made many decades ago on *Drosophila* polytene chromosomes and on the lampbrush chromosomes of amphibians. However, new methods that allow the organization of interphase chromosomes to be determined in single cells reveal that—unlike the situation in those special cases—most of the loops in a typical interphase chromosome are unstable: although those loops form in favored locations, their positions can rapidly fluctuate. In addition, loops can form within loops (see Figure 4-54). Thus, during interphase a typical eukaryotic chromosome has a structure that is highly dynamic.

Euchromatin and Heterochromatin Separate Spatially in the Nucleus

The same Hi-C experiments that reveal the boundaries of looped domains (see Figure 4-54) also detect a lower frequency of nearest neighbor contacts between DNA segments that can be either on different chromosomes or far away from

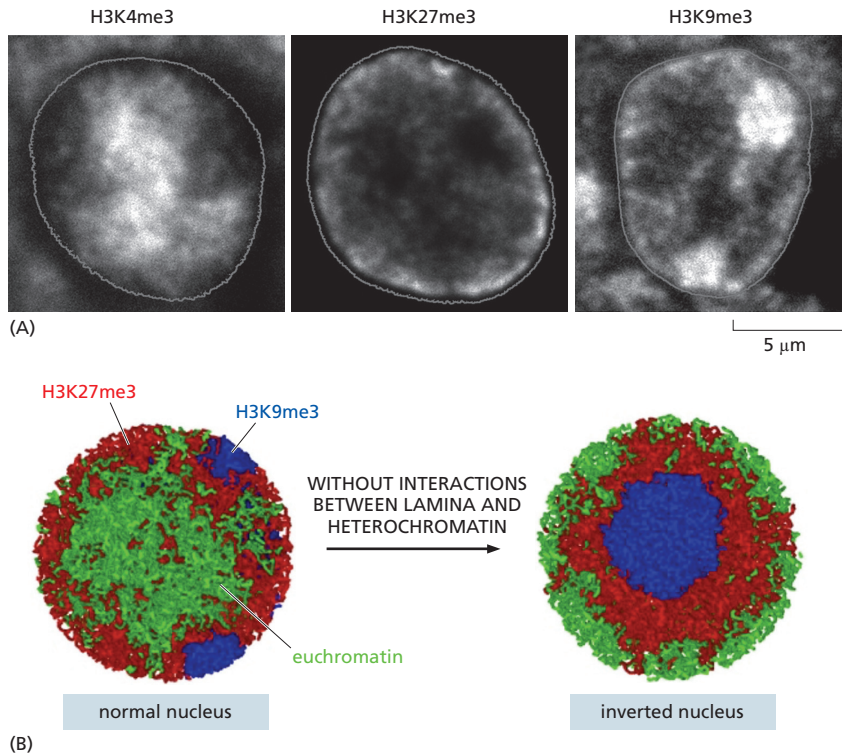


Figure 4-58 The distribution of heterochromatin and euchromatin in an interphase nucleus. (A) In this nucleus from a mouse fibroblast, the chromatin has been stained with antibodies to histones that contain three different covalent modifications: one recognizes the H3K4me3 mark in the active genes in euchromatin (see Table 4-2), and the others recognize either the H3K27me3 or the H3K9me3 silencing mark in heterochromatin (see Table 4-2). This segregation of heterochromatin toward the nuclear periphery is found in nearly all animal cells. (B) Polymer modeling of chromatin can explain the near-universal arrangement of the three types of chromatin in A, as well as the unusual arrangement that is observed in the special inverted nuclei of the photoreceptor cells of nocturnal animals, by assuming that the only difference in the latter nuclei is the absence of lamina–heterochromatin interactions. These models assume that chromatin–chromatin interactions are strongest between heterochromatin of the H3K9me3 type, weaker between the heterochromatin of the H3K27me3 type, and nonexistent between euchromatin. (A, courtesy of Irina Solovoi; B, adapted from M. Falk et al., *Nature* 570:395–399, 2019. Reproduced with permission of SNGSC.)

each other with respect to DNA sequence on the same chromosome. Analysis of these segments reveals that every chromosome can be subdivided into two compartments, within which loci preferentially interact with each other but avoid interactions with the other compartment, regardless if from the same or another chromosome. It has been shown that these two compartments closely match euchromatin and heterochromatin with regard to their gene richness, gene expression, and replication timing.

This finding is consistent with the images obtained by treating cells with antibody probes that preferentially stain these different types of chromatin; this staining produces a striking picture of the interphase nucleus, in which H3K9me3 heterochromatin is preferentially located near the nuclear periphery, H3K27me3 heterochromatin interior to that, and the open chromatin clustered in more interior regions (Figure 4-58A).

The tendency of heterochromatin to self-associate can be mimicked in experiments with reconstituted chromatin fragments that contain the abundant, heterochromatin-specific protein HP1 (see Figure 4-40). These experiments have led to a proposal that a phenomenon that resembles phase separation holds heterochromatin together, mediated by many fluctuating weak interactions (Figure 4-59). This self-associating heterochromatin is then tethered to the nuclear periphery through a set of proteins that link heterochromatic

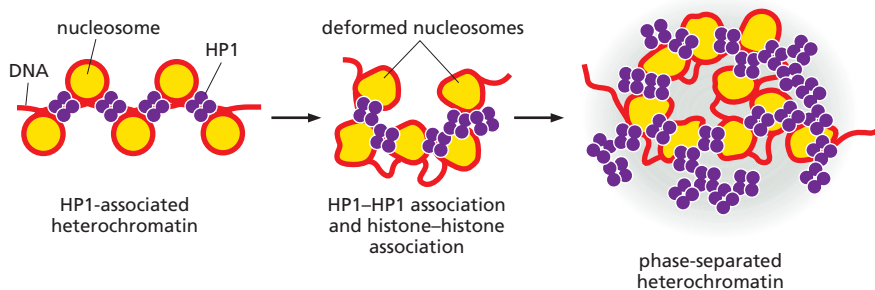


Figure 4-59 A model for the compaction and clustering of heterochromatin through fluctuating, weak associations that drive phase separation. According to this proposal, both associations between HP1 molecules and between transiently exposed histone amino acids in deformed nucleosomes contribute to the multiple weak interactions that condense heterochromatin. For a discussion of phase separation, see p. 173. (Adapted from S. Sanulli et al., *Cold Spring Harb. Symp. Quant. Biol.* 84:217–225, 2019.)

regions marked with specific histone modifications to the nuclear lamina. And the lamina is in turn anchored to the inner rim of the nuclear envelope by a set of transmembrane proteins embedded in the inner nuclear membrane (see Figure 16–67).

The functional relevance of heterochromatin positioning at the nuclear periphery remains unclear. However, this nuclear arrangement of chromatin prevails in eukaryotes with only one known exception—the so-called *inverted* nuclei of the rod cells of nocturnal mammals, in which euchromatin and heterochromatin have exchanged positions. The very center of the rod nucleus is occupied by highly condensed inert chromatin, whereas active euchromatin with ongoing gene expression is squeezed to the nuclear periphery (Figure 4–58B). Surprisingly, despite the inversion, the rod nuclei remain fully functional and are highly transcriptionally active.

Mitotic Chromosomes Are Highly Condensed

Having discussed the dynamic structure of interphase chromosomes, we now turn to **mitotic chromosomes**. The chromosomes from nearly all eukaryotic cells become readily visible by light microscopy during mitosis, when they coil up to form highly condensed structures. This condensation reduces the length of a typical interphase chromosome by about tenfold, and it produces a dramatic change in chromosome appearance.

A typical mitotic chromosome at the metaphase stage of mitosis was shown in Figure 4–18 (for the stages of mitosis, see Panel 17–1, pp. 1048–1049). The two DNA molecules produced by DNA replication during interphase of the cell-division cycle are separately folded to produce two *sister chromatids* held together at their centromeres. These chromosomes are normally covered with a variety of molecules, including large amounts of RNA–protein complexes. Once this covering has been stripped away, each chromatid can be seen in electron micrographs to be organized into loops of chromatin emanating from a central scaffolding (Figure 4–60).

Recent experiments have begun to decipher how interphase chromosomes condense to form mitotic chromosomes. The process is intimately connected with the progression of the cell cycle that will be the focus of Chapter 17. It begins in early M phase when gene expression shuts down, and specific modifications are made to histones that may help to reorganize the chromatin. Most of the ring-shaped *cohesin* proteins that organize the interphase chromosomes make way for their **condensin** relatives, whose ATP-driven movements—aided by the DNA cutting-and-rejoining enzyme topoisomerase II—drive the compaction and form a linear chromosome axis (Figure 4–61). This axis is highly enriched in both condensin II and topoisomerase II, and it is the combined action of these two enzymes that is believed to separate each chromosome into two sister chromatids.

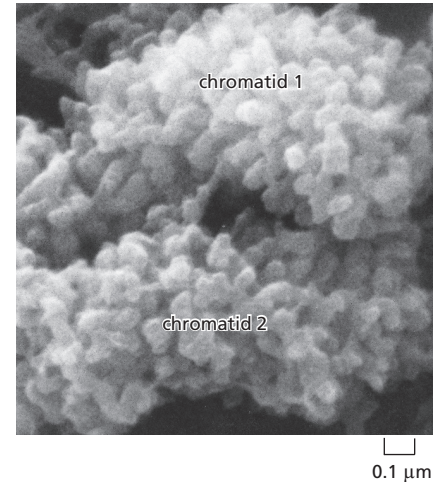
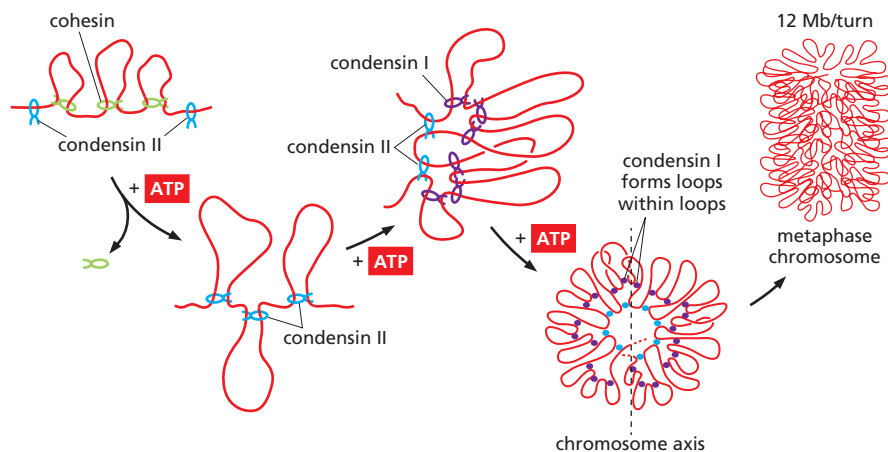


Figure 4–60 A scanning electron micrograph of a region near one end of a typical mitotic chromosome. Each knoblike projection is believed to represent the tip of a separate looped domain. Note that the two identical paired chromatids can be clearly distinguished. (From M.P. Marsden and U.K. Laemmli, *Cell* 17:849–858, 1979. With permission from Elsevier.)

Figure 4–61 How chromosomes fold in M phase. There are two types of condensins in mammalian cells, designated as condensins I and II. As cells enter mitosis, the interphase organization of the chromatin loops that is created by cohesin (see Figure 4–57) is lost within minutes. A different SMC protein, condensin II, now begins to form very large new chromatin loops by a similar ATP-driven mechanism. These new loops are organized radially by a central chromosome axis, and as the condensin II loops grow ever larger in size, a second set of loops is formed by condensin I inside them. This “loops within loops” organization, when combined with an ever-tighter winding of the chromatin loops around the mitotic chromosome axis, creates the compact chromatin that is observed in the final metaphase chromosome. Not shown are the special cohesin molecules that hold the two sister chromatids together; as described in Chapter 17, these cohesins are released only at anaphase. (Adapted from J.H. Gibcus et al., *Science* 359:eaao6135, 2018.)

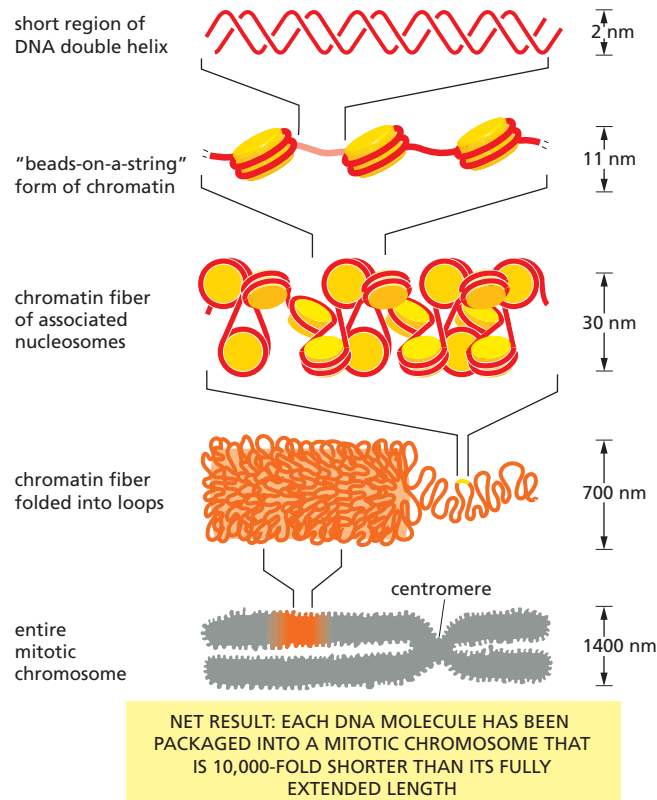


Figure 4–62 Chromosome organization. This model shows some of the levels of chromatin packaging that give rise to the highly condensed mitotic chromosome.

Mitotic chromosome condensation can be thought of as the final level in a hierarchy of chromosome packaging (Figure 4–62). This final DNA folding serves at least two important purposes. First, when condensation is complete (in metaphase), sister chromatids have been disentangled from each other and lie side by side. Thus, the sister chromatids can easily separate when the mitotic apparatus begins pulling them apart. Second, the compaction of chromosomes protects the relatively fragile DNA molecules from being broken as they are pulled to separate daughter cells by the mitotic spindle. These and other critical details of mitosis will be fully discussed in Chapter 17.

Summary

Chromosomes are generally decondensed during interphase, so that the details of their structure are difficult to visualize. Notable exceptions are the specialized lampbrush chromosomes of vertebrate oocytes and the polytene chromosomes in the giant secretory cells of insects. Studies of these two types of interphase chromosomes reveal that each long DNA molecule in a chromosome is divided into a large number of discrete domains organized as long loops of chromatin that are compacted by further folding. When genes contained in a loop are expressed, the loop unfolds and allows the cell's machinery access to the DNA. New studies using sensitive biochemical techniques show that the interphase chromosomes in a wide range of eukaryotes are also folded into a series of looped domains, and they reveal a central folding role for ring-shaped, cohesin protein complexes.

Interphase chromosomes occupy discrete territories in the cell nucleus; that is, they are not extensively intertwined. Much of the chromatin in interphase exists as loosely folded fibers of nucleosomes. However, this open chromatin is interrupted by stretches of heterochromatin, in which the nucleosomes are subjected to additional packing that renders the DNA resistant to gene expression. Heterochromatin has the important property that its type of nucleosome packing can be directly inherited during chromosome replication, thereby generating an epigenetic form of cell memory. Two major classes of heterochromatin contain different

covalent histone marks propagated by distinct reader-writer enzyme complexes. Constitutive heterochromatin is formed on repeated DNA sequences; marked by H3K9me3, it is found in large blocks that remain unchanged in and around centromeres and near telomeres. This heterochromatin blocks both gene expression and genetic recombination. Facultative heterochromatin is present at many other positions on chromosomes, where its presence can be controlled so as to have a critical role in regulating genes. H3K27me3 heterochromatin is always facultative; in contrast, only some of the H3K9me3 heterochromatin is facultative, depending on its location.

The interior of the nucleus is highly dynamic, with heterochromatin often positioned near the nuclear envelope and loops of chromatin moving away from their chromosome territory when genes are very highly expressed. This reflects the existence of nuclear subcompartments, where different sets of biochemical reactions are facilitated by an increased concentration of selected proteins and RNAs.

During mitosis, gene expression shuts down and all chromosomes adopt a highly condensed, organized conformation in a process that begins early in M phase. This condensation packages the two DNA molecules of each replicated chromosome as two separately folded chromatids, and it is catalyzed by the ATP-driven movements of ring-shaped, condensin protein complexes.

HOW GENOMES EVOLVE

In this final section of the chapter, we provide an overview of some of the ways that genes and genomes have evolved over time to produce the vast diversity of modern-day life-forms on our planet. The complete sequencing of the genomes of thousands of organisms is revolutionizing our view of the process of evolution, uncovering an astonishing wealth of information about not only family relationships among organisms, but also the molecular mechanisms by which evolution has proceeded.

Much of life's chemistry is shared between all organisms, from bacteria to humans, and it is thus not surprising that genes with similar functions can be found in a diverse range of living things. But the great revelation of the past 40 years has been the extent to which the actual nucleotide sequences of many genes have been conserved. Homologous genes—that is, genes that are similar in both their nucleotide sequence and function because of a common ancestry—can often be recognized across vast phylogenetic distances. Unmistakable homologs of many human genes are present in organisms as diverse as nematode worms, fruit flies, yeasts, plants, and even bacteria. In many cases, the resemblance is so close that, for example, the protein-coding portion of a yeast gene can be substituted with its human homolog—even though humans and yeast are separated by more than a billion years of evolutionary history.

Now that human genome sequencing is being carried out at the population level, with more than a million genomes thus far available for comparison, the extent to which natural selection constrains our own DNA sequence is being analyzed in intimate detail. The results are useful for revealing exactly which bits of our genome are functional. In addition, the data are advancing the field of *precision medicine*, with health implications both for the individuals whose genomes are sequenced and for their future generations.

As emphasized in Chapter 3, the recognition of sequence similarity has become a major tool for inferring gene and protein function. Although a sequence match does not guarantee similarity in function, it has proved to be an excellent clue. Thus, it is often possible to predict the function of genes in humans for which no biochemical or genetic information is available simply by comparing their nucleotide sequences with the sequences of genes that have been characterized in other more readily studied organisms.

In general, the sequences of individual genes are much more tightly conserved than is overall genome structure. Features of genome organization such as genome size, number of chromosomes, order of genes along chromosomes, abundance and size of introns, and amount of repetitive DNA are found to differ

greatly when comparing distant organisms, as does the number of genes that each organism contains.

Genome Comparisons Reveal Functional DNA Sequences by Their Conservation Throughout Evolution

A first obstacle in interpreting the sequence of the 3.1 billion nucleotide pairs in the human genome is the fact that the nucleotide sequence of most of it (about 90%) is probably functionally unimportant. The regions of the genome that code for the amino acid sequences of proteins (the exons) are typically found in short segments (median size about 130 nucleotide pairs), small islands in a sea of DNA whose exact nucleotide sequence is thought to be mostly of little consequence. This arrangement can make it difficult to identify all the exons in a stretch of DNA, and it is often hard also to determine exactly where a gene begins and ends.

One very important approach to deciphering our genome is to search for DNA sequences that are closely similar between different species, on the principle that DNA sequences that have a function are much more likely to be conserved than those without a function. For example, humans and mice are thought to have diverged from a common mammalian ancestor about 90×10^6 years ago, which is long enough for the majority of nucleotides in their genomes to have been changed by random mutational events. Consequently, the only regions that will have remained closely similar in the two genomes are those in which mutations would have impaired function and put the animals carrying them at a disadvantage, resulting in their elimination from the population by natural selection. Such closely similar pieces of DNA sequence are known as *conserved DNA regions*. In addition to revealing those DNA sequences that encode functionally important exons and RNA molecules, these conserved regions will include regulatory DNA sequences as well as DNA sequences with functions that are not yet known but which are inferred to be somehow important. In contrast, most *nonconserved DNA regions* will reflect DNA whose sequence is much less likely to be critical for function.

The power of this method can be increased by including in such comparisons the genomes of large numbers of species whose genomes have been sequenced, such as rat, chicken, fish, dog, and chimpanzee, as well as mouse and human. By revealing in this way the results of a very long natural “experiment,” lasting for hundreds of millions of years, such comparative DNA sequencing studies have highlighted some of the most interesting regions in our genome. The comparisons reveal that about 4.5% of the human genome consists of *multispecies conserved sequences*. To our great surprise, only about one-fourth of these sequences code for proteins (see Table 4–1, p. 194). Most of the remaining conserved sequences consist of DNA that is thought to contain clusters of protein-binding sites involved in gene regulation, while others produce RNA molecules that are not translated into protein but are important for other reasons.

When the DNA sequences of hundreds of thousands of individual humans are compared, an additional 5% of our genome shows a reduced variation in the human population, which implies that the sequences in this 5% of the genome are also important. Taken together, these analyses suggest that only about 10% of the human genome contains nucleotide sequences that truly matter.

The important question of how much of the DNA sequence of the human genome is functionally relevant was briefly confused by a set of high-profile publications that appeared in 2012 from a large, federally funded US genome project named ENCODE. These publications, which reported the results of a massive survey using sensitive assays that can detect the presence of RNA molecules in cells at extremely low levels, reported that 76% of the total DNA sequence in human cells is transcribed to produce RNA molecules. Even though many of these transcripts were found at levels of less than a single RNA molecule per cell, the ENCODE scientists used such data to assert that most of human DNA is functional, with very little “junk.” This claim received widespread publicity, along with their belief that our genome contains tens of thousands of previously undetected genes that produce RNA molecules that do not code for protein.

As previously stated, there is a strong scientific consensus that most of the human genome consists of DNA whose nucleotide sequence is not relevant to biological function—being the so-called junk. This conclusion rests on the finding that natural selection fails to preserve these sequences in the face of the inevitable random changes to genomes that occur over time, as can be seen both when different species are compared and from detailed analyses of human variation. The fact that these DNA sequences nevertheless produce an occasional RNA molecule can be explained by the occurrence of background “noise” in gene expression. Although gene expression is very accurate, it is not perfect, and biochemical errors occasionally occur. Such errors are to be expected, and so long as they are kept at a low level, they are thought to have little or no consequence for the cell.

Genome Alterations Are Caused by Failures of the Normal Mechanisms for Copying and Maintaining DNA, as Well as by Transposable DNA Elements

Evolution depends on accidents and mistakes followed by nonrandom survival. Most of the genetic changes that occur result simply from failures in the normal mechanisms by which genomes are copied or repaired when damaged, although the movement of transposable DNA elements (discussed shortly) also plays an important part. As we will explain in Chapter 5, the mechanisms that maintain DNA sequences are remarkably precise—but errors will occur. DNA sequences are inherited with such extraordinary fidelity that typically, along a given line of descent, only about one nucleotide pair in a thousand is randomly changed in the human germ line every million years. Even so, in a population of 10,000 diploid individuals, every possible nucleotide substitution will have been “tried out” on about 20 occasions in the course of a million years—a short span of time in relation to the evolution of species.

Errors in DNA replication, DNA recombination, or DNA repair can lead either to simple local changes in DNA sequence—so-called *point mutations* such as the substitution of one base pair for another—or to large-scale genome rearrangements such as deletions, duplications, inversions, and translocations of DNA from one chromosome to another. In addition to these rare failures of the genetic machinery, genomes contain mobile DNA elements that are an important source of genomic change (see Table 5–3, p. 286). These transposable DNA elements (*transposons*) are parasitic DNA sequences that can spread within the genomes they colonize. In the process, they often disrupt the function or alter the regulation of existing genes. On occasion, they have created altogether novel genes through fusions between transposon sequences and segments of existing genes. Over long periods of evolutionary time, DNA transposition events have profoundly affected genomes, so much so that nearly half of the DNA in the human genome consists of recognizable relics of past transposition events (Figure 4–63). Even more of our genome is thought to have been derived from transpositions occurring so long ago ($>10^8$ years) that, due to the accumulation of mutations, the sequences can no longer be traced to transposons.

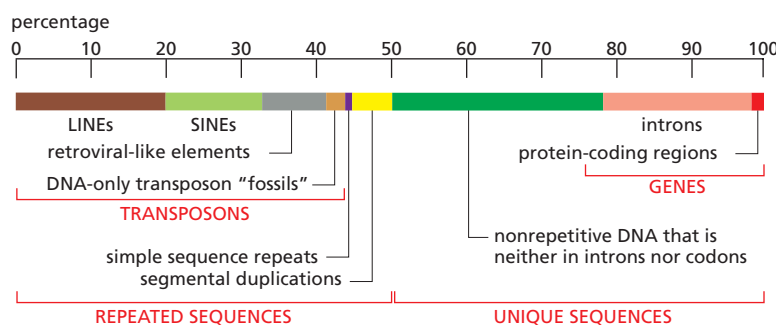


Figure 4–63 A representation of the nucleotide sequence content of the sequenced human genome. The LINEs (long interspersed nuclear elements), SINEs (short interspersed nuclear elements), retroviral-like elements, and DNA-only transposons are mobile genetic elements that have multiplied in our genome by replicating themselves and inserting the new copies in different positions. These mobile genetic elements are discussed in Chapter 5 (see Table 5–3, p. 286). Simple sequence repeats are short nucleotide sequences (fewer than 14 nucleotide pairs) that are repeated again and again for long stretches. Segmental duplications are large blocks of DNA sequence (1000–200,000 nucleotide pairs) that are present at two or more locations in the genome. Most of the highly repeated blocks of DNA in heterochromatin have not yet been completely sequenced; therefore, about 10% of human DNA sequences are not represented in this diagram. (Data courtesy of E. Margulies.)

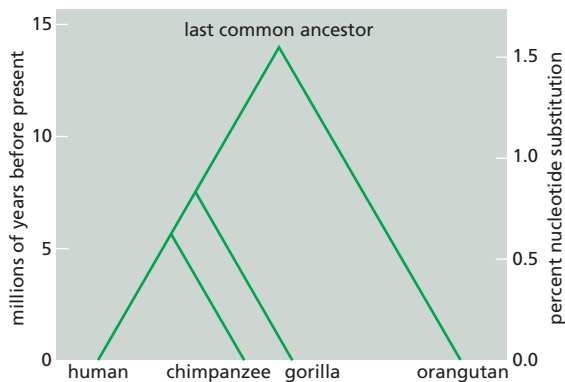


Figure 4-64 A phylogenetic tree showing the relationship between humans and the great apes based on nucleotide sequence data. As indicated, the sequences of homologous portions of the genomes of all four species are estimated to differ from the sequence of the genome of their last common ancestor by a little over 1.5%. Because changes occur independently on both diverging lineages, pairwise comparisons will reveal twice the sequence divergence from the last common ancestor. For example, human–chimpanzee comparisons show divergences of approximately 1.2%. (Modified from F.C. Chen and W.H. Li, *Am. J. Hum. Genet.* 68:444–456, 2001.)

The Genome Sequences of Two Species Differ in Proportion to the Length of Time Since They Have Separately Evolved

The differences between the genomes of all species alive today have accumulated over more than 3 billion years. Although we lack a direct record of changes over time, scientists can reconstruct the process of genome evolution from detailed comparisons of the genomes of contemporary organisms.

The basic organizing framework for comparative genomics is the phylogenetic tree. A simple example is the tree describing the divergence of humans from the great apes (Figure 4-64). The primary support for this tree comes from comparisons of genome sequences. For example, comparisons between the sequences of human genes or proteins and those of the great apes typically reveal the fewest differences between human and chimpanzee and the most between human and orangutan.

For closely related organisms such as humans and chimpanzees, it is relatively easy to reconstruct the gene sequences of the extinct, last common ancestor of the two species (Figure 4-65). The close similarity between human and chimpanzee

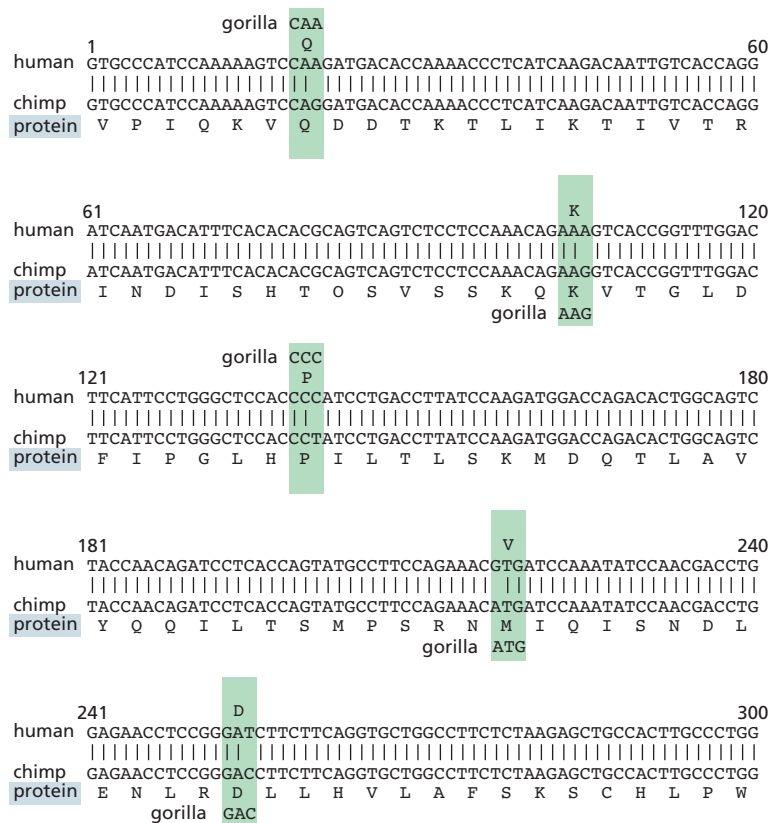


Figure 4-65 Tracing the ancestral sequence from a sequence comparison of the coding regions of human and chimpanzee leptin genes. Reading left to right and top to bottom, a continuous 300-nucleotide segment of a leptin-coding gene is illustrated. Leptin is a hormone that regulates food intake and energy utilization in response to the adequacy of fat reserves. As indicated by the codons boxed in green, only 5 nucleotides (of 441 total) differ between the two species. Moreover, in only one of the five positions does the difference in nucleotide lead to a difference in the encoded amino acid. For each of the five variant nucleotide positions, the corresponding sequence in the gorilla is also indicated. In two cases, the gorilla sequence agrees with the human sequence, while in three cases it agrees with the chimpanzee sequence.

What was the sequence of the leptin gene in the last common ancestor? The most economical assumption is that evolution has followed a pathway requiring the minimum number of mutations consistent with the data. Thus, it seems likely that the leptin sequence of the last common ancestor was the same as the human and chimpanzee sequences when they agree. When they disagree, the gorilla sequence can be used as a tiebreaker, a conclusion that should be tested by including the sequences from other great apes. For convenience, only the first 300 nucleotides of the leptin-coding sequences are given. The remaining 141 are identical between humans and chimpanzees.

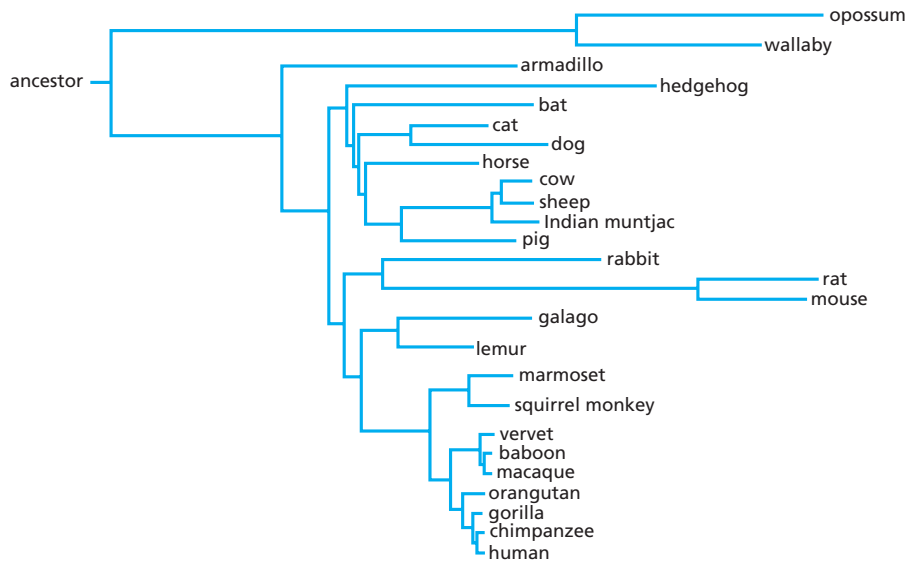


Figure 4-67 A phylogenetic tree showing the evolutionary relationships of some present-day mammals. The length of each line is proportional to the number of *neutral substitutions*; that is, nucleotide changes at sites where there is assumed to be no purifying selection. (Adapted from G.M. Cooper et al., *Genome Res.* 15:901–913, 2005. With permission from Cold Spring Harbor Laboratory Press.)

Categories of DNA for which the clock runs fast are most informative for recent evolutionary events; the mitochondrial DNA clock has been used, for example, to chronicle the divergence of the Neanderthal lineage from that of modern *Homo sapiens*. To study ancient evolutionary events, one must examine DNA for which the clock runs unusually slowly; thus, the divergence of the major branches of the tree of life—bacteria, archaea, and eukaryotes—has been deduced from study of the sequences specifying ribosomal RNA.

In general, molecular clocks, appropriately chosen, have a finer time resolution than that of the fossil record, and they are a more reliable guide to the detailed structure of phylogenetic trees than are classical methods of tree construction, which are based on family resemblances in anatomy and embryonic development. For example, the precise family tree of great apes and humans was not settled until sufficient molecular sequence data accumulated in the 1980s to produce the pedigree shown previously in Figure 4-64. And with huge amounts of DNA sequence now determined from a wide variety of mammals, much better estimates of our relationship to them are being obtained (**Figure 4-67**).

A Comparison of Human and Mouse Chromosomes Shows How the Structures of Genomes Diverge

As would be expected, the human and chimpanzee genomes are much more alike than are the human and mouse genomes, even though all three genomes are roughly the same size and contain nearly identical sets of genes. Mouse and human lineages have had approximately 90 million years to diverge through accumulated mutations, versus 6 million years for humans and chimpanzees. In addition, rodent lineages (represented by the rat and the mouse in Figure 4-67) have unusually fast molecular clocks and have diverged from the human lineage more rapidly than otherwise expected.

While the way that the genome is organized into chromosomes is almost identical between humans and chimpanzees, this organization has diverged greatly between humans and mice. According to rough estimates, a total of about 180 chromosome breakage-and-rejoining events have moved large blocks of DNA sequence in the human and mouse lineages since they last shared a common ancestor. As a result, although the number of chromosomes is similar in the two species (23 per haploid genome in the human versus 20 in the mouse), their overall structures differ greatly. Even so, there are many large blocks of DNA in which the gene order is the same in the human and the mouse. These stretches of conserved gene order in chromosomes are referred to as regions of *synteny*. **Figure 4-68** illustrates the extent of this synteny by showing how segments of

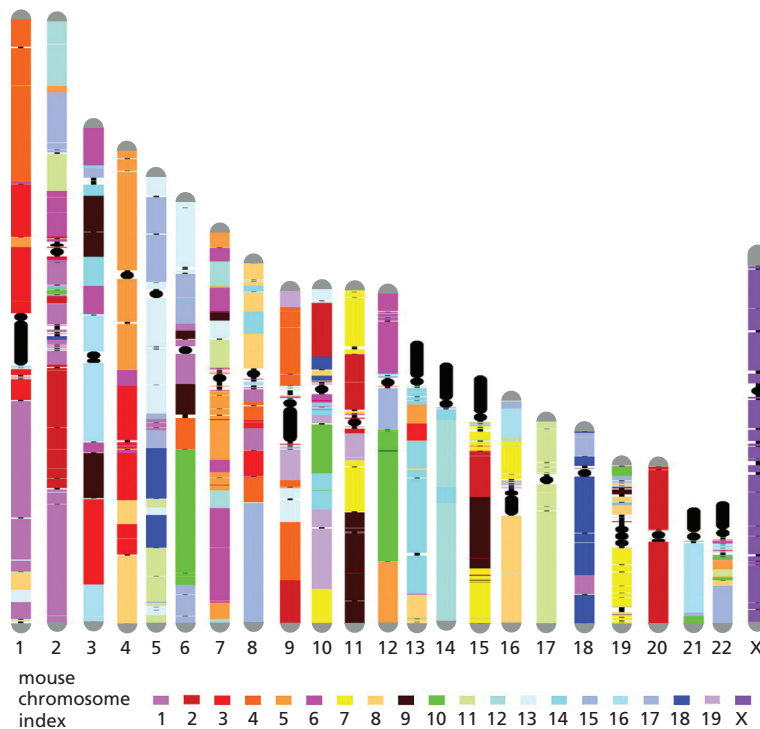


Figure 4-68 Synteny between human and mouse chromosomes. In this diagram, the human chromosome set is shown, with each part of each chromosome colored according to the mouse chromosome with which it is syntenic. The color coding used for each mouse chromosome is shown at the bottom of the figure.

Heterochromatic highly repetitive regions (such as centromeres) that are difficult to sequence cannot be mapped in this way; these are colored *black*. (Adapted from E.E. Eichler and D. Sankoff, *Science* 301:793–797, 2003. With permission from AAAS.)

mouse chromosomes map onto the human chromosome set. For much more distantly related vertebrates, such as chicken and human, the number of breakage-and-rejoining events has been much greater, and the regions of synteny are much shorter; in addition, these regions are often hard to discern because of the divergence of the DNA sequences they contain.

An unexpected conclusion from a detailed comparison of the complete mouse and human genome sequences, confirmed by subsequent comparisons between the genomes of other vertebrates, is that small blocks of DNA sequence are being deleted from and added to genomes at a surprisingly rapid rate. Thus, if we assume that our common ancestor had a genome of human size (about 3.1 billion nucleotide pairs), mice would have lost a total of about 45% of that genome from accumulated deletions during the past 90 million years, while humans would have lost about 25%. However, substantial sequence gains from many small chromosome duplications and from the multiplication of transposons have compensated for these deletions. As a result of this series of gains and losses, the size of our genome is thought to be practically unchanged from that of the last common ancestor of humans and mice, while the mouse genome is smaller by only about 0.4 billion nucleotides.

Good evidence for the loss of DNA sequences in small blocks during evolution can be obtained from a detailed comparison of regions of synteny in the human and mouse genomes. The comparative shrinkage of the mouse genome can be clearly seen from such comparisons, with the net loss of sequences scattered throughout the long stretches of DNA that are otherwise homologous (**Figure 4-69**).

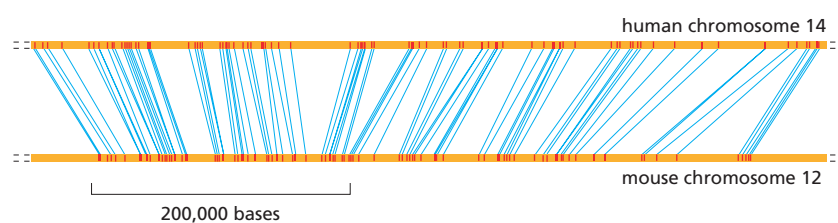


Figure 4-69 Comparison of a syntenic portion of mouse and human genomes.

About 90% of the two genomes can be aligned in this way. Note that while there is an identical order of the matched index sequences (*red marks*), there has been a net loss of DNA in the mouse lineage that is interspersed throughout the entire region. This type of net loss is typical for all such regions, and it accounts for the fact that the mouse genome contains 14% less DNA than does the human genome. (Adapted from Mouse Genome Sequencing Consortium, *Nature* 420:520–562, published 2002 by Nature Publishing Group. Reproduced with permission of SCS.

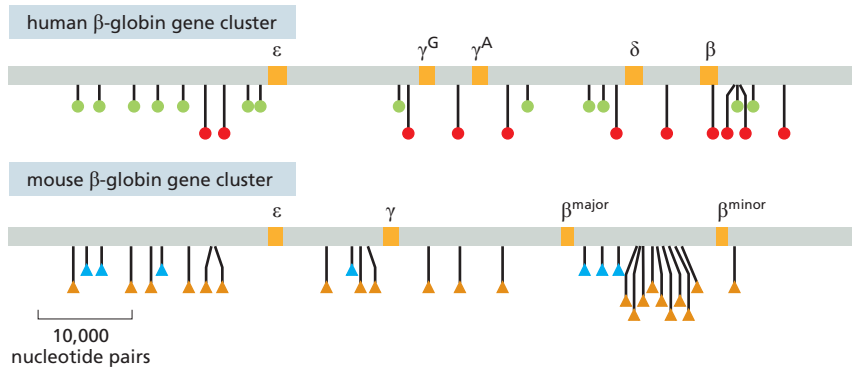


Figure 4-70 A comparison of the β -globin gene cluster in the human and mouse genomes, showing the locations of transposable elements. This stretch of the human genome contains five functional β -globin-like genes (orange); the comparable region from the mouse genome has only four. The positions of the human *Alu* sequences are indicated by green circles, and the positions of the human *L1* sequences are indicated by red circles.

The mouse genome contains different but related transposable elements: the positions of B1 elements (which are related to the human *Alu* sequences) are indicated by blue triangles, and the positions of the mouse *L1* elements (which are related to the human *L1* sequences) are indicated by orange triangles. The absence of transposable elements from the globin structural genes can be attributed to purifying selection, which would have eliminated any insertion that compromised gene function. (Courtesy of Ross Hardison and Webb Miller.)

DNA is added to genomes both by the spontaneous duplication of chromosomal segments that are typically tens of thousands of nucleotide pairs long (as will be discussed shortly) and by insertion of new copies of active transposons. Most transposition events are duplicative, because the original copy of the transposon stays where it was when a copy inserts at the new site; see, for example, Figure 5-59. Comparison of the DNA sequences derived from transposons in the human and the mouse readily reveals some of the sequence additions (Figure 4-70). In contrast, the nucleotide sequences and positions of the transposons in human and chimpanzee are very similar, indicating that their movement occurred before the two species diverged.

It remains a mystery why all mammals have maintained genome sizes of roughly 3 billion nucleotide pairs that contain nearly identical sets of genes, even though 90% of this DNA appears not to be under sequence-specific functional constraints.

The Size of a Vertebrate Genome Reflects the Relative Rates of DNA Addition and DNA Loss in a Lineage

In more distantly related vertebrates, genome size can vary considerably, apparently without a drastic effect on the organism or its number of genes. Thus, the chicken genome, at 1 billion nucleotide pairs, is only about one-third the size of the mammalian genome, even though it contains nearly the same number of genes. An extreme example is the puffer fish, *Fugu rubripes* (Figure 4-71A), which has a tiny genome for a vertebrate (0.4 billion nucleotide pairs compared to 1 billion or more for most other fish). The small size of the *Fugu* genome is largely due to the small size of its introns and intergenic regions. Specifically, *Fugu* introns, as well as other noncoding segments of the *Fugu* genome, lack the repetitive DNA that makes up a large portion of the genomes of most well-studied vertebrates. Nevertheless, the positions of the *Fugu* introns between the exons of each gene are almost the same as in mammalian genomes (Figure 4-72).

While initially a mystery, we now have a simple explanation for such large differences in genome size between similar organisms: because all vertebrates experience a continual process of DNA loss and DNA addition, the size of a genome merely depends on the balance between these opposing processes

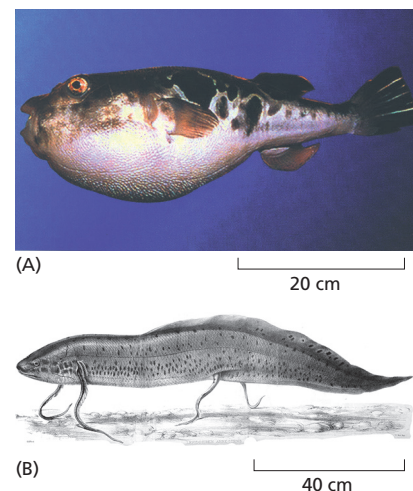


Figure 4-71 Two fish with very different genome sizes. The puffer fish, *Fugu rubripes* (A), has a genome size that is 300 times smaller than that of the West African lungfish, *Protopterus annectens* (B). (A, Courtesy of Byrappa Venkatesh; B, History and Art Collection/Alamy Stock Photo.)

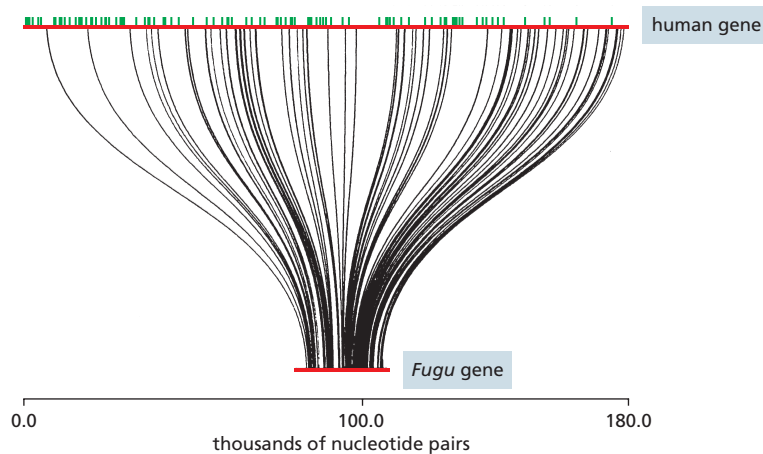


Figure 4-72 Comparison of the genomic sequences of the human and *Fugu* genes encoding the protein huntingtin. Both genes (indicated in red) contain 67 short exons that align in 1:1 correspondence to one another; these exons are connected by curved lines. The human gene is 7.5 times larger than the *Fugu* gene (180,000 versus 24,000 nucleotide pairs). The size difference is entirely due to larger introns in the human gene. The larger size of the human introns is due in part to the presence of retrotransposons (discussed in Chapter 5), whose positions are represented by green vertical lines; the *Fugu* introns lack retrotransposons. In humans, mutation of the huntingtin gene causes Huntington's disease, an inherited neurodegenerative disorder. (Adapted from S. Baxendale et al., *Nat. Genet.* 10:67–76, published 1995 by Nature Publishing Group. Reproduced with permission of SCS.

acting over millions of years. Suppose, for example, that in the lineage leading to *Fugu*, the rate of DNA addition happened to slow greatly. Over long periods of time, this would result in a major “cleansing” from this fish genome of those DNA sequences whose loss could be tolerated. The result is an unusually compact genome, relatively free of junk and clutter, but retaining through purifying selection the vertebrate DNA sequences that are functionally important. This makes *Fugu*, with its 400 million nucleotide pairs of DNA, a valuable resource for genome research aimed at understanding humans.

At the other end of the scale, some fish—such as the primitive-looking lungfish—have enormous genomes, more than 300 times the size of that of *Fugu* and 30 times the size of that of humans (Figure 4-71B). Most of the extra DNA consists of transposons and other repeated DNA sequences, suggesting that genome additions have greatly exceeded losses in this lineage.

Multispecies Sequence Comparisons Identify Many Conserved DNA Sequences of Unknown Function

The mass of DNA sequence now in freely accessible databases (hundreds of billions of nucleotide pairs) provides a rich resource that scientists can mine for many purposes. This information can be used not only to unscramble the evolutionary pathways that have led to modern organisms, but also to provide insights into how cells and organisms function. Perhaps the most remarkable discovery in this realm comes from the observation that a striking amount of DNA sequence that does not code for protein has been conserved during mammalian evolution (see Table 4-1, p. 194). This is most clearly revealed when we align and compare DNA synteny blocks from many different species, thereby identifying large numbers of so-called *multispecies conserved sequences*: some of these code for protein, but most of them do not.

Many of the conserved sequences that do not code for protein are now known to produce untranslated RNA molecules, such as the thousands of *long noncoding RNAs* (*lncRNAs*) that are thought to have important functions in regulating gene transcription. As we shall also see in Chapter 7, many others are regions of DNA scattered throughout the genome that directly bind proteins

involved in gene regulation. But the function of much of the conserved noncoding DNA remains a mystery. This enigma highlights how much more we need to learn about the fundamental biological mechanisms that operate in animals and other complex organisms, and its solution is certain to have profound consequences for medicine.

How can cell biologists tackle the mystery of noncoding conserved DNA? Traditionally, attempts to determine the function of a puzzling DNA sequence begin by looking at the consequences of its experimental disruption. But many DNA sequences that are crucial for an organism in the wild can be expected to have no noticeable effect on its phenotype under laboratory conditions: what is required for a mouse to survive in a laboratory cage is very much less than what is required for it to succeed in nature. Moreover, calculations based on population genetics reveal that just a tiny selective advantage—less than a 0.1% difference in survival—can be enough to strongly favor retaining a particular DNA sequence over evolutionary time spans. One should therefore not be surprised to find that many conserved DNA sequences can be deleted from the mouse genome without any noticeable effect on that mouse in a laboratory.

Changes in Previously Conserved Sequences Can Help Decipher Critical Steps in Evolution

Given genome sequence information, we can tackle another intriguing question: What alterations in our DNA have made humans so different from other animals—or for that matter, what makes any individual species so different from its relatives? For example, as soon as both the human and the chimpanzee genome sequences became available, scientists began searching for DNA sequence changes that might account for the striking differences between us and chimpanzees. With 3.1 billion nucleotide pairs to compare in the two species, this might seem an impossible task. But the job was made much easier by confining the search to 35,000 clearly defined multispecies conserved sequences (a total of about 5 million nucleotide pairs), representing parts of the genome that are most likely to be functionally important. Though these sequences are conserved strongly, they are not conserved perfectly, and when the version in one species is compared with that in another they are generally found to have drifted apart by a small amount corresponding simply to the time elapsed since the last common ancestor. In a small proportion of cases, however, one sees signs of a sudden evolutionary spurt. For example, some DNA sequences that have been highly conserved in other mammalian species are found to have accumulated nucleotide changes exceptionally rapidly during the 6 million years of human evolution since we diverged from the chimpanzees. These *human accelerated regions* (HARs) are thought to reflect functions that have been especially important in making us different in some useful way.

About 50 such sites were identified in one study, one-fourth of which were located near genes associated with neural development. The sequence exhibiting the most rapid change (18 changes between human and chimpanzee, compared to only two changes between chimpanzee and chicken) was examined further and found to encode a 118-nucleotide noncoding RNA molecule, HAR1F (human accelerated region 1F), that is produced in the human cerebral cortex at a critical time during brain development. The function of this HAR1F RNA is not yet known, but findings of this type are stimulating research studies that may shed light on crucial features of the human brain.

A related approach in the search for the important mutations that contributed to human evolution likewise begins with DNA sequences that have been conserved during mammalian evolution, but rather than screening for accelerated changes in individual nucleotides, it focuses instead on chromosome sites that have experienced deletions in the 6 million years since our lineage diverged from that of chimpanzees. More than 500 such sequences—conserved among other species but deleted in humans—have been discovered. Each deletion removes an average of 95 nucleotides of DNA sequence. Only one of these deletions

affects a protein-coding region: the rest are thought to alter regions that affect how nearby genes are expressed, an expectation that has been experimentally confirmed in a few cases. A large proportion of the presumed regulatory regions identified in this way lie near genes that affect neural function and/or near genes involved in steroid signaling, suggesting that changes in the nervous system and in immune or reproductive functions have played an especially important role in human evolution.

Mutations in the DNA Sequences That Control Gene Expression Have Driven Many of the Evolutionary Changes in Vertebrates

The vast hoard of genomic sequence data now being accumulated can be explored in many other ways to reveal events that happened even hundreds of millions of years ago. For example, one can attempt to trace the origins of the regulatory elements in DNA that have played critical parts in vertebrate evolution. One such study began with the identification of nearly 3 million noncoding sequences, averaging 28 base pairs in length, that have been conserved in recent vertebrate evolution while being absent in more ancient ancestors. Each of these special noncoding sequences is likely to represent a functional innovation peculiar to a particular branch of the vertebrate family tree, and most of them are thought to consist of regulatory DNA that governs the expression of a neighboring gene. Given full genome sequences, one can identify the genes that appear most likely to have fallen under the sway of these novel regulatory elements. By comparing many different species, with known divergence times, one can also estimate when each such regulatory element came into existence as a conserved feature.

The findings suggest remarkable evolutionary differences between the various functional classes of genes (**Figure 4-73**). Conserved regulatory elements that originated early in vertebrate evolution—that is, more than about 300 million years ago, which is when the mammalian lineage split from the lineage leading to birds and reptiles—seem to be mostly associated with genes that code for transcription regulatory proteins and for proteins with roles in organizing embryonic development. Then came an era when the regulatory DNA innovations arose next to genes coding for receptors for extracellular signals. Finally, over the course of the past 100 million years, the regulatory innovations seem to have been concentrated in the neighborhood of genes coding for proteins (such as protein kinases) that function to modify other proteins post-translationally.

Many questions remain to be answered about these phenomena and what they mean. One possible interpretation is that the logic—the circuit diagram—of the gene regulatory network in vertebrates was established early, and that more recent evolutionary change has mainly occurred through the tuning of quantitative parameters. This could help to explain why, among the mammals, for

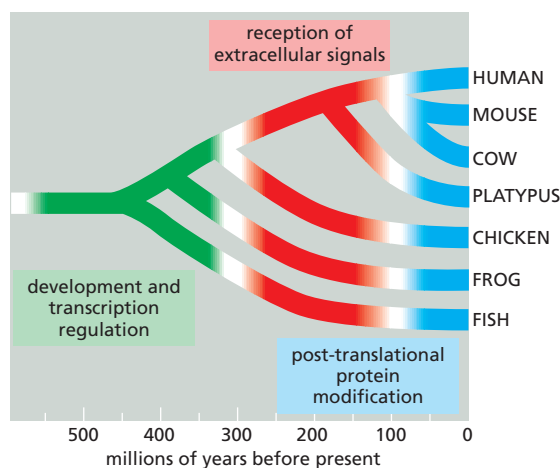


Figure 4-73 The types of changes in gene regulation inferred to have predominated during the evolution of our vertebrate ancestors. To produce the information summarized in this plot, wherever possible the type of gene regulated by each conserved noncoding sequence was inferred from the identity of its closest protein-coding gene. The time when each conserved sequence became fixed in the vertebrate lineage was then used to derive the conclusions shown. (Based on C.B. Lowe et al., *Science* 333:1019–1024, 2011.)

example, the basic body plan—the topology of the tissues and organs—has been largely conserved.

Gene Duplication Also Provides an Important Source of Genetic Novelty During Evolution

Evolution depends on the creation of new genes, as well as on the modification of those that already exist. How does this occur? When we compare organisms that seem very different—a primate with a rodent, for example, or a mouse with a fish—we rarely encounter genes in the one species that have no homolog in the other. Genes without homologous counterparts are relatively scarce even when we compare such divergent organisms as a mammal and a worm. On the other hand, we frequently find gene families that have different numbers of members in different species. To create such families, genes have been repeatedly duplicated, and the copies have then diverged to take on new functions that often vary from one species to another.

Gene duplication occurs at high rates in all evolutionary lineages, contributing to the vigorous process of DNA addition discussed previously. In a detailed study of spontaneous duplications in yeast, duplications of 50,000–250,000 nucleotide pairs were commonly observed, most of which were tandemly repeated. These appeared to result from DNA replication errors that led to the inexact repair of double-strand chromosome breaks. A comparison of the human and chimpanzee genomes reveals that, since the time that these two organisms diverged, such *segmental duplications* have added about 5 million nucleotide pairs to each genome every million years, with an average duplication size being about 50,000 nucleotide pairs (although there are some duplications five times larger). In fact, if one counts nucleotides, duplication events have created more differences between our two species than have single-nucleotide substitutions.

Duplicated Genes Diverge

What is the fate of newly duplicated genes? In most cases, there is presumed to be little or no selection—at least initially—to maintain the duplicated state because either copy can provide an equivalent function. Hence, many duplication events are likely to be followed by loss-of-function mutations in one or the other gene. This cycle would functionally restore the one-gene state that preceded the duplication. Indeed, there are many examples in contemporary genomes where one copy of a duplicated gene can be seen to have become irreversibly inactivated by multiple mutations. Over time, the sequence similarity between such a **pseudogene** and the functional gene whose duplication produced it would be expected to be eroded by the accumulation of many mutations in the pseudogene—the homologous relationship eventually becoming undetectable.

An alternative fate for gene duplications is for both copies to remain functional, while diverging in their sequence and pattern of expression, thus taking on different roles. This process of *duplication and divergence* almost certainly explains the presence of large families of genes with related functions in biologically complex organisms, and it is thought to play a critical role in the evolution of increased biological complexity. An examination of many different eukaryotic genomes suggests that the probability that any particular gene will undergo a duplication event that spreads to most or all individuals in a species is approximately 1% every million years.

Whole-genome duplications offer particularly dramatic examples of the duplication–divergence cycle. A whole-genome duplication can occur quite simply: all that is required is one round of genome replication in a germ-line cell lineage without a corresponding cell division. Initially, the chromosome number simply doubles. Such abrupt increases in the ploidy of an organism are common, particularly in fungi and plants. After a whole-genome duplication, all genes exist as duplicate copies. However, unless the duplication event occurred so recently that there has been little time for subsequent alterations in genome structure,

the results of a series of segmental duplications—occurring at different times—are hard to distinguish from the end product of a whole-genome duplication. In mammals, for example, the role of whole-genome duplications versus a series of piecemeal duplications of DNA segments is quite uncertain. Nevertheless, it is clear that a great deal of gene duplication has occurred in the distant past.

Analysis of the genome of the zebrafish, in which at least one whole-genome duplication is thought to have occurred hundreds of millions of years ago, has cast some light on the process of gene duplication and divergence. Although many duplicates of zebrafish genes appear to have been lost by mutation, a significant fraction—perhaps as many as 30–50%—have diverged functionally while both copies have remained active. In many cases, the most obvious functional difference between the duplicated genes is that they are expressed in different tissues or at different stages of development. One attractive theory to explain such an end result imagines that different, mildly deleterious mutations occur quickly in both copies of a duplicated gene set. For example, one copy might lose expression in a particular tissue as a result of a regulatory mutation, while the other copy loses expression in a second tissue. After such an occurrence, both gene copies would be required to provide the full range of functions that were once supplied by a single gene; hence, both copies would now be protected from loss through inactivating mutations. Over a longer period, each copy could then undergo further changes through which it could acquire new, specialized features.

The Evolution of the Globin Gene Family Shows How DNA Duplications Contribute to the Evolution of Organisms

The globin gene family provides an especially good example of how DNA duplication generates new proteins, and its evolutionary history has been worked out particularly well. The unmistakable similarities in amino acid sequence and structure among the present-day globins indicate that they all must derive from a common ancestral gene, even though some are now encoded by widely separated genes in the mammalian genome.

We can reconstruct some of the past events that produced the various types of oxygen-carrying hemoglobin molecules by considering the different forms of the protein in organisms at different positions on the tree of life. A molecule like hemoglobin was necessary to allow multicellular animals to grow to a large size, because large animals cannot simply rely on the diffusion of oxygen through the body surface to oxygenate their tissues adequately. But oxygen plays a vital part in the life of nearly all living organisms, and oxygen-binding proteins homologous to hemoglobin can be recognized even in plants, fungi, and bacteria. In animals, the simplest oxygen-carrying molecule is a globin polypeptide chain of about 150 amino acids that is found in many marine worms, insects, and primitive fish. The hemoglobin molecule in more complex vertebrates, however, is composed of two kinds of globin chains. It appears that about 500 million years ago, just before fish and mammals diverged from their common ancestor, a series of gene mutations and duplications occurred. These events established two slightly different globin genes in the genome of each individual, coding for α -globin and β -globin chains that associate to form a hemoglobin molecule consisting of two α chains and two β chains (Figure 4-74). The four oxygen-binding sites in the $\alpha_2\beta_2$ molecule interact, allowing a cooperative allosteric change in the molecule as it binds and releases oxygen, which enables hemoglobin to take up and release oxygen more efficiently than can the single-chain version.

Still later, during the evolution of mammals, the β -chain gene apparently underwent duplication and mutation to give rise to a second β -like chain that

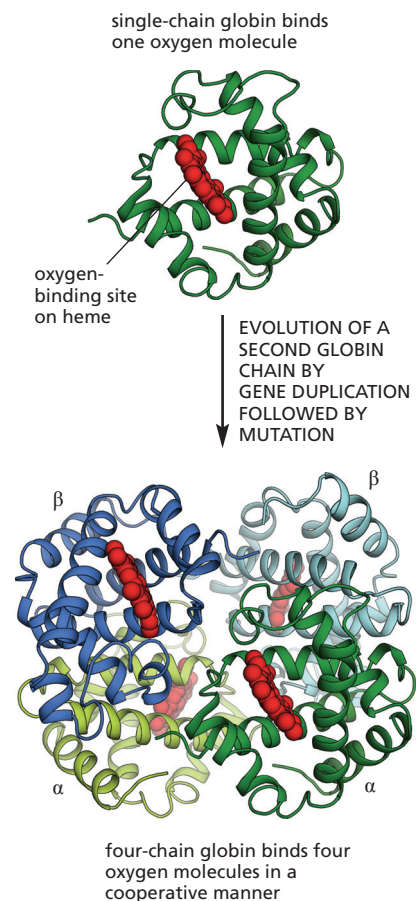


Figure 4-74 A comparison of the structure of one-chain and four-chain globins. The four-chain globin shown is hemoglobin, which is a complex of two α -globin and two β -globin chains. The one-chain globin present in some primitive vertebrates represents an intermediate in the evolution of the four-chain globin. With oxygen bound it exists as a monomer; without oxygen it dimerizes. (PDB code: 2DHB.)

is synthesized specifically in the fetus. The resulting hemoglobin molecule has a higher affinity for oxygen than that of adult hemoglobin and thus helps in the transfer of oxygen from the mother to the fetus. The gene for the new β -like chain subsequently duplicated and mutated again to produce two new genes, ϵ and γ , the ϵ chain being produced earlier in development (to form $\alpha_2\epsilon_2$) than the fetal γ chain, which forms $\alpha_2\gamma_2$. A duplication of the adult β -chain gene occurred still later, during primate evolution, to give rise to a δ -globin gene and thus to a minor form of hemoglobin ($\alpha_2\delta_2$) that is found only in adult primates (Figure 4-75).

Each of these duplicated genes has been modified by point mutations that affect the properties of the final hemoglobin molecule, as well as by changes in regulatory regions that determine the timing and level of expression of the gene. As a result, each globin is made in different amounts at different times of human development.

The history of these gene duplications is reflected in the arrangement of hemoglobin genes in the genome. In the human genome, the genes that arose from the original β gene are arranged as a series of homologous DNA sequences located within 50,000 nucleotide pairs of one another on a single chromosome. A similar cluster of human α -globin genes is located on a separate chromosome. Not only other mammals, but birds too have their α -globin and β -globin gene clusters on separate chromosomes. In the frog *Xenopus*, however, they are together, suggesting that a chromosome translocation event in the lineage of birds and mammals separated the two gene clusters about 300 million years ago, soon after our ancestors and those of amphibians diverged (see Figure 4-75).

There are several duplicated globin DNA sequences in the α -globin and β -globin gene clusters that are not functional genes but pseudogenes. These have a close sequence similarity to the functional genes but have been disabled by mutations that prevent their expression as functional proteins. The existence of such pseudogenes makes it clear that, as expected, not every DNA duplication leads to a new functional gene. Indeed, the human genome is thought to contain more pseudogenes than genes.

Genes Encoding New Proteins Can Be Created by the Recombination of Exons

The role of DNA duplication in evolution is not confined to the expansion of gene families. It can also act on a smaller scale to create single genes by stringing together short duplicated segments of DNA. The proteins encoded by genes generated in this way can be recognized by the presence of repeating similar protein domains, which are covalently linked to one another in series. The immunoglobulins (Figure 4-76), for example, as well as most fibrous proteins (such as collagens) are encoded by genes that have evolved by repeated duplications of a primordial DNA sequence.

In genes that have evolved in this way, as well as in many other genes, each separate exon often encodes an individual protein folding unit, or domain. It is believed that the organization of DNA coding sequences as a series of such exons separated by long introns has greatly facilitated the evolution of new proteins. The duplications necessary to form a single gene coding for a protein with repeating domains, for example, can easily occur by breaking and rejoining the DNA anywhere in the long introns on either side of an exon. Without introns there would be only a few sites in the original gene at which a recombinational exchange between DNA molecules could duplicate the domain and not disrupt it. Moreover, introns often contain sequences that are repeated many times in a genome, facilitating recombination between different introns. By enabling recombination at many potential sites rather than just a few, introns increase the probability that a duplication event will produce a new protein.

More generally, we know from genome sequences that the various parts of genes—both their individual exons and their regulatory elements—have served as modular elements that have been duplicated and moved about the genome to create the great diversity of living things. Thus, for example, many present-day

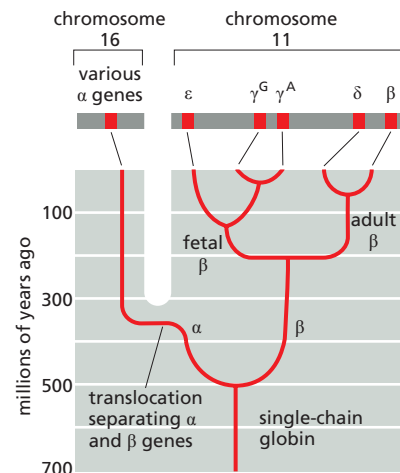


Figure 4-75 An evolutionary scheme for the globin chains that carry oxygen in the blood of animals. The scheme emphasizes the β -like globin gene family. A relatively recent gene duplication of the γ -chain gene produced γ^G and γ^A , which are fetal β -like chains of identical function. The location of the globin genes in the human genome is shown at the top of the figure.

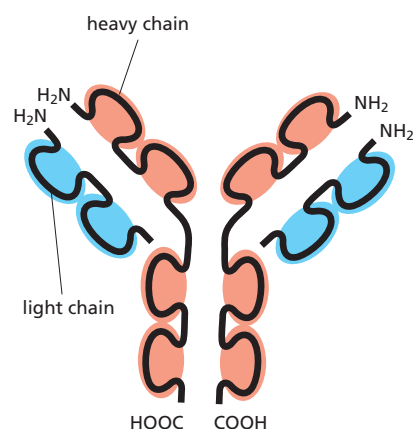


Figure 4-76 Schematic view of an antibody (immunoglobulin) molecule. This molecule is a complex of two identical heavy chains (orange) and two identical light chains (blue). Each heavy chain contains four similar, covalently linked domains, while each light chain contains two such domains. Each of these domains is encoded by a separate exon, and all of the exons are thought to have evolved by the serial duplication of a single ancestral exon.

proteins are formed as a patchwork of domains from different origins, reflecting their complex evolutionary history (see Figure 3–15).

Neutral Mutations Often Spread to Become Fixed in a Population, with a Probability That Depends on Population Size

In comparisons between two species that have diverged from one another by millions of years, it makes little difference which individuals from each species are compared. For example, typical human and chimpanzee DNA sequences differ from one another by about 1%. In contrast, when the same region of the genome is sampled from two randomly chosen humans, the differences are typically about 0.1%. For more distantly related organisms, the interspecies differences outshine intraspecies variation even more dramatically. However, each “fixed difference” between the human and the chimpanzee (in other words, each difference that is now characteristic of all or nearly all individuals of each species) started out as a new mutation in a single individual. If the size of the interbreeding population in which the mutation occurred is N , the initial allele frequency for a new mutation would be $1/(2N)$ for a diploid organism. How does such a rare mutation become fixed in the population, and hence become a characteristic of the species rather than of a few scattered individuals?

The answer to this question depends on the functional consequences of the mutation. If the mutation has a significantly deleterious effect, it will simply be eliminated by purifying selection and will not become fixed. (In the most extreme case, the individual carrying the mutation will die without producing progeny.) Conversely, the rare mutations that confer a major reproductive advantage on individuals who inherit them can spread rapidly in the population. Because humans reproduce sexually and genetic recombination occurs each time a gamete is formed (discussed in Chapter 5), the genome of each individual who has inherited the mutation will be a unique recombinational mosaic of segments inherited from a large number of ancestors. The selected mutation along with a modest amount of neighboring sequence—ultimately inherited from the individual in which the mutation occurred—will simply be one piece of this huge mosaic.

The great majority of mutations that are not harmful are not beneficial either. These selectively *neutral* mutations can also spread and become fixed in a population. In fact, neutral mutations make a large contribution to evolutionary change in genomes. For example, as we saw earlier, they account for most of the DNA sequence differences between apes and humans. The spread of neutral mutations is not as rapid as the spread of the rare strongly advantageous mutations. It depends on a random variation in the number of mutation-bearing progeny produced by each mutation-bearing individual: through a sort of “random walk” process, the mutant allele may eventually become extinct or it may become commonplace. This can be modeled mathematically for an idealized interbreeding population, on the assumption of constant population size and random mating. While neither of these assumptions matches human population history, this idealized case reveals the general principles.

When a new neutral mutation occurs in a population of constant size N that is undergoing random mating, the probability that it will ultimately become fixed is approximately $1/(2N)$. This is because there are $2N$ copies of the gene in the diploid population, and each of them has an equal chance of becoming the predominant version in the long run. For those mutations that do become fixed, the mathematics shows that the average time to fixation is approximately $4N$ generations. Detailed analyses of data on human genetic variation have suggested an ancestral population size of approximately 10,000 at the time when the current pattern of genetic variation was largely established. With a population that has reached this size, the probability that a new, selectively neutral mutation would become fixed is small ($1/20,000$), while the average time to fixation would be on the order of 800,000 years (assuming a 20-year generation time). Thus, while we know that the human population has grown enormously since the development

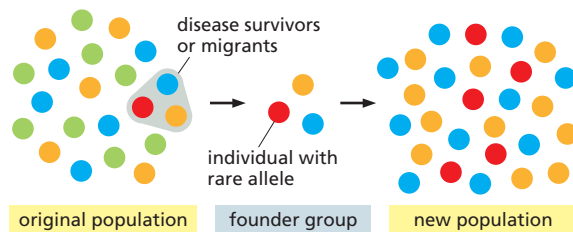


Figure 4–77 How founder effects can determine the set of genetic variants in a population of individuals belonging to the same species. In this diagram, each dot is used to represent an individual, with different colors used to denote the different variants of a particular gene. This example illustrates how, by a chance event that greatly reduces population size, an allele that is initially rare in a large population (*red*) can become established at high frequency in a new population, even though the mutation that produced it has no selective advantage—or is even mildly deleterious.

of agriculture approximately 15,000 years ago, most of the present-day set of common human genetic variants reflects a mixture of variants that was already present long before this time, when the human population was still small.

Similar arguments explain another phenomenon with important practical implications for genetic counseling. In an isolated community descended from a small group of founders, such as the people of Iceland or the Jews of Eastern Europe, a variant form of a gene that is rare in the human population as a whole can often be present at a high frequency, even if that variant is mildly deleterious (**Figure 4–77**).

We Can Trace Human History by Analyzing Genomes

The genomes of ancestral organisms can be inferred, but most can never be directly observed. DNA is very stable compared with most organic molecules, but it is not perfectly stable, and its progressive degradation, even under the best circumstances, means that it is extremely difficult to extract sequence information from fossils that are more than a million years old. Although a modern organism such as the horseshoe crab looks remarkably similar to fossil ancestors that lived 200 million years ago, there is every reason to believe that the horseshoe-crab genome has been changing during all that time in much the same way as in other evolutionary lineages, and at a similar rate. Selection must have maintained key functional properties of the horseshoe-crab genome to account for the morphological stability of the lineage. However, comparisons between different present-day organisms show that the fraction of the genome subject to purifying selection is small; hence, it is fair to assume that the sequence of the genome of the modern horseshoe crab, while preserving features critical for function, must differ greatly from that of its extinct ancestors, known to us only through the fossil record.

It is possible to get direct sequence information by examining DNA samples from ancient materials if these are not too old. In recent years, technical advances have allowed DNA sequencing from exceptionally well-preserved bone fragments that date from more than 100,000 years ago. Although any DNA this old will be imperfectly preserved, genome sequences can be reconstructed from many millions of short DNA sequences. In 2010, investigators completed their analysis of the first Neanderthal genome, obtained from DNA that was extracted from a fossilized bone fragment found in a cave in Croatia. The average difference in DNA sequence between humans and Neanderthals shows that our two lineages diverged somewhere between 270,000 and 440,000 years ago, well before the time that humans are believed to have migrated out of Africa. Neanderthals are one of our closest evolutionary relatives, and they lived side by side with the ancestors of modern humans in Europe and Western Asia. By comparing the Neanderthal genome sequence with those of people from different parts of the world, these studies revealed that many modern humans—particularly those from Europe and Asia—share about 2% of their genomes with Neanderthals. This genetic overlap indicates that our ancestors mated with Neanderthals—before outcompeting or actively exterminating them—on the way out of Africa. These and other ancient relationships still being discovered have thus left a permanent mark in the human genome (**Figure 4–78**).

But what about deciphering the genomes of much older ancestors, those for which no useful DNA samples can be isolated? For organisms that are as closely

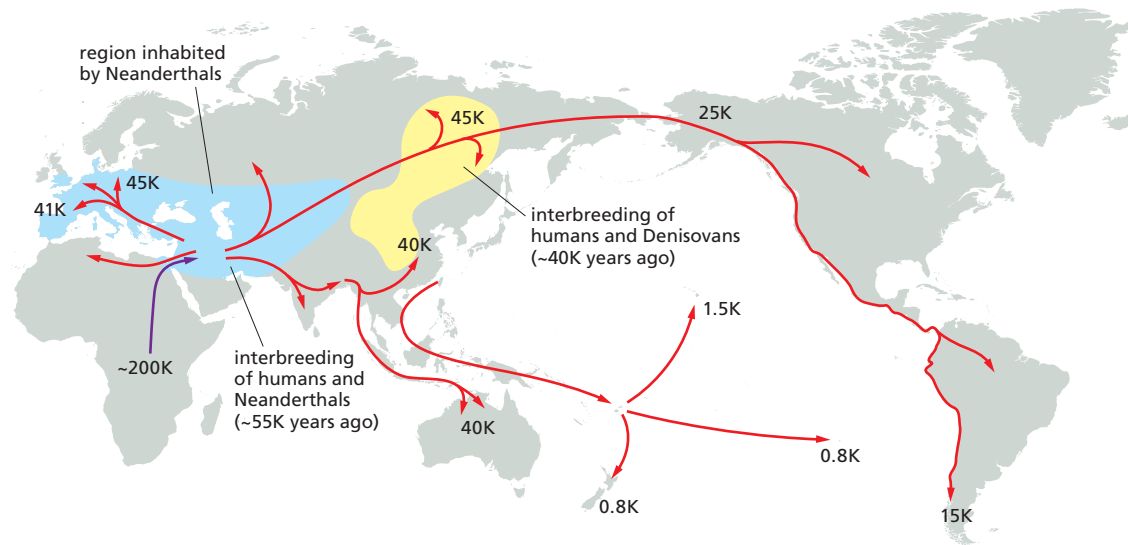


Figure 4-78 Tracing the course of human history by analyses of genome sequences. Modern humans arose in Africa approximately 200,000–300,000 years ago, where the earliest fossils of *Homo sapiens* are found. One small group—perhaps as few as 10,000 individuals—migrated northward, and their descendants spread across the globe. As these ancestral humans left Africa, around 60,000–80,000 years ago (purple arrow), they encountered Neanderthals who inhabited the region indicated in blue. As a result of interbreeding, the humans who subsequently spread throughout Europe and Asia (red arrows) carried with them traces of Neanderthal DNA. Many of those moving eastward would later also interbreed with Denisovans (yellow), a second type of ancient human. Later, ancestral humans continued their global spread to the New World, reaching North America approximately 25,000 years ago and the southern regions of South America 10,000 years later. This scenario is based on many types of data, including fossil records, anthropological studies, and the genome sequences of Neanderthals, Denisovans, and modern humans from around the world. (Adapted from M.A. Jobling et al., *Human Evolutionary Genetics*, 2nd ed. New York: Garland Science, 2014.)

related as human and chimpanzee, we saw that this may not be difficult: reference to the gorilla sequence can be used to sort out which of the few sequence differences between human and chimpanzee are inherited from our common ancestor some 6 million years ago (see Figure 4-65). And for an ancestor that has generated a large number of different species alive today, the DNA sequences of the existing species can be compared simultaneously to unscramble much of the ancestral sequence, allowing scientists to derive DNA sequences much farther back in time. For example, from the genome sequences obtained for dozens of modern placental mammals, it should be possible to infer much of the genome sequence of their 100-million-year-old common ancestor—the precursor of species as diverse as dog, mouse, rabbit, armadillo, and human (see Figure 4-67).

Once the amino acid sequence of an ancestral protein has been inferred, that protein can be readily produced in pure form using the recombinant DNA methods described in Chapter 8. By thus “resurrecting” the extinct protein, its biochemical properties can be compared to those of its modern counterparts. These procedures can be repeated at several branch points in the protein’s evolutionary history, allowing that protein’s stepwise evolution over many millions of years to be directly measured. This approach provides scientists with a powerful way to visualize evolution in action. It can also help us to understand why modern proteins look and behave the way they do.

The Sequencing of Hundreds of Thousands of Human Genomes Reveals Much Variation

Even though the common variant alleles among modern humans originate from the variants present in a comparatively tiny group of ancestors, the total number of variants that exist among modern humans is huge. This is because unavoidable errors occur in the replication and maintenance of our DNA sequences, as will be explained in Chapter 5. As a result, every person is born with 50–100 new changes,

termed *de novo* mutations, in the DNA sequences that they inherit from their parents. Thus, new mutations are constantly occurring and accumulating in human populations. But the vast majority of the variants they create are extremely rare in the human population as a whole, because they arose among a huge number of relatively recent births.

Modern DNA sequencing technologies have made it possible to determine the genome sequences for millions of humans, at costs that have now dropped below \$1000 per genome. Before introducing what these data show, it is important to define a few basic terms. A *variant* is typically defined as a genomic location at which two or more versions of a sequence exist in the human population, while an *allele* refers to the specific form that each such variant takes in a given person. Because humans are diploid organisms, harboring two haploid genomes in their cells (one from their mother and one from their father), if alleles A and B exist for a variant X, an individual's genotype can be either "A/A," "A/B," or "B/B" at that site.

Each human genome sequence is compared to a "reference human genome assembly" that was produced by the public Human Genome Project on the basis of the initial sequences from several individuals. For this comparison, sequence "reads" for millions or billions of DNA fragments obtained from an individual are aligned by a computer to the reference assembly (which is a haploid representation and includes only one allele at each location). The locations at which the sequenced individual displays "alternative" alleles relative to the reference are then identified.

When this is done for many individuals, single-nucleotide variants (SNVs) are found to compose the vast majority of differences among humans. SNVs are points in the genome sequence where some haploid human genomes have one nucleotide, while some have another. They have been extensively characterized because of their abundance, their relative ease of detection, and their value for genetic studies. Historically, the term single-nucleotide polymorphism, or SNP, was reserved for SNVs present at a frequency above 1% in the human population. Today, however, no precise threshold is used to differentiate "rare" and "common" variations, which span frequencies ranging from a singleton "private" variation (present in only one haploid genome within one person) to as high as 50% (a variant with two alleles that exist at equal frequency among human genomes).

Importantly, SNVs are only one type of variation that exists among humans. *Structural variants*, such as deletions, duplications, inversions, or rearrangements of a variety of lengths, are also prevalent features in our genomes. Additional types of variation—including mobile-element insertions and low-complexity repeat expansions or contractions—are also important components of human genetic diversity.

Two haploid human genomes sampled from the modern world population will differ from one another at roughly 1 per 1,000 nucleotides. This number, however, varies considerably between human populations, and it is affected by historical factors such as population bottlenecks, breeding patterns, and ancestral population sizes. And the more closely related two individuals are to one another, the more similar their genomes will be. The above number does not include the considerable structural diversity present among human genomes. Indeed, because they often affect tens or even hundreds of thousands of nucleotides, structural variants are a large contributor to the total number of nucleotide-level differences between any two human genomes (Table 4-3).

Most of the Variants Observed in the Human Population Are Common Alleles, with at Most a Weak Effect on Phenotype

A comparison of any one human genome with the reference human genome will typically find 3–5 million locations at which the sequenced individual harbors at least one allele different from that seen in the reference assembly. Some of these variants are heterozygous within that person, which is to say the individual harbors two distinct alleles, typically one that matches the reference and one "alternative" allele. More than 95% of the variant alleles observed within

TABLE 4–3 Typical Differences Between Any One Human Being’s Genome Sequence and the Reference Human Genome

Type of difference	Size in nucleotide pairs	Differences per genome
Single-nucleotide variation (SNV)	1	3–4 million
Small deletion or insertion (indel)	1–49	0.4–0.5 million
Low-complexity simple sequence repeats (microsatellite and satellite DNA repeats)	1–200	100,000
Mobile-element insertion (SINE, LINE)	300–7000	2000
Structural variation (deletions, duplications, and inversions)	50 to >1,000,000	Tens of thousands; length is inversely correlated with frequency
Karyotypically visible abnormalities (e.g., aneuploidies)	Chromosome scale	Very rare; most are lethal

Courtesy of Greg Cooper and Rick Myers, HudsonAlpha Institute for Biotechnology, Huntsville, AL; based on H.J. Abel et al., *Nature* 583:83–88, 2020; gnomAD (<https://www.nature.com/immersive/d42859-020-00002-x/index.html>); and <https://www.internationalgenome.org>).

any given person are relatively common, reflecting their ancient origin. A small fraction of variant alleles will be rare; some of these, as described earlier, represent *de novo* mutations and are extremely rare, potentially existing within only that individual.

As will be described in Chapter 8, those SNVs that are frequent in the human genome have been extremely useful for genetic mapping analyses, known as genome-wide association studies (GWAS). Here, one attempts to associate specific human traits (phenotypes) with a large set of alleles that are relatively common in the human population (see p. 526). Most of these frequent SNVs have little or no effect on human fitness. This is as expected, as the deleterious SNVs will have been selected against during human evolution and should therefore be rare. Importantly, this property is generally true of all variation, including structural variation. Indeed, common structural variants also tend to not be deleterious because, if they were, they would not have arisen to appreciable frequencies. Across all forms of variation then, allele frequency is of crucial importance: common variant alleles tend to have at most weak effects on human phenotypes, and large-effect variants tend to be rare. On the other hand, it is important to note that just because a variant is rare does not mean it influences a phenotype; most rare variants, in fact, have little to no effect.

In recent years, GWAS research has identified many associations between common traits, including common diseases, and common alleles. This is a result of the fact that while common alleles tend to have at most weak effects, the combined effects of many common alleles can ultimately produce a large impact on phenotype.

Forensic Analyses Exploit Special DNA Sequences with Unusually High Mutation Rates

While mutation rates in humans are generally quite low, typically estimated to be on the order of 10^{-8} (that is, roughly one mutation every 100 million base pairs), certain sequences with exceptionally high mutation rates stand out. A dramatic

example is provided by CACA repeats, which are abundant in the human genome and in the genomes of other eukaryotes. Segments of the genome with the motif $(CA)_n$ are replicated with relatively low fidelity because a slippage occurs between the template and the newly synthesized strands during DNA replication; hence, the precise value of n can vary over a considerable range from one genome to the next. These repeats make ideal DNA-based genetic markers because most humans are heterozygous at each of these loci, having inherited one repeat length (n) from their mother and a different repeat length from their father. While the value of n changes sufficiently rarely that most parent–child transmissions propagate CACA repeats faithfully, the changes are sufficiently frequent to maintain high levels of heterozygosity in the human population. Because of this, these and other simple repeats that display exceptionally high variability provide a *DNA fingerprint* that is widely used for identifying individuals by DNA analysis in crime investigations, paternity suits, and other forensic applications (see Figure 8–37).

An Understanding of Human Variation Is Critical for Improving Medicine

While most variations in the human genome are thought to have weak effects on phenotype, a subset of our genome sequence variations must be responsible for the heritable aspects of human individuality. Because it is now possible to sequence individual genomes cheaply and rapidly, we can link even rare alleles with specific phenotypes. We know that even a single nucleotide change that alters one amino acid in a protein can cause a serious disease, as for example in sickle-cell disease, which, although not rare, is caused by such a mutation in hemoglobin (**Movie 4.3**). We also know that gene dosage—a doubling or halving of the copy number of some genes—can have a profound effect on development by altering the level of gene product, as can a change in one of the many regulatory DNA sequences that are dispersed throughout the vast expanse of noncoding DNA in the human genome (see Chapter 7).

Some of the many differences between the genomes of any two human beings will have substantial effects on human health, physiology, behavior, and physique. A major challenge in modern human genetics is to discriminate those differences from the majority that are of little consequence. Significant progress is being made; for instance, we now know the nucleotide changes that give rise to thousands of rare inherited traits and diseases. These and many other results are greatly expanding our understanding of human biology. This understanding is critical for the new field of precision medicine, in which both disease prevention and disease treatments will be tailored to take account of individual genetic differences.

Summary

Comparisons of the nucleotide sequences of present-day genomes have revolutionized our understanding of gene and genome evolution. Because of the extremely high fidelity of DNA replication and DNA repair processes, random errors in maintaining the nucleotide sequences in genomes occur rarely (for example, only 50–100 new mutations for every person born). Not surprisingly, therefore, a comparison of human and chimpanzee chromosomes—which are separated by about 12 million years of evolution (double the ~6 million years since humans and chimps shared a common ancestor)—reveals fixed changes at only ~1% of base pairs. Not only are our protein-coding sequences highly similar, but also their order on each chromosome is almost identical. Although a substantial number of segmental duplications and segmental deletions have occurred, even the positions of the transposable elements that make up a major portion of our noncoding DNA are mostly unchanged.

When one compares the genomes of two more distantly related organisms—such as a human and a mouse, separated by about 90 million years—one finds

many more changes. While selection leaves detectable signatures at all time scales, including for example through comparisons of human genomes to one another or through comparisons of the closely related human and chimp genomes, at greater distances the actions of selection are more pronounced. Comparison of human and mouse genomes, for example, reveal that most coding sequences (exons) and many regulatory regions have been highly conserved by the actions of selection, while huge fractions of our respective genomes have been altered to such an extent that one can detect no similarity at all among them.

Because of purifying selection, the comparison of the genome sequences of multiple related species is an especially powerful way to find DNA sequences with important functions. While it has been estimated that between 8 and 10% of the human genome has been conserved as a result of purifying selection, the function of the majority of this DNA (tens of thousands of sequence elements conserved across mammalian evolution) remains mysterious. Experiments that characterize their functions continue to teach us many new lessons about vertebrate biology.

Other sequence comparisons show that a great deal of the genetic complexity of present-day organisms is due to the expansion of ancient gene families. DNA duplication followed by sequence divergence has clearly been a major source of genetic novelty during evolution. On a more recent time scale, the diploid genomes of any two humans will differ from each other both because of nucleotide substitutions (single-nucleotide variations) and because of inherited DNA gains and DNA losses (structural variations). Deciphering the effects of these differences will improve both medicine and our understanding of human biology.

PROBLEMS

Which statements are true? Explain why or why not.

4-1 Human females have 23 different chromosomes, whereas human males have 24.

4-2 The four core histones are relatively small proteins with a very high proportion of positively charged amino acids; the positive charge helps the histones bind tightly to DNA, regardless of its nucleotide sequence.

4-3 Nucleosomes bind DNA so tightly that they cannot move from the positions where they are first assembled.

4-4 The long linear DNA molecule in an interphase chromosome is organized into loops of chromatin that appear to emanate from a central axis.

4-5 In a comparison between the DNAs of related organisms such as humans and mice, identifying the conserved DNA sequences facilitates the search for functionally important regions.

4-6 Gene duplication and divergence are thought to have played a critical role in the evolution of increased biological complexity.

Discuss the following problems.

4-7 DNA isolated from the bacterial virus M13 contains 25% A, 33% T, 22% C, and 20% G. Do these results

strike you as peculiar? Why or why not? How might you explain these values?

4-8 A segment of DNA from the interior of a single strand is shown in **Figure Q4-1**. Should this sequence be written as ACT or TCA? Why?

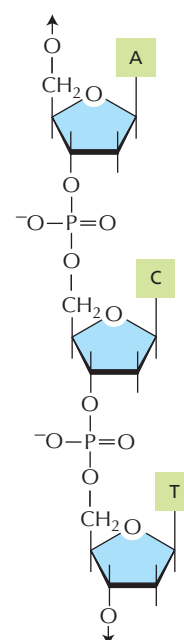


Figure Q4-1 Three nucleotides from the interior of a single strand of DNA (Problem 4-8). Arrows at the ends of the DNA strand indicate that the structure continues in both directions.

4-9 Human DNA contains 20% C on a molar basis. What are the mole percents of A, G, and T?

4-10 In contrast to histone acetylation, which always correlates with gene activation, histone methylation can lead to either transcriptional activation or repression. How do you suppose that the same modification—methylation—can mediate different biological outcomes?

4-11 Why is a chromosome with two centromeres (a dicentric chromosome) unstable? Would a backup centromere not be a good thing for a chromosome, giving it two chances to form a kinetochore and attach to microtubules during mitosis? Surely, a backup centromere would ensure that the chromosome did not get left behind at mitosis.

4-12 Look at the two yeast colonies in **Figure Q4-2**. Each of these colonies contains about 100,000 cells descended from a single yeast cell, originally somewhere in the middle of the clump. A white colony arises when the *Ade2* gene is expressed from its normal chromosomal location. When the *Ade2* gene is moved near a telomere, however, it is packed into heterochromatin and inactivated in most cells, giving rise to colonies that are mostly red. In these largely red colonies, white sectors fan out from the middle of the colony. In both the red portions and the white sectors, the *Ade2* gene is still located near telomeres. Explain why white sectors have formed near the rim of the red colony. On the basis of the patterns observed, what can you conclude about the propagation of the transcriptional state of the *Ade2* gene from mother to daughter cells in this experiment?

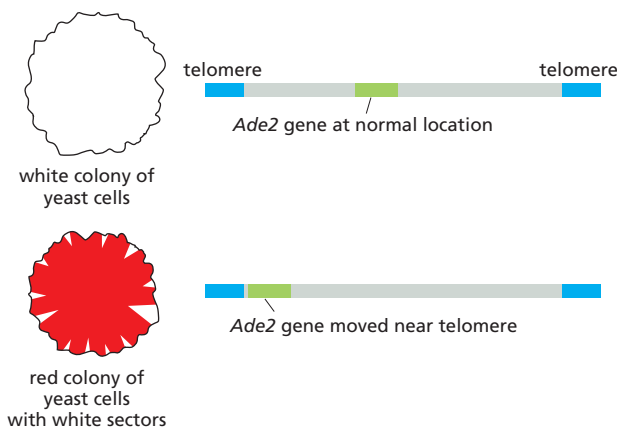


Figure Q4-2 Position effect on expression of the yeast *Ade2* gene (Problem 4-12). The *Ade2* gene codes for one of the enzymes for adenosine biosynthesis, and the absence of the *Ade2* gene product leads to the accumulation of a red pigment. Therefore, a colony of cells that express *Ade2* is *white*, and one composed of cells in which the *Ade2* gene is not expressed is *red*.

4-13 Chromosomes from different amphibians form typical lampbrush chromosomes when injected into oocytes as demembrated sperm heads. When the sperm heads from *Rana pipiens* (northern leopard frog), which

forms large loops in its own oocyte chromosomes, were injected into *Xenopus laevis* oocytes, the resulting lampbrush chromosomes had the small loops typical of those found in *X. laevis* oocytes. Similarly, when the sperm heads from *X. laevis* were injected into *Notophthalmus viridescens* (red-spotted newt) oocytes, the resulting lampbrush chromosomes had the very large loop structure typical of *N. viridescens*.

Do these heterologous injection experiments support the idea that loop structure is a fixed property of a chromosome? Why or why not?

4-14 Mobile pieces of DNA—transposable elements—that insert themselves into chromosomes and accumulate during evolution make up more than 40% of the human genome. Transposable elements of four types—long interspersed nuclear elements (LINEs), short interspersed nuclear elements (SINEs), long terminal repeat (LTR) retrotransposons, and DNA transposons—are inserted more or less randomly throughout the human genome. These elements are conspicuously rare at the four homeobox gene clusters, *HoxA*, *HoxB*, *HoxC*, and *HoxD*, as illustrated for *HoxD* in **Figure Q4-3**, along with an equivalent region of chromosome 22, which lacks a *Hox* cluster. Each *Hox* cluster is about 100 kilobases (kb) in length and contains 9–11 genes, whose differential expression along the anteroposterior axis of the developing embryo establishes the basic body plan for humans (and for other animals). Why do you suppose that transposable elements are so rare in the *Hox* clusters?

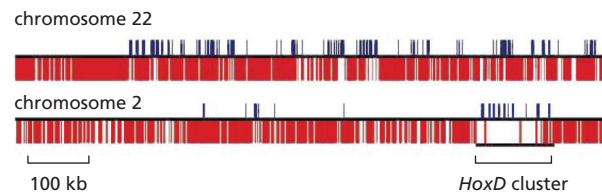


Figure Q4-3 Transposable elements and genes in 1-megabase regions of chromosomes 2 and 22 (Problem 4-14). *Blue lines* that project *upward* indicate exons of known genes. *Red lines* that project *downward* indicate transposable elements; they are so numerous (constituting more than 40% of the human genome) that they merge into nearly a solid block outside the *Hox* clusters. (Adapted from E. Lander et al., *Nature* 409:860–921, 2001. With permission from Springer Nature.)

4-15 Chromosome 3 in orangutans differs from chromosome 3 in humans by two inversion events that occurred in the human lineage (**Figure Q4-4**). Draw the

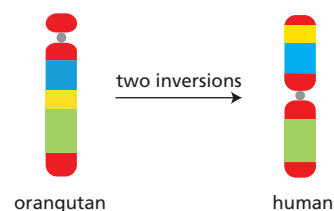


Figure Q4-4 Chromosome 3 in orangutans and humans (Problem 4-15). Differently colored blocks indicate segments of the chromosomes that are homologous in DNA sequence.

intermediate chromosome that resulted from the first inversion and explicitly indicate the segments included in each inversion.

4–16 There has been a colossal snafu in the maternity ward at your local hospital. Four sets of male twins, born within an hour of each other, were inadvertently shuffled in the excitement occasioned by that unlikely event. You have been called in to set things right. As a first step, you want to get the twins matched up. To that end, you analyze a small blood sample from each infant using a hybridization probe that detects differences in the numbers of simple sequence repeats such as $(CA)_n$ located in widely scattered regions of the genome. The results are shown in **Figure Q4–5**.

- A.** Which infants are brothers? Are they all identical twins?
B. How will you match brothers to the correct parents?



Figure Q4–5 DNA fingerprint analysis of shuffled twins (Problem 4–16).

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